ALAGAPPA UNIVERSITY

(A State University Established in 1985)
Karaikudi - 630003, Tamil Nadu, India

2017 — Accredited with A+ Grade by NAAC (CGPA : 3.64)

2018 — MHRD Govt. of India; UGC — Graded as Category - 1 & Granted Autonomy

2018 — MHRD Government of India — Swachh Campus Rank : 4

2019 — nirf National Institutional Ranking Framework — Rank : 28

2019 — QS — India Rank : 20, BRICS Rank : 104, Asia Rank : 216

DEPARTMENT OF WOMEN'S STUDIES

M.A., INTEGRATED HOME SCIENCE

[Choice Based Credit System (CBCS)]

[For the candidates admitted from the academic year 2019-2020]

ALAGAPPA UNIVERSITY
DEPARTMENT OF WOMEN'S STUDIES
(A State University Established in 1985)
KARAIKUDI - 630 003, Tamil Nadu, India
www.alagappauniversity.ac.in

2017	2018	2018	2018	2019
Accredited with A+ Grade by NAAC (CGPA : 3.64)	Graded as Category-1 & Granted Autonomy	Rank : 27	Swachh Campus Rank : 4	India Rank : 20 BRICS Rank: 104 Asia Rank : 216

Minutes of the Broad Based Board of Studies Meeting for M.A. Integrated Home Science

Broad Based Board of Studies meeting for M.A. Integrated Home Science was held on 26.05.2019 &27.05.2019 at 10.00 am in the Department of Women's Studies. The following members were present.

1.	Dr.K.Manimekalai Professor & Head Department of Women's Studies Alagappa University Karaikudi	Chair person
2.	Dr. Meera Kaur, Associate Professor, Department of Family Medicine, University of Manitoba, 2MA05-2300 Mcphillips Street, Winnipeg MB R2V 3M3 Canada	Foreign Subject Expert
3.	Dr. S. Lakshmi Manokari, Professor & Head, Department of Textiles and Apparel Design, Periyar University, Salem.	Subject Expert
4.	Dr. K. Sangeetha, Professor and Head, Textiles and Apparel Design, Bharathiar University, Coimbatore	Subject Expert
5.	Mr. G. Somasundaram, South Zone Head (Textiles and Clothing), ILFS Clusters Development Initiative Limited, 7/1, 2nd Floor, 6th Cross, Dharmaram College Post, Bangalore	Co-opted Member from Industry
6.	Dr.E. Kannapiran Dean i/c College Development Council Alagappa University, Karaikudi	Ex officio Member

7. Dr. P. Veeramani, Member
Assistant Professor
Department of Women's Studies
Alagappa University,
Karaikudi

Dr.K.Manimekalai welcomed the members and explained the agenda for the BBOS meeting. The Board members discussed the agenda and the following resolutions were made;

1. The Board has reviewed the revised syllabus meant for M.A Integrated Home Science

2. Foreign expert Prof. Dr. Meera Kaur, Department of Family Medicine, University of Manitoba, Canada Associate Professor, has expressed her satisfaction about the syllabus and she has made changes to enrich the syllabus so as to have world class curriculum.

3. The credit system for the under graduation of the affiliated colleges under Alagappa University have been adopted and approved by the board

4. The credit system for the post graduation of our Alagappa University have been adopted and approved by the board

5. The syllabus for the subjects Tamil and English prescribed to the affiliated colleges under Alagappa University have been adopted and approved by the board

6. The purchased books and Journals for the discipline Home Science have been approved by the board

Members	**Signature**
Dr.K.Manimekalai	K. Mani 27.5.19
Dr. Meera Kaur	27-5-19
Dr. S. Lakshmi Manokari	27/5/19.
Dr. K. Sangeetha	
Mr. G. Somasundaram	
Dr.E. Kannapiran	E. Kannapiran 12/6/19
Dr. P. Veeramani	P. Veeramani

ALAGAPPA UNIVERSITY, KARAIKUDI – 630 003

CHOICE BASED CREDIT SYSTEM (CBCS)

&

CREDIT FRAMEWORK FOR M.A. INTEGRATED HOME SCIENCE

REGULATIONS

(Applicable to all the candidates admitted from the academic year 2019-20 onwards)

1.PROGRAMME GENERAL OBJECTIVES

The general objective of Home Science is to contribute towards national development by training students to take up leadership roles in extension and community outreach programs. The hall mark of education in Home Science is to familiarise the students with the use of newer technologies, methods in family and community linkages, and sustainable use of resources for human development. As a discipline Home Science integrates the ingredients of the sciences, social sciences and technology to facilitate the study of and enhance the quality of human life.

2. PROGRAMME SPECIFIC OBJECTIVES

- To develop the capabilities and knowledge of students in the areas of
- Food Science, Gender Studies, Nutrition through Life Cycle, Human Physiology, Food Microbiology, Gender and Development, Extension ,Environmental Management, Housekeeping and Front Office Management
- Food Service Management, Therapeutic Nutrition and Practical
- Fashion Technology and Garment Construction, Apparel Merchandising, Wet Processing, Fashion Clothing and Psychology
- Research Methodology
- Computer Application in Nutrition and Dietetics/ Textiles Clothing

3. PROGRAMME OUTCOMES

On successful completion of the programme

- The students will be able to promote capacity building training programme for individuals and communities for social and economic empowerment.
- The student will gain and provide employment in research organizations, food and textile industries, dietetic practice, education and child development domains, accreditation of green buildings, strategic planning and communication technologies.

4. ELIGIBILITY:

i) For Admission: A pass in the Higher Secondary Examination (Academic / Vocations Stream) conducted by the Government of Tamil Nadu.

5. DURATION:

The course is for a period of five years. Each academic year shall comprise of two semester viz. Odd and Even semesters. Odd semesters shall be from June / July to October / November and Even Semesters shall be from November / December to April /May. There shall be not less than 90 working days which shall comprise 450 teaching clock hours for each semester. (Exclusive of the days for the conduct of University end-semester examinations)

i) The M.A. Integrated Home Science course is for a period of five years.

The M. A. Integrated Home Science course has single entry and double exit points. Thus, the students can opt to leave (if passes the examinations) in the following stages with appropriate B.A Home Science /M.A. Integrated Home Science as indicated in

Table 1:

Part I Credits	Part II **Credits**	Part III Credits	Part IV **Credits**	Part V **Credits**	Total Credits for Award	Normal Duration	Exit Points/ Awards
12	12	95	20	1	140	Six Semesters	B.A. Home Science
-	-	90 (Core, Elective& Non Major Elective)	-	-	90	Four Semesters	M.A. Integrated Home Science

For the Degree (B.A. Home Science): The candidates shall have subsequently undergone the prescribed course of study for a period of not less than three academic years, passed the examinations prescribed and fulfilled such conditions as have been prescribed therefore.

For the Degree (M.A Integrated Home Science): The candidates shall have subsequently undergone the prescribed course of study for a period of not less than five academic years, passed the examinations prescribed and fulfilled such conditions as have been prescribed therefore.

6. THE CBCS SYSTEM:

All programmes (named after the core subject) mentioned earlier shall be run on Choice Based Credit System (CBCS). It is an instructional package developed to suit the needs of students to keep pace with the developments in higher education and the quality assurance expected of it in the light of liberalization and globalization in higher education.

7. COURSES IN PROGRAMMES:

A. Courses in UG Programmes

The UG & PG programme consists of a number courses. The term 'course' is applied to indicate a logical part of the subject matter of the programme and is invariably equivalent to the subject matter of a 'paper' in the conventional sense. The following are the various categories of courses suggested for the UG programmes.

Part I – Language Courses (LC) (Tamil special subjects designed in lieu of the above).

Part II – English Language courses (ELC) or special subject designed in lieu of.

The Language courses and English Language Courses are 4 each in number and the LC and ELC are meant to develop the students' communicative skill at the UG level. Core courses are the basic courses compulsorily required for each of the programme of study.

Part III includes Core Course (CC), Allied Course (AC) and Elective Course (EC).

I. Core courses are the basic courses compulsorily required for each of the program of study. These will be related to the subject of programme in which the candidate gets his / her degree. The number of Core Courses shall be 15.
II. Allied Courses cover preferably two disciplines that are generally related to the main subject of the programme. Each discipline shall provide 4 Allied Courses.
III. Elective Courses are three in number for the UG programme. The programme shall provide three set of Elective Courses (each set contain 2 Elective courses). Out of which, a student is required to choose three Elective courses from the options given in the respective Discipline. Six Elective Courses are given to the students. A student shall choose three Elective Courses from the list of Elective Courses offered at their respective disciplines.

Selection of students to the EC:

The Department Committee shall follow a selection procedure on a first come first served basis, fixing the maximum number of students, giving counselling to the students etc. to avoid overcrowding to particular course (s) at the expense of some other courses. The Colleges shall provide all information relating to the ECs in each programme to all the students so as to enable them to choose their ECs.

Part IV: It consists of five categories:

I. Those who have not studied Tamil upto XII standard and taken a Non-tamil language under Part I shall take Tamil comprising of two courses (level will be at 6th Standard).Those who have studied Tamil upto XII standard and taken a non-tamil language under Part I shall take advanced Tamil comprising of two core subjects.

II. Two Non –Major Elective in Third and Fourth Semester

III. Skill-based Subjects: In order to enhance the employability skills of the students Skill Based Subjects are included from third semesters.

IV. UGC sponsored Ability enhancement compulsory course(AECC): Environmental studies course in Second Semester

V. Value Education in First Semester

Part V Extension and Extra Curricular Activities: These should be carried out outside the class hours

SELF LEARNING COURSES (SLC) under MOOCs in the Semester III and IV

SEMESTERS:

An academic year is divided into two Semesters. In each semester, courses are offered in 15 teaching weeks and the remaining 5 weeks are to be utilized for conduct of examinations and evaluation purposes. Each week has 30 working hours spread over 5 / 6 days a week.

B. Courses in PG programme

The following are the various categories of courses suggested for the PG programmes.

I. **CORE COURSES (CC) 13** papers for project pattern

II. **ELECTIVE COURSES (EC) 3** in the Semester I, II and III

III. **NON - MAJOR ELECTIVE COURSES (NME)** in the Semester III and IV

IV. **SELF LEARNING COURSES (SLC) under MOOCs** in the Semester II and III

V. **Project Work (PW**) in the Semester IV

i) A candidate who does not pass the examination in any course(s) may be permitted to appear in such failed course(s) in the subsequent examinations to be held in October / November or April / May. However candidates who have arrears in Practical shall be permitted to take their arrear Practical examination only along with Regular Practical examination in the respective semester.

ii) A candidate should get registered for the first semester examination. If registration is not possible owing to shortage of attendance beyond condonation limit / regulation prescribed OR belated joining OR on medical grounds, the candidates are permitted to move to the next semester. Such candidates shall re-do the missed semester after completion of the programme.

iii) Viva-Voce: Each candidate shall be required to appear for Viva-Voce Examination (in defense of the Project only).

iv) For the Project Report, the maximum marks will be 75 percent for project report evaluation and for the Viva-Voce it is 25 percent (if in some programmes, if the project is equivalent to more than one course, the project marks would be in proportion to the number of equivalent courses).

v) The results of all the examinations will be published through the University Department where the student underwent the course as well as through University Website.

8. CREDITS:

The term 'credit' refers to the weight age given to a course, usually in relation to the instructional hours to it. For instance, a six hour course per week is assigned to five/ four credits, four / five hour course per week is assigned to four / three credits and two hour course per week is given to two credits. However, in no instance the credits of a course can be greater than the hours allotted to it.

The total minimum credits, required for completing a UG programme is 140. The details of credits for individual components and individual courses are given in Table – 2.

The total minimum credits, required for completing a PG programme is 90.The details of the number of courses and credits per course for PG programmes are givenin Table -3

9. COURSE:

Each course is to be designed variously under lectures / tutorials / laboratory or field work / seminar / practical training / Assignments / Term paper or Report writing etc., to meet effective teaching and learning needs.

10. CONDONATION:

Students must have 75% of attendance in each course for appearing the examination. Students who have 74% to 70% of attendance shall apply for condonation in the prescribed form with the prescribed fee. Students who have 69% to 60% of attendance shall apply for condonation in prescribed form with the prescribed fee along with the Medical Certificate.

Students who have below 60% of attendance are not eligible to appear for the examination. They shall re-do the semester(s) after completion of the programme.

11. QUESTION PAPER PATTERN:

Core & Allied Papers

Part A

Ten questions (No choice) 10 x 2 = 20 marks

Two questions from each unit

Part B

Five questions (either or type) 5 x 5 = 25 marks

One question from each unit

Part C

Three questions out of five 3 x 10 = 30 marks

One question from each unit

Distribution of marks between Theory, Practical and Project:

Core / Skill Based	Int.	Ext.	Total
Theory papers	25	75	100
Practical papers	25	75	100
Internship	25	75	100
Project work	25	75	100

12. EVALUATION:

The performance of a student in each course is evaluated in terms of percentage of marks with a provision for conversion to grade points. Evaluation for each course shall be done by a continuous internal assessment by the concerned course teacher as well as by an end semester examination and will be consolidated at the end of the course. The components for continuous internal assessment are:

Two tests	-	15 marks (third / repeat test for genuine absentees)
Seminar / Quiz	-	5 marks
Assignments	-	5 marks
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Total	-	25 Marks

Attendance need not be taken as a component for continuous assessment, although the students should put in a minimum of 75% attendance in each course. In addition to continuous evaluation component, the end semester examination, which will be a written type examination of at least 3 hours duration, would also form an integral component of the evaluation. The ratio of marks to be allotted to continuous internal assessment and to end semester examination is 25 : 75. The evaluation of laboratory component, wherever applicable will also be based on continuous internal assessment and on an end-semester practical examination.

13. PASSING MINIMUM:

The passing minimum for CIA shall be 40% out of 25/15* marks (i.e.10/6* marks) in Theory papers and 40% out of 40/10* marks (i.e. 16/4* marks) in Practical Examinations.

Failed candidates in the Internal Assessment are permitted to improve their Internal Assessment marks in the subsequent semesters. (2 chances will be given) by writing test and by submitting Assignments.

The passing minimum for University Examinations shall be 40% out of 75 marks (i.e. 30 marks) for Theory papers and 40% out of 60 marks (i.e. 24 marks) for Practical papers.

14. GRADING

Once the marks of the CIA and end-semester examinations for each of the courses are available, they will be added. The marks thus obtained will then be graded as per details provided in Table 4.

Computation of SGPA and CGPA: Following procedure to compute the Semester Grade Point Average (SGPA) and Cumulative Grade Point Average (CGPA) may be adopted:

- The SGPA is the ratio of sum of the product of the number of credits with the grade points scored by a student in all the course components taken by a student and the sum of the number of credits of all the courses undergone by a student, i.e.

  $$SGPA(Si) = \sum (Ci x Gi) / \sum Ci$$

  Where 'Ci' is the number of credits of the ith course component and 'Gi' is the grade point scored by the student in the ith course component.

- The CGPA is also calculated in the same manner taking into account all the courses undergone by a student over all the semesters of the programme, i.e.

  $$CGPA = \sum (Ci x Si) / \sum Ci$$

  Where 'Si' is the SGPA of the ith semester and 'Ci' is the total number of credits in that semester.

- The SGPA and CGPA shall be rounded off to 2 decimal points and reported in the transcripts.

- UGC guidelines on Choice Based Credit System (CBCS) may be referred for further illustration on computation of SGPA, CGPA etc.

15. CLASSIFICATION OF FINAL RESULTS (TABLE – 4)

For each of the three parts, there shall be separate classification on the basis of CGPA as indicated in Table – 5.

16. CONFERMENT OF THE DEGREE

A candidate shall be eligible for the conferment of the Degree of Bachelor of Arts only if he / she has earned the minimum required credits for the programme prescribed therefore (i.e.140 credits).

A candidate shall be eligible for the conferment of the Degree only after he / she has earned the minimum required credits for the programme prescribed therefore (i.e.90 credits).

17. RANKING: UNIVERSITY EXAMINATION

The top scorer in the Semester Examinations would be declared as University Rank Holders. Rank Certificate will be issued for a Programme.

18. REVISION OF REGULATIONS AND CURRICULUM

The University may from time to time revise, amend and change the Regulations and Curriculum, if found necessary.

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Table – 2

Details on the number of courses and credits per course in the UG programme

Study Components		Number of Courses	Credits per Course	Total Credits	Total hours / 180 weekly hours
Language Course (LC)		4	3	12	20
English Language Course (ELC)		4	3	12	20
Core Course (CC) Theory		11	5	55	55
Core Course (CC) Practical		3	4	12	34
Internship		1	3	3	
Allied Course (AC)		4	4	16	16
Elective Courses (EC)		3	3	9	9
Part IV Courses: a) Non-major Elective		2	2	4	6
b) Skill based subjects	Theory	4	2	12	8
	Practical	2	2	-	8
c) Environmental Studies		1	2	2	2
d) Value Education		1	2	2	2
Part V: Extension activities		1	1	1	-
TOTAL				140	180

Table – 3.a

Details on the number of courses and credits per course in the PG programme with specialization Nutrition and Dietetics

Semester	I	II	III	IV	Total hours/120 weekly hour
CC (5credits) Theory	3	2	3	2	53
CC (5credits) Practical	1	-	-	1	20
EC (4credits)	1	1	1		14
NME (2 credits)	-	1	1	-	6
Internship (CC)(8 credits)	-	1	-	-	10
Project (6 credits)	-	-	-	1	10
Library	-	-	-	-	1
Library, Yoga & Career guidance	-	-	-	-	6
SLC	-	Extra credits	Extra credits	-	-
Credits per Semester	24	24	21	21	**120**
Total credits	**90 +** Extra credits				

Table – 3.b

Details on the number of courses and credits per course in the PG programme with specialization Textiles and Clothing

Semester	I	II	III	IV	Total hours/120 weekly hour
CC (5credits) Theory	3	2	2	2	45
CC (5credits) Practical	1	-	1	1	30
EC (4credits)	1	1	1	-	12
NME (2 credits)	-	1	1	-	6
Internship (CC)(8 credits)	-	1	-	-	10
Project (6 credits)	-	-	-	1	10
Library	-	-	-	-	1
Library, Yoga & Career guidance	-	-	-	-	6
SLC		Extra credits	Extra credits		
Credits per Semester	24	24	21	21	**120**
Total credits	**90 +** Extra credits				

Table 4

Grading of the Courses

Marks	Grade Point	Letter Grade
96 and above	10	S^{+}
91 – 95	9.5	S
86 – 90	9.0	D^{++}
81 – 85	8.5	D^{+}
76 – 80	8.0	D
71 – 75	7.5	A^{++}
66 – 70	7.0	A^{+}
61 – 65	6.5	A
56 – 60	6.0	B^{+}
51 – 55	5.5	B
46 – 50	5.0	C^{+}
40 – 45	4.5	C
Below 40	0	F

Table 5

Final Result

CGPA	Letter Point	Classification of Final Result
9.51 and above	S^{+}	First Class – Exemplary
9.01 – 9.50	S	
8.51 – 9.00	D^{++}	First Class – Distinction
8.01 – 8.50	D^{+}	
7.51 – 8.00	D	
7.01 – 7.50	A^{++}	First Class
6.51 – 7.00	A^{+}	
6.01 – 6.50	A	
5.51 – 6.00	B^{+}	Second Class
5.01 – 5.50	B	
4.51 – 5.00	C^{+}	Third Class
4.00 – 4.50	C	
Below 4.00	F	Fail

Credit based weighted Mark System is adopted for individual semesters and cumulative semesters in the column 'Marks Secured' (for 100).

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B.A. HOME SCIENCE

S.No	Part	Code	I Semester	Credit	Hrs/ week	Marks		Total Marks
						Int	Ext	
1.	I	4531T1	**Tamil-** பொதுத்தமிழ் தாள்-1 தற்காலக் கவிதையும் உரைநடையும்	3	5	25	75	100
2.	II	4531E1	**English-**Course – I - English For Enrichment – I	3	5	25	75	100
3.	III	4531C1	**Core I.** Principles of Food Science (T)	5	6	25	75	100
4.		4531C2	**Core II**. Principles of Food Science (P)	4	8	25	75	100
5.		4531A1	**Allied:** I. Basics of Textiles and Clothing (T)	4	4	25	75	100
6.	IV	4531VE	**Value Education**: I. Introduction to Gender Studies	2	2	25	75	100
Total Credits				**21**	**30**	**---**	**---**	**600**

S.No	Part	Code	II Semester	Credit	Hrs/ Week	Marks		Total Marks
						Int	Ext	
7.	I	4532T2	**Tamil** –பொதுத்தமிழ்தாள்-2 இடைக்கால இலக்கியமும் சிறுகதையும்	3	5	25	75	100
8.	II	4532E2	**English –** Course - II – English For Enrichment – II	3	5	25	75	100
9.	III	4532C1	**Core: III.** Principles of child development	5	6	25	75	100
10.		4532C2	**Core: IV.** Fabric Surface ornamentation(P)	4	8	25	75	100
11.		4532A2	**Allied:** II. Principles of Biochemistry (T)	4	4	25	75	100
12.	IV	4532ES	**Ability Enhancement Compulsory Courses (AECC)-**Environmental Studies (T)	2	2	25	75	100
Total Credits				**21**	**30**	**---**	**---**	**600**

S.No	Part	Code	III Semester	Credit	Hrs/ week	Marks		Total Marks
						Int	Ext	
13.	I	4533T3	**Tamil -**பொதுத் தமிழ் தாள் - 3 காப்பியமும் புதினமும்	3	4	25	75	100
14.	II	**4533E3**	**English -** Course – III - English For Enrichment – III	3	4	25	75	100
15.		4533C1	**Core: V.** Principles of Nutrition (T)	5	6	25	75	100
16.		4533C2	**Core:VI.** Human Physiology (T)	5	5	25	75	100
17.		4533A3	**Allied: III.** Fashion Designing (T)	4	4	25	75	100

	III								
18.	IV		**Non Major Elective: I**		2	3	25	75	100
19.		**453S01**	**Skill Based Subject I.**Fabric Analysis(P)	Any one	2	4	25	75	100
		453S02	Garment Designing and Construction (P)						
20.	V		**Extension activities**		1	--	100	--	100
21.			**SLC - MOOCs**		EC				
Total Credits					**25**	**30**	**--**	**--**	**800**

S.No	Part	Code	IV Semester		Credit	Hrs/ week	Marks		Total Marks
							Int	Ext	
22.	**I**	**4534T4**	**Tamil -** பொதுத்தமிழ் தாள்- 4 பண்டைய இலக்கியமும் நாடகமும		3	4	25	75	100
23.	**II**	**4534E4**	**English-**Course – IV- English For Enrichment – IV		3	4	25	75	100
24.	III	4534C1	**Core: VII**. Community Nutrition (T)		5	6	25	75	100
25.		4534C2	**Core:VIII**. Public Health Nutrition (T)		5	6	25	75	100
26.		4534A4	**Allied**: **IV.** Interior design(T)		4	5	25	75	100
27.	IV	453S03	**Skill Based Subject**: **II.** Food processing & Preservation (P)	Any one	2	2	25	75	100
		453S04	Fundamentals of Apparel Designing (P)						
28.			**Non Major Elective: II**		2	3	25	75	100
29.			**SLC - MOOCs**		Extra credits				
Total Credits					24	30	---	---	700

S. No	Part	Code	V Semester		Credit	Hrs/ week	Marks Int	Marks Ext	Total Marks
1.	III	4535C1	**Core: I**X Clinical Nutrition (T)		5	5	25	75	100
2.		4535C2	**Core:X** Clinical Nutrition (P)		4	8	25	75	100
3.		4535C3	**Core: XI** Food Microbiology(T)		5	5	25	75	100
4.		453E1A	**Elective: I**(A) Pattern Making (P)	Any One	3	5	25	75	100
		453E1B	(B) Fashion CAD and Portfolio (P)						
		453E2A	**Elective:II:** A) Family Resource management (T)	Any One	3	3	25	75	100
		453E2B	(B) Genetically Modified and Organic Foods (T)						
5.	IV	453S05	**Skill Based Subject: III.** A. Bakery and confectionary (T)	Any One	2	2	25	75	100
		453S06	Home Based Catering(T)						
		453S07	**Skill Based Subject: III.**B Basics of Wet processing (T)	Any One	2	2	25	75	100
		453S08	Training for Community Development (T)						
Total Credits					**24**	**30**	**--**	**-**	**700**

S.No	Part	Code	VI Semester		Credit	Hrs /	Marks Int	Marks Ext	Total Marks
1.	III	4536C1	**Core: XII.** Diet Therapy (T)		5	6	25	75	100
2.		4536C2	**Core:XIII.** Diet Therapy Internship(P)		3	6	25	75	100
3.		4536C3	**Core:XIV**. Extension Education in Home Science (T)		5	6	25	75	100
4.		4536C4	**Core:XV.** Food Service Management (T)		5	5	25	75	100
5.		453E3A	**Elective: III:** A) Food Sanitation and Hygiene (T).	Any One	3	3	25	75	100
		453E3B	**III** B) Regulations of Food Product Development (T)						
6.	IV	453S09	**Skill Based Subject**: **IV.** A) Quality Management in Food Processing (T)	Any One	2	2	25	75	100
		453S10	Fundamentals of Tourism and Hospitality Management (T)						
		453S11	**Skill Based Subject**: **IV.** B) Office Automation (T)	Any One	2	2	25	75	100
		453SB12	NGO and Corporate Social Responsibility (T)						
Total Credits					**25**	**30**	**-**	**-**	**700**
Grand Total					**140**	**180**	**4100**		

M.A. INTEGRATED HOME SCIENCE (NUTRITION AND DIETETICS)

Semester	Code	Course Name		Credit	Hrs/ week	Marks		Total Marks
						Int.	Ext.	
I	453N01	Advanced Food Science		5	5	25	75	100
	453N02	Advanced Food Science (P)		5	10	25	75	100
	453N03	Community Nutrition and Education		5	5	25	75	100
	453N04	Food Product Development and Marketing		5	5	25	75	100
	453EC1	(A) Home Textiles	Any One	4	4	25	75	100
	453EC2	(B) Women and Society						
Library					1	--	---	
TOTAL				**24**	**30**			**500**
II	453N05	Advanced Nutrition And Intermediary Metabolism		5	5	25	75	100
	453N06	Therapeutic Nutrition		5	5	25	75	100
	453N07	Therapeutic Nutrition Internship		8	10	25	75	100
	453EC3	(A) Women and Health	Any One	4	4	25	75	100
	453EC4	(B) Food Service Management						
	NME -1			2	3	25	75	100
	SLC -1	MOOCs		Extra credit	-	-	-	-
Library, Yoga And Career Guidance					3	-		-
TOTAL				**24+Extra credit**	**30**			**500**
III	453N08	Advanced Biochemistry		5	**6**	25	75	100
	453N09	Food microbiology		5	**6**	25	75	100
	453N10	Research Methodology and statistical Analyses		5	**6**	25	75	100
	453EC5	(A) Fabric Sourcing	Any One	4	**6**	25	75	100
	453EC6	(B) Functional foods and Nutraceuticals						
	NME-II			2	**3**	25	75	100
	SLC-II	MOOCs		Extra Cr		-	-	-
Library, Yoga And Career Guidance					**3**	-	-	-
TOTAL				**21Extra credit**	**30**	-	-	**500**
IV	453N11	Macro and Micro Nutrients		5	5	25	75	100
	453N12	Nutrition Through Life Cycle		5	5	25	75	100
	453N13	Computer Application in Nutrition and Dietetics (P)		5	10	25	75	100
	453N14	Project/Dissertation		6	10	25	75	100
TOTAL				**21**	**30**	**-**	**-**	**400**
GRAND TOTAL				**90+Extra credit**	**120**	**-**	**-**	**1900**

M.A. INTEGRATED HOME SCIENCE (TEXTILES & CLOTHING)

Semester	Core/ Elective	COURSE NAME		Credit	Hrs/ week	Marks		Total
						Int.	Ext.	
I	453T01	Fashion and Clothing Psychology		5	5	25	75	100
	453T02	Care for textiles and clothing		5	5	25	75	100
	453T03	Advanced Wet Processing		5	5	25	75	100
	453T04	Advanced Wet Processing (P)		5	10	25	75	100
	453EC1	(A) Home Textiles	Any one	4	4	25	75	100
	453EC2	(B)Women and Society						
	Library			-	1	-	-	-
			TOTAL	**24**	**30**	**-**	**-**	**500**
II	453T05	Apparel Merchandising		5	5	25	75	100
	453T06	Advanced Garment Construction		5	5	25	75	100
	453T07	One month Internship in garment /textiles /Dyeing Industries		8	10	25	75	100
	453EC3	(A)Women and Health	Any one	4	4	25	75	100
	453EC4	(B)Food Service Management						
	NME-I			2	3	25	75	100
	SLC-I	MOOCs		Extra credit		-	-	-
	Library, yoga and carrier guidance				3	-	-	
			TOTAL	**24+ Extra Credit**	**30**	**-**	**-**	**500**
III	453T08	Textile testing and quality control		5	5	25	75	100
	453T09	Research methodology and statistical Analyses		5	5	25	75	100
	453T10	Computer application in textiles and clothing (P)		5	10	25	75	100
	453EC5	(A) Fabric Sourcing		4	4	25	75	100
	453EC6	(B) Functional Foods and Nutraceuticals						
	NME-II			2	3	25	75	100
	SLC-II	MOOCs		Extra credit	-	-	-	-
	Library, yoga and carrier guidance				3	-	-	-
TOTAL				**21+ Extra credit**	**30**	**-**	**-**	**500**
IV	453T11	Technical Textiles		5	5	25	75	100
	453T12	Fashion Marketing		5	5	25	75	100
	453T13	Portfolio presentation (P)		5	10	25	75	100
	453T14	Project work		6	10	25	75	100
		TOTAL		21	30	-	-	400
		GRAND TOTAL		**90+ Extra credits**	**120**	**-**	**-**	**1900**

Semester -I			
Coursecode: 4531T1	**பொதுத்தமிழ் தாள்- 1** **தற்காலக் கவிதையும் உரைநடையும்**	**Credits:3**	**Hours:5**
நோக்கம்	நம் தமிழிலக்கியங்கள் பழம்பெரும் பெருமை கொண்டதாகும். இதனடிப்படையில் காலத்திற்கேற்ற நிலையில் இலக்கியங்களின் மாற்றம் அடைந்தன. இவ்வாறு மாற்றமடைந்த வடிவங்களில் குறிப்பிடத்தக்கது ஆங்கிலேயர்களின் வரவால் ஏற்பட்ட தற்கால உரைநடை, கட்டுரை, கவிதைகள் போன்றவைகளாகும். இவற்றை இலக்கணப்பிழைகளின்றி இலக்கிய, இலக்கண நயத்தோடு படிக்கவும், படைக்கவும் செய்வது இந்த தாளின் நோக்கமாகும்.		
Unit - I	**அ. மரபுக் கவிதை** 1. பாரதி - நிலாவும் வான்மீனும் காற்றும் (முழுமையும்) 2. பாரதிதாசன் - தோழனே! உன்னிடம் சொல்வேன்! 3. நாமக்கல் கவிஞர் - உலகம் வாழ்க! 4. ஜீவானந்தம் - கோடிக்கால் பூதமடா 5. முடியரசன் - தலைமை வகிப்போம் (பாடுங்குயில், ப.8) 6. கண்ணதாசன் - புதியதோர் உலகு செய்வோம் (ஏழாவது தொகுதி) **ஆ. புதுக்கவிதை** 7. மு.மேத்தா - தேசப்பிதாவிற்கு ஒரு தெருப் பாடகனின் அஞ்சலி (கண்ணீர் பூக்கள்) 8. கவிக்கோ அப்துல்ரகுமான் - மானுடத்தின் மகுடாபிஷேகம் (பால்வீதி) 9. மீரா - காதல் என்ன கத்திரிக்காயா? (ஊசிகள்) 10. வைரமுத்து - மரங்களைப் பாடுவேன் (இந்தப் பூக்கள் விற்பனைக்கு அல்ல)		
Unit - II	1. **எண்ணங்கள்** - எம்.எஸ். உதயமூர்த்தி.		
Unit - III	**இலக்கணம்** எழுத்திலக்கணம், எண், பெயர், முறை, பிறப்பு, வடிவம், மாத்திரை, மொழி முதல் எழுத்துக்கள், மொழி இறுதி எழுத்துக்கள், இடைநிலை மெய்ம்மயக்கம், மொழி, பகுபத உறுப்பு, வடமொழி எழுத்து, (ஆ.சிவலிங்கனார், தமிழ் இலக்கண உணர்வுகள், பக்கம் 26 முதல் 69 வரை, கபிலன் பதிப்பகம், புதுச்சேரி)		
Unit – IV	**இலக்கிய வரலாறு** அலகு 1, அலகு 2ல் உள்ள பாடம் தொடர்பான இலக்கிய வகைகள் தொடர்பான இலக்கிய வரலாறு.		
Unit - V	**படைப்பாற்றல்** பொதுக்கட்டுரை படைத்தல்.		

பார்வை நூல்கள்

ஆறுமுகநாவலர் (1994). *நன்னூல் எழுத்ததிகாரம் காண்டிகை உரை.*

ஆ.சிவலிங்கனார் (2005).*தமிழ் இலக்கண உணர்வுகள்.*

பாக்கியமேரி (2013).*வகைமை நோக்கில் தமிழிலக்கிய வரலாறு* (5வது பதிப்பு).

தமிழண்ணல் (2017).*புதிய நோக்கில் தமிழிலக்கிய வரலாறு* (36வது பதிப்பு).

எம்.எஸ்.உதயமூர்த்தி (2018).*எண்ணங்கள்.*

(அலகு-1 இல் உள்ள கவிஞர்களின் கவிதை நூல்கள்)

Course
Teacher
Dept. of Tamil

Semester -I			
Coursecode:4531E1	**Course – I - ENGLISH FOR ENRICHMENT – I**	**Credits: 3**	**Hours: 5**
Objectives	➢ To offer the learners a first-hand knowledge of the prose works. ➢ To make them familiar with grammar.		
Unit - I	**Prose** 1. Education for New India – C.Rajagopalachari. 2. All about a Dog – A.G.Gardiner 3. I have a Dream – Martin Lutherking		
Unit - II	**Prose** 1. How I Became a Public Speaker – G.B. Shaw 2. With the Photographer – Stephen Leacock 3. Early Influences: Dr. APJ. Abdul Kalam		
Unit - III	**Poetry** 1. Gitanjali (Songs : 1-2) Rabindranath Tagore 2. Shall I Compare thee to a Summer's Day(Sonnet 18)–William Shakespeare 3. On his Blindness – John Milton.		
Unit – IV	**Grammar** Noun, Pronoun, Verb, Adverb		
Unit - V	**Composition** Informal Letter, Comprehension, Dialogue Writing, Hints Developing		
Reference and Textbooks: Narayan, R. K (2008*). Swami and Friends*.Chennai: Indian Thought Publications. Krishnaswamy,N.(2009).Modern English – *A Book of Grammar Usage and Composition*. Macmillan Publishers. Nagarajan,V. et al(2015).*Six Short Stories*, Ed. By the Board of Editors, Harrows Publications, Chennai. Nagarajan,V. et al(2015*).One Act Plays*, Ed. By the Board of Editors, Harrows Publications, Chennai. Nagarajan,V. et al(2016)Gate Way to English – *An Anthology of Prose and Poetry* Ed. By the Board of Editors, Harrows Publications, Chennai.			
Outcomes	➢ Can understand the prose works of great writers across the world. ➢ Can frame sentences in English.		

Course Teacher
Dept of English& Foreign Languages

Semester -I			
Coursecode:4531C1	**Core I: PRINCIPLES OF FOODSCIENCE**	**Credits:5**	**Hours: 6**

Objectives	➢ To obtain knowledge of different food groups and their nutritional value. ➢ To gain experience in the preparation of food.
Unit - I	**Introduction to Food Science** a. Definitions - Food, Nutrients. b. Classification of foods and nutrients, based on their functions – energy yielding, body building and regulative. c. Basic food groups - IV, V (ICMR) & VII, Balanced Diet.
Unit - II	**Food Preparation Techniques** a. Preliminary techniques – Cleaning, Peeling & Stringing, Cutting & Grating, Soaking, Processing, Sieving, Coating, Blanching, Marinating, Roasting, Grinding and Sprouting. b. Moist heat – Boiling, Simmering, Poaching, Stewing, Steaming, Pressure Cooking; Dry Methods - Baking, Roasting, Toasting, Grilling c. Microwave and Solar cooking d. Different methods of cooking and their influence on nutrient retention
Unit - III	**Cereals Pulses Nuts and Oil seeds** a. Cereals – Rice & Wheat-– Structure, Milling, Parboiling, by-products; nutritive value, storage. Role of cereals in cookery b. Pulses - Structure, nutritive value, sprouting, presence of toxic substances, storage. Role of pulses in cookery c. Nuts and oilseeds – nutritive value and its importance in meals.
Unit – IV	**Fruits Vegetables spices and condiments** a. Fruits - Classification, nutritive value, role of fruits in human nutrition. b. Vegetables - Classification according to structure, nutritive value, effect of heat, acid and alkali, storage and conservation of nutrients, role of vegetables in cookery. c. Spices and condiments – uses and abuses, role of spices and condiments in cookery.
Unit - V	**Animal Food** a. Milk and Milk products, nutritive value, physical properties, processing of milk, role of milk in cookery. b. Flesh Foods – meat, fish and poultry – classification, nutritive value –methods of cooking, role of flesh foods in meal c. Egg, structure, composition, uses in diet.

Reference and Textbooks:

Brow, A. (2000).*Understanding Food*, Thomson Learning Publications, Wadsworth.

Mehas, K.Y. and Rodgers, S.L. (2000). *Food Science and You*, McMillan McGraw Company, New York.

Parker, R. (2000) *Introduction to food Science*, Delmer, Thomson Learning Co., Delma.

Usha Chandrasekhar,(2002).*Food Science and Application in Indian Cookery*, Phoenix Publishing House P. Ltd., New Delhi,

Srilakshmi,B.(2010) *Food Science*, New Age International Publishers, New Delhi.

Outcomes	Enable the students to ➢ Understand about Experimental Cookery on Healthy Foods ➢ Gain knowledge in the preparation of Healthy Foods

Course Teacher
P.Venkateswari
Prof.K. Manimekalai

Semester -I			
Coursecode:4531C2	**Allied I: PRINCIPLES OF FOOD SCIENCE (P)**	**Credits:4**	**Hours:8**
Objectives	➢ To gain basic knowledge on cooking methods. ➢ To prepare various food without nutrient loss**.**		
	Practical **Beverages**: Milk shake/ Lassi, Fruit based Beverages **Cereals**: Boiled rice, Pulao, Chapatti, Paratha, Puri, pastas **Pluses**: processing - Malting, Roasting, Whole, Dehusked **Vegetable**s: Dry Preparations, Dehydration Boiling **Milk and milk products**: processing –Pasteurization, spring, drying , Dehydration of milk Meat, Fish and poultry preparations **Egg Preparations**: Boiled, Poached, Fried, Scrambled, Omelets, Egg Pudding **Soups**: Broth, Plain and Cream Soups **Snacks**: Pakodas, Cutlets, Samosas, Upma, Sandwiches **Salads**: Salads and Salad Dressings		
Reference and Textbooks: Khanna K, Gupta S, Seth R, Mahna R, Rekhi, T. (2004). *The Art and Science of Cooking*: A Practical Manual, Revised Edition. Elite Publishing House Pvt Ltd. Wardlaw and Insel MG, Insel PM. (2004). *Perspectives in Nutrition* (6th ed.).Mosby. Srilakshmi (2007). *Food Science* (4th ed.). New Age International Ltd. Bamji MS, Krishnaswamy K, Brahmam GNV. (2009). *Textbook of Human Nutrition*, (3rd ed.). Oxford and IBH Publishing Co. Pvt. Ltd. Chadha,R.Mathur, P. (2015). *Nutrition A Lifecycle Approach*. Orient Blackswan, Delhi.			
Outcomes	➢ Understand the basic principles and practices of cleaning and sanitation in food preparation. ➢ Know about the cooking methods, principles of menu planning and food presentation		

Course Teacher
P.Venkateswari
Dr.P.Veeramani

Semester -I			
Coursecode:4531A1	**BASICS OF TEXTILES AND CLOTHING**	**Credits: 4**	**Hours: 4**
Objectives	➢ To understand the basics of Textiles and Clothing ➢ To impart knowledge on different textile finishes.		
Unit - I	**Introduction to the Field of Textiles** - Major Goals - Classification of Fibres - Natural & Man Made - Primary and Secondary Characteristics of Textile Fibres		
Unit - II	**Manufacturing Process**, Properties and uses of Natural Fibres - Cotton, Linen, Jute, Pineapple, Hemp, Silk, Wool, Hair Fibres, Man-Made Fibres-Viscose Rayon, Nylon, Polyester.		
Unit - III	**Spinning** - Definition, Classification. Spinning - Blending, Opening, Cleaning, Doubling, Carding, Combing, Drawing, Roving. Yarn - Definition, Classification - Simple and Fancy Yarns, Sewing Threads and its importance.		
Unit – IV	**Wovens** - Basic Weaves - Plain, Twill, Satin. Fancy Weaves - Pile, Double Cloth, Leno, Swivel, Lappet, Dobby and Jacquard.Knitting –Warp and Weft Knitting.		
Unit - V	**Non-wovens** - Felting, Fusing, Bonding, Lamination, Netting, Braiding & Calico, Tatting and crotcheting		

Reference and Textbooks:

Corbmann, B.P (1985). *Textiles - Fibre to Fabric*, International Students edition, Me Grew HillBook Co, Singapore.

Joseph J Prctal. (1990). *Fabric Science* (5thed). Fairchild Publications, New York

Readers. (1993).*Sewing and Knitting*- A Reader's Digest step-by-step guide. Reader'sDigest. New York.

Humuga Rai. (2000). *Fabric forming* .New age International.

Anbumani. (2006).*Knitting Manufacture Technology*, New Age International, Chennai. .

Outcomes	Enable the students to ➢ Get basic knowledge on basic textiles and manufacturing processing ➢ Gain in depth knowledge on cloths, fibres and yarn.

Course Teacher
Dr.A.K.P.Dhanakodi
Prof.K. Manimekalai

Semester -I			
Coursecode:4531VE	**INTRODUCTION TO GENDER STUDIES**	**Credits: 2**	**Hours: 2**

Objectives	➢ To introduce women's studies as an academic discipline ➢ To explain changing trend in the status of Women
Unit - I	**Gender studies**: Definition - Scope of gender studies - Interdisciplinary nature of Gender studies, Gender studies Vs Women's studies, Need for Gender as an academic category.
Unit - II	**Gender concepts**: Gender Vs Sex – Understanding patriarchy – Gender roles- Stereotypes- Equity Vs Equality-– Gender discrimination-Gender violence - Gender awareness – Gender sensitization and training.
Unit - III	**Women's Status in the Family**; Patriarchal- Matriarchal –Joint –Nuclear Family –Socialization –Internalization.
Unit – IV	**Gender and Education** –Enrolment Pattern – Primary to Higher Education – literacy Rates - Drop-out Rates - Educational Policy in India
Unit - V	**Gender Division of Labour** – Women Triple Role - Unpaid labour- Women in Organised and Unorganised Sectors

Reference and Textbooks:

Brike, Lynda (1987) *Women, Feminism and Biology*, The Feminist Challenge, Brighton; Harvester press.

Robinson, Victoria and Diane, Richardson. (1993). *Introducing Women's Studies*: *Feminist Theory and Practice*. London: Macmillan.

Suryakumari(1993).*Women's Studies An Emerging Discipline* New Delhi Gyan Publising House.

Mittal Mukta (1995).*Women power in India*. New Delhi: Indian Association of Women's Studies.

Krishnaraj, Maithreyi(1995).*Remaking Society for Women*: Visions Past and Present. New Delhi: Indian Association for Women's Studies.

Outcomes	Enable the students to ➢ Gain knowledge on gender studies and gender concepts ➢ Aware about the gender gap in education and work force participation

Course Teacher
Prof.K. Manimekalai
Dr.P.Veeramani

Semester -II			
Coursecode: 4532T2	**பொதுத்தமிழ் தாள் -2 இடைக்கால இலக்கியமும் சிறுகதையும்**	**Credits: 3**	**Hours: 5**
நோக்கம்	தமிழ் இலக்கியங்களைக் கால அடிப்படையில் பண்டைய இலக்கியம், இடைக்கால இலக்கியம், இக்கால இலக்கியம் எனக் கூறலாம். இதனடிப்படையில் இடைக்கால இலக்கியங்களாகப் பக்தி இலக்கியம், காப்பிய இலக்கியம், சிற்றிலக்கியம் போன்றவை காணப்படுகின்றன. இவற்றில் பக்தி இலக்கியம் என்பது நம் தமிழ்ச் சமூகம் பக்தியில் எவ்வாறு தன்மை ஈடுபடுத்திக் கொண்டனர் என்பதை விளக்குகின்றது. இப்பக்தி நெறியையும், சிறுகதைப் பற்றியும், சொல்லிலக்கணம் பற்றியும், சிறுகதை படைத்தல் பற்றியும் மாணவர்களுக்கு தெளிவாய் உணர்த்துவது இத்தாளின் நோக்கமாகும்.		
Unit - I	**அ. திருஞானசம்பந்தர்** 1. திருவாடானை - "மாதோர் கூறு" எனத் தொடங்கும் பாடல். 2. திருப்புனவாசல் - "மின்னியல் செஞ்சடை" எனத் தொடங்கும் பாடல். 3. திருக்கொடுங்குன்றம் - "வானிற் பொலிவெய்தும்" எனத் தொடங்கும் பாடல். **ஆ. திருநாவுக்கரசர்** 1. திருப்புத்தூர் - "மின்காட்டும்" எனத் தொடங்கும் பாடல். 2. திருஇராமேச்சுரம் - "பாசமும்" எனத் தொடங்கும் முதல் பாடல். 3. திருப்பூவணம் - "வடியேறு" எனத் தொடங்கும் பாடல். **இ. சுந்தரர்** 1. திருக்கானப்பேர் - "தொண்டர் அடித் தொழலும்" எனத் தொடங்கும் பாடல். 2. திருச்சுழியல் - "ஊனாய் உயிர் உகலாய்" எனத் தொடங்கும் பாடல். **ஈ. மாணிக்கவாசகர் - திருவாசகம்** 1. திருப்பெருந்துறை இன்பம் பெருக்கி எனத் தொடங்கும் பாடல்.(திருவெண்பா.11) 2. திரு உத்தரகோசமங்கை நீத்தல் விண்ணப்பம்,இருதலைக்கொள்ளி என்று தொடங்கும் பாடல். **உ. திருமூலர் - திருமந்திரம்** 1. அன்பும் சிவமும் எனத் தொடங்கும் பாடல். 2. எட்டிப் பழுத்த எனத் தொடங்கும் பாடல். 3. படமாடக் கோயில் எனத் தொடங்கும் பாடல். **ஊ. திருமங்கை ஆழ்வார்** திருப்புல்லாணி - ஒன்பதாம் பத்து நாலாம் திருமொழி "காவார் மடல் பெண்ணை" தொடங்கும் ஒன்றாம் பாடல் முதல் "வில்லாள் இலங்கை" எனத் தொடங்கும் ஐந்தாம் பாடல் வரை (மொத்தம் ஐந்து பாடல்கள்) **எ. சிற்றிலக்கியம்** 1. அபிராமி அந்தாதி - உதிக்கின்ற செங்கதிர் எனத் தொடங்கும் முதற்பாடல் தொடங்கி அதனைத் தொடர்ந்து வரும் 9 பாடல்கள் (ஆக மொத்தம் 10 பாடல்கள்). 2. தமிழ்விடு தூது - 17 ஆம் கண்ணி முதல் 27 ஆம் கண்ணி வரை. 3. திருக்குற்றாலக்குறவஞ்சி, வசந்தவள்ளி பந்தடித்தல். 4. பாடுவார் முத்தப்பர், செயங்கொண்டார் சதகம் முதல் இரு பாடல்கள்.		
Unit - II	**சிறுகதை** சிறுகதைகள் 10 ஆசிரியர் குழு, அறிவுப் பதிப்பகம்.		
Unit - III	**இலக்கணம்** சொல்லிலக்கணம்சொல்வகை, பெயர்ச்சொல், வினைச்சொல்,இடைச்சொல், உரிச்சொல்,இலக்கணம், வேற்றுமை, மயக்கம், ஆகுபெயர், (ஆ.சிவலிங்கனார், தமிழ் இலக்கண உணர்வுகள் - கபிலன் பதிப்பகம், புதுச்சேரி).		
Unit – IV	**இலக்கிய வரலாறு** அலகு 1, அலகு 2ல் உள்ள பாடம் தொடர்பான இலக்கிய வகைகள் தொடர்பான இலக்கிய வரலாறு.		
Unit - V	**படைப்பாற்றல்** சிறுகதை படைத்தல்.		

பார்வை நூல்கள்

பன்னிரு திருமுறைகள் (1971)- கழக வெளியீடு.
நாலாயிரத் திவ்ய பிரபந்தம் (1973) - கழக வெளியீடு.
ஆறுமுகநாவலர்(2001).*நன்னூல் சொல்லதிகாரம் காண்டிகை உரை.*
ஆ.சிவலிங்கனார்(2005).*தமிழ் இலக்கண உணர்வுகள்.*
பாக்கியமேரி(2013).*வகைமை நோக்கில் தமிழிலக்கிய வரலாறு* (5வது பதிப்பு).
தமிழண்ணல்(2017).*புதிய நோக்கில் தமிழிலக்கிய வரலாறு* (36வது பதிப்பு).
(மேலும் சிற்றிலக்கியங்கள், சிறுகதைகள் தொடர்பான நூல்கள்)

Course Teacher
Dept of Tamil
Prof.K. Manimekalai

Semester -II			
Coursecode:4532E2	**Course - II -English For Enrichment-II**	**Credits: 3**	**Hours: 5**

Objectives	➢ To help them to write letters, resume and general English. ➢ To initiate the learners to varied genres of English Literature.
Unit - I	**Prose** 1. My Greatest Olympic Prize – Jesse Owens 2. Voluntary Poverty – Mahatma Gandhi 3. Helen Kellar – Ishbel Ross
Unit - II	**Prose** 1. Coffee Worries – R.K. Narayan 2. A Night Among the Pines – R.L. Stevenson 3. Spoon Feeding – W.R.Inge
Unit - III	**Poetry** 1. Daffodils - Wordsworth 2. Mending Wall – Robert Frost 3. A River – A.K.Ramanujan
Unit – IV	**Grammar** Adjective, Preposition, Conjunction and Interjection.
Unit - V	**Composition** Formal Letters, Resume Writing, Precise Writing and General Essays.

Reference and Textbooks:

Narayan, R. K (2008*). Swami and Friends.*Chennai: Indian Thought Publications.

N.Krishnaswamy.(2009).Modern English – *A Book of Grammar Usage and Composition.* Macmillan Publishers.

Nagarajan,V et al(2015).*Six Short Stories*, Ed. By the Board of Editors, Harrows Publications, Chennai.

Nagarajan,V et al(2015*).One Act Plays*, Ed. By the Board of Editors, Harrows Publications, Chennai.

Nagarajan,V et al(2016)Gate Way to English – *An Anthology of Prose and Poetry* Ed. By the Board of Editors, Harrows Publications, Chennai.

Outcomes	➢ Can familiar with prose and poetry. ➢ Can write general English, resume and letters

Course Teacher
Dept of English
Dr.P.Veeramani

Semester -II			
Coursecode:4532C1	**CORE III: PRINCIPLES OF CHILD DEVELOPMENT**	**Credits: 5**	**Hours: 6**
Objectives	➢ To become knowledgeable about the basic concepts in Child Development ➢ To inculcate the growth processes taking place from conception to Late childhood		
Unit - I	**Pre-natal Development**: the beginning, conception, period of prenatal development, factors affecting & hazards of prenatal period. Infancy: The newborn, characteristics, major adjustment of infants, factors influencing adjustment to post natal life, developmental trends & hazards.		
Unit - II	**Baby hood**: Characteristic & developmental tasks of babyhood. Physical, mental, motor, speech, emotional, social, moral and personality development, during baby hood period. Early interest in play, family relationship development in babyhood period, Hazards & happiness of baby hood period.		
Unit - III	**Early childhood**: Characteristic & developmental tasks of early childhood. Physical, mental, motor, speech, emotional, social moral & personality development during early childhood period. Play & common interest in early childhood period, Family relationship in early childhood, Hazards & happiness in early childhood period.		
Unit – IV	**Late childhood**: Characteristic & developmental tasks of late childhood. Physical growth, motor skills, speech development, emotional development, expressions pattern of socialization, play interest, moral, personality development& Development of understanding during late childhood period. Cmmon interest in family relationship during late childhood. Hazards & happiness in late childhood period.		
Unit - V	**Children with special needs:** Definition, classification, characteristics of special children. Physical, emotional and social development of children with special needs and rehabilitation.		

Reference and Textbooks:

Devadas, R.P. and Jaya, N. (1996). *A Text Book on Child Development*. New Delhi: Macmillan India Limited.

Pankajam,G. (2005). *Know Your Child*. New Delhi: Concept Publishing Company.

Hurlock, E.B. (2007). *Developmental Psychology A Life-Span Approach*. New Delhi: Tata Mcgraw Hill Publishing Company Limited.

Suriakanthi. A (2009).*Child Development – An Introduction.*Gandhigram: Kavitha Publications

Santrock, J.W. (2010) *Child Development: An Introduction*. (*12th Edition*) Newyork: McGraw hill.

Outcomes	Enable the students to ➢ Understand the growth processes taking place from conception to Late childhood ➢ Know about the development of children with special needs and rehabilitation

Course Teacher
P.Archana
Prof.K. Manimekalai

Semester -II			
Coursecode:4532C2	**Core IV: FABRIC SUFACE ORNAMENTATION (P)**	**Credits:4**	**Hours: 8**
Objectives	➢ To become knowledgeable on traditional embroidery types in India ➢ To learn about the preparation for fabric surface embellishment		
	Introduction to Embroidery and Indian Traditional embroidery motifs (only theoretical inputs). Theoretical/ Tutorial and Self-study inputs **Andhra Pradesh:** Banjara Embroidery – Bihar: Kantha Embroidery Sujani Embroidery **Tamil Nadu:** Conventional Embroidery - Toda Embroidery (Pukhoor) **Uttar Pradesh:** Chikankari Embroidery – Mukaish/ Badla Embroidery - Patti ka kaam Embroidery (Aligarh) **West Bengal:** Kantha Embroidery **Gujarat:** Ahir Embroidery - Banni/ Heer Bharat Embroidery - Jat Embroidery - Kanbi Embroidery - Kathi Embroidery - Khaarek Embroidery - Kutch Embroidery - Mirror Embroidery - Ari Embroidery - Mutwa Embroidery - Pakko Embroidery - Rabari Embroidery - Suf Embroidery - Bead Embroidery		
	Himachal Pradesh: Chamba Rumal Embroidery **Jammu & Kashmir:** Kashidakari Embroidery - Crewel Embroidery - Sozni/ Dorukha Embroidery **Karnataka:** Lambani Embroidery - Kasuti Embroidery **Manipur:** Shamilami Embroidery - Hijaya Mayek Embroidery **Odisha:** Pipli Applique Embroidery **Punjab:** Phulkari Embroidery - Bagh Embroidery **Rajasthan:** Dabka Embroidery - Danka Embroidery - Gota Patti Embroidery - Karchobi Embroidery - Meo Embroidery – Pichwai		
	Practical: Embroidery and Handwork a. Running and its variations b. Back stitch c. Stem stitch d. Chain stitch and its variations e. Lazy daisy stitch f. Button hole and its variations g. Herring bone and its variations h. Knots and their variations i. Straight stitch		
	Practical: Embroidery and Handwork a. Feather stitch b. Couching c. Appliqué work d. Cut work e. Patch work f. Beads Work g. Sequins Work h. Ribbon work		

Reference and Textbooks:	
Anne Morrell (1995). *The Techniques of Indian Embroidery*, Interweave Press Inc. Shailaja D.Naik. (1996). *Traditional Embroideries of India*, S.B. Nangia for A.P.H. Publishing Corporation. Bhatnagar, P. (2004). *Traditional Indian Costumes & Textiles*, Abhishek Publications, Chandigarh Gokarneshan. N. (2009), *Fabric Structure and Design* - New Age International. Lehri, R.M. (2011).*Indian Embroidery Ethnic and Beyond*. Super Book House, Mumbai.	
Outcomes	Enable the students to ➢ Assimilate with various Traditional embroidery types in India ➢ Understand the preparation for fabric surface embellishment

Course Teacher
M.Priyanka
Prof.K. Manimekalai

Semester -II			
Coursecode:4532A2	**ALLIED II:PRINCIPLES OF BIOCHEMISTRY**	**Credits: 4**	**Hours: 4**

Objectives	➢ To understand the basic concepts in food biochemistry ➢ To study about the biochemical pathways and their relevance to the life
Unit - I	**Carbohydrates** - Classification, monosaccharide's, oligosaccharides, polysaccharides, structure and configuration of aldose, ketoses, triose, tetrose, pentose, hexose.Homo polysaccharides-starch, inulin, glycogen, cellulose, Heteropolysaccharides
Unit - II	**Lipids** - Classification of lipids, fatty acids, essential fatty acids, triglycerols, phospholipids, glycolipids, lipoprotein, sterols, amphipathic lipids, fatty acid oxidation (β-oxidation). Functions, sources and requirements.
Unit - III	**Amino acids and Proteins**: Amino acid, classification, properties, essential amino acid, structure of proteins- primary, secondary, tertiary, quaternary structure of proteins. Protein denaturation.
Unit – IV	**Enzymes and Vitamins Enzymes**: Nomenclature and classification, kinetics of enzymic reactions, types of enzymes, enzyme inhibition, reversible inhibition, irreversible inhibition, conditions affecting enzymatic reactions, co-enzymes. Vitamins: classification, functions, requirement and deficiency conditions, vitamin A, D, E, K, Ascorbic acid, Thiamine, Riboflavin, Niacin, Pyridoxine, Folic acid, Pantothenic acid, Cyanocobalamine.
Unit - V	**Metabolism of Carbohydrates and Proteins:** Glycolysis and Tricarboxylic acid (TCA) cycle, HMP shunt, Gluconeogenesis Transamination, Deamination, Urea cycle.

Reference and Textbooks:

Das, D. (1992). *Biochemistry* (7th ed.).Academic publishers, Calcutta.

Devlin. T.M. (1997). *Text book of Clinical Biochemistry*. New York, John Wiley and Sons.

Jain, J.L. (2001). *Fundamentals of Biochemistry* (5th ed). S.Chand and Company Ltd.

Satyanarayana,U. and Chakrapani,U.(2006). *Biochemistry* (*3rd ed.*). Books and Allied Pvt Ltd, Kolkata.

Berg, J.M., Tymocezko,J. L.(2007) *Biochemistry(6th ed.).* W H Freeman and Company,New York.

Outcomes	Enable the students to ➢ Understand the basic concepts in food biochemistry ➢ Know about the classification of macro nutrients and its enzymatic reactions

Course Teacher
R.Ramya
Dr.P.Veeramani

Semester -II			
Coursecode:4532ES	**AECC -ENVIRONMENTAL STUDIES**	**Credits:2**	**Hours:2**
Objectives	➢ To gain knowledge on the various types of natural resources, its uses and degradation. ➢ To understand the terms of pollution and pollutant and effect of pollution on environment and health		
Unit - I	**The Multidisciplinary Nature of Environmental Studies** Definition, Scope and importance Need for public awareness		
Unit - II	**Natural Resources** Renewable and non-renewable resources a) Forest Resources: Use and over-exploitation, deforestation, case studies, Timber extraction, mining, dams and their effect on forests and tribal people b) Water Resources: Use and over-Utilization of surface and ground water, floods, drought, conflicts over water, dams- benefits and problems. c) Mineral resources: Use and exploitation, experimental effects of extracting and using mineral resources, case studies. d) Food resources: world food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies. e) Energy resources: Growing energy needs, renewable and non-renewable energy sources, use of alternate energy resources, Case studies. f) Land resources: Land as a resource, land degradation, main induced landsides, soil-erosion and desertification • Role of individual in conservation of natural resources • Equitable use of resources for sustainable lifestyle		
Unit - III	**Ecosystems, Bio-diversity and its conservation** Ecosystems • Concept of an Ecosystem • Structure and function of an Ecosystem • Energy Flow in the Ecosystem • Food Chains, Food Webs and Ecological Pyramids Biodiversity and its conservation • Introduction- Definition: Genetic, Species and Ecosystem Diversity • Bio-Geographical Classification of India • Value of Biodiversity: Consumptive Use, Productive Use, Social Ethical, Aesthetic and Option Values. • Biodiversity at Global, National and Local Levels • India as a Mega-Diversity Nation • Hot Spots of Biodiversity • Threats to Biodiversity: Habitat Loss, Poaching of Wildlife, Man-Wildlife Conflicts • Endangered and Endemic Species of India • Conservation of Biodiversity in-Situ and Ex-Situ Conservation of Biodiversity		
Unit – IV	**Environmental Pollution** • Causes, Effects and Control measures of:- a) Air Pollution b) Water pollution c) Soil pollution d) Marine pollution e) Noise pollution		

	f) Thermal pollution g) Nuclear hazards
Unit - V	**Field Work** • Visit to a local area to document environmental assets–river/ forest/ grassland/ hill/ mountain • Visit to a local polluted site- Urban/Rural/Industrial/Agricultural • Study of common Plants, insects, birds • Study of simple ecosystem-pond, River, Hill slopes, etc

Reference and Textbooks:

Hawlinks R.E (1987).*Encyclopedia of Indian Natural History*, Bombay Natural History Society, Bombay.

Burner,R.C.(1989). *Hazardous Waste Inclineration*. McGraw Hill Inc.

Heywood,V.H&Watson,R.T.(1995).*Global biodiversity Assesment*. Cambridge Univ.Press.

Agarwal, K.C. (2001). *Environmental Biology*, Nidi Publ.Ltd., Bikane

Cunnigham, W.P.Cooper, T.H.Gorhani, E&Hepworth, M.T (2001). *Environmental Encylopedia*, Jaico Publ. House, Mumbai, 1196p.

Outcomes	Enable the students to ➢ Gain knowledge about Renewable and non-renewable resources ➢ Understand knowledge about Bio-diversity and its conservation.

Course Teacher
Dr.S.Poulpunitha
Prof.K. Manimekalai

Semester -III			
Coursecode:4533T3	**பொதுத் தமிழ் தாள் - 3 - காப்பியமும் புதினமும்**	**Credits:3**	**Hours: 4**
நோக்கம்	தமிழ் இலக்கியங்களின் கால அடிப்படையில் காப்பிய இலக்கிய இடைக்காலத்திலும், புதினகங்கள் இக்காலத்திலும் காணப்படுகின்றன. காப்பியங்கள் பெரும்பாண்மையாக அறம், பொருள், இன்பம், வீடு ஆகிய நான்கு உறுதிப்பொருட்களையும் வெளிப்படுத்துவதாய் அமைந்துள்ளன. இக்கால இலக்கிய வரவில் புதினமும் ஒன்று. புதினங்கள் பெரும்பாலும் சமூகங்களில் நிகழும் அவலங்களையும், அநீதிகளையும் எடுத்துக் கூறுவதாய்த் திகழ்கிறது. இவற்றை மாணவர்களுக்கு உணர்த்தும் விதமாவும், தமிழ் இலக்கணங்களான எழுத்து, சொல், பொருள், யாப்பு, அணி ஆகியவற்றில் யாப்பு, அணி இலக்கணங்கள் பற்றி அறியும் வண்ணமாகவும், மாணவர்களின் மரபு, புதுக்கவிதை படைப்புத்திறன்களை வெளிக்கொணரும் வண்ணமும் இத்தாள் அமைக்கப்பெற்றுள்ளது.		
Unit - I	1. சிலப்பதிகாரம் - மங்கல வாழ்த்துப்பாடல். 2. மணிமேகலை - பாத்திர மரபு கூறிய காதை. 3. கம்பராமாயணம் - சேது பந்தனப்படலம். 4. பெரியபுராணம் - கோச்செங்கட்சோழ நாயனார் புராணம். 5. தேம்பாவணி - கோலியாத் படலம். 6. சீறாப்புராணம் - மானுக்குப் பிணை நின்ற படலம்		
Unit - II	**புதினம்** வேரில் பழுத்தபலா - சு.சமுத்திரம்.		
Unit - III	**இலக்கணம்** யாப்பும் அணியும் செய்யுள் உறுப்புகள், எழுத்து, அசை, சீர், தளை, அடி, தொடை ஆகியன பற்றிய விளக்கம். பாவகை, வெண்பா, ஆசிரியப்பா ஆகியவற்றின் பொது இலக்கணங்கள்.அணி, வகைகள், உவமை, உருவகம், வேற்றுமை, பின்வருநிலை, சிலேடை அணிகள்.		
Unit – IV	**இலக்கிய வரலாறு** அலகு 1, அலகு 2ல் உள்ள பாடம் தொடர்பான இலக்கிய வகைகள் தொடர்பான இலக்கிய வரலாறு.		
Unit - V	**படைப்பாற்றல்** மரபுக் கவிதை - புதுக்கவிதை படைத்தல்.		

பார்வை நூல்கள்

தண்டி (1926).*தண்டியலங்காரம்.*

அமிர்தசாகர் (2009.)*யாப்பருங்கலக்காரிகை.*

பாக்கியமேரி (2013).*வகைமை நோக்கில் தமிழிலக்கிய வரலாறு* (5வது பதிப்பு).

சு.சமுத்திரம் (2015).*வேரில் பழுத்த பலா*

தமிழண்ணல் (2017).*புதிய நோக்கில் தமிழிலக்கிய வரலாறு* (36வது பதிப்பு).

(மேலும் அலகு-1-ல் குறிப்பிடப்பெற்றுள்ள காப்பியப் பகுதிகள்)

Course Teacher
Dept of Tamil
Prof.K. Manimekalai

Semester -III			
Coursecode: 4533E3	**COURSE – III - ENGLISH FOR ENRICHMENT – II**	**Credits: 3**	**Hours: 4**
Objectives	➢ To introduce the learners the short stories and one act plays. ➢ To teach tenses, voices, degrees of comparison, descriptive writings etc.		
Unit - I	**Short Stories** 1. Two Old Men – Leo Tolstoy 2. The Diamond Necklace – Guy de Maupassant 3. The Verger – Somerset Maugham 4. The Postmaster – Rabindranath Tagore.		
Unit - II	**One Act Plays** 1. Riders to the Sea – J.M.Synge 2. The Rising of the Moon – Lady Gregory		
Unit - III	**One Act Plays** 1. A Kind of Justice – Margaret Wood 2. The Refugee – Asif Currimbhoy		
Unit – IV	**Grammar** Tenses, Voices, Degrees of Comparison		
Unit - V	**Composition** Agenda, Minutes, Notice, Descriptive Writing		
Reference and Textbooks: Narayan, R. K (2008*). Swami and Friends.*Chennai: Indian Thought Publications. Krishnaswamy.N(2009).Modern English – *A Book of Grammar Usage and Composition*. Macmillan Publishers. Nagarajan,V et al(2015).*Six Short Stories*, Ed. By the Board of Editors, Harrows Publications, Chennai. Nagarajan,V et al(2015*).One Act Plays*, Ed. By the Board of Editors, Harrows Publications, Chennai. Nagarajan,V et al(2016)Gate Way to English – *An Anthology of Prose and Poetry* Ed. By the Board of Editors, Harrows Publications, Chennai.			
Outcomes	➢ Can familiar with short stories and one act plays of great writers. ➢ Can speak and write English without committing mistakes.		

Course Teacher
Dept of English
Dr.P.Veeramani

Semester -III			
Coursecode:4533C1	**CORE: V PRINCIPLES OF NUTRITION**	**Credits:5**	**Hours:6**

Objectives	➢ To obtain an insight into the major nutrients in food and the current trends in nutrition. ➢ To understand the nutritional deficiencies and recommended dietary allowances to acquaint about nutritional requirement in special conditions.
Unit - I	**Introduction to Nutrition -** History of Nutrition- Brief History of Nutritional Status, Definition of Nutrition and Malnutrition. Scope of Nutritional Studies.Requirements and RDA - Formulation of RDA and Dietary Guidelines, Reference Man and Reference Women.
Unit - II	**Energy and Carbohydrates -** Energy Balance, Assessment of Energy Requirements, Deficiency and Excess. Carbohydrates- Definition, Classification and functions, Food sources, Digestion, Absorption, Metabolism, RDA and deficiency. Dietary fiber – Classification and Nutritional significance.Glycemic index of foods.
Unit - III	**Proteins -** Definition, Classification and functions, Food sources, Digestion, Absorption, Metabolism, RDA and deficiency. Factors affecting protein quality; Protein quality evaluation methods; Quality difference between vegetable and animal protein sources;
Unit – IV	**Lipids -** Definition, Classification, Functions and Sources of Lipids, Digestion and Absorption, Metabolism,RDA and deficiency. Types such as saturated and unsaturated; Essential Fatty Acids (EFA): Definition and functions; PUFA; Role of n-3, n-6 fatty acids in health and disease; Trans fatty acids and its association to cardiovascular diseases.
Unit - V	**Vitamins,Minerals and Water -** Vitamins- Functions, Requirements, Sources, Deficiency and Excess (Fat soluble and water soluble)Macro and Micro Minerals - Functions, Requirements, Sources, Deficiency and Excess (Calcium, Phosphorus, Magnesium, Sodium, Potassium and Chloride, Iron, Copper, Manganese, Fluoride, Zinc and Iodine).Interrelationship between vitamins and minerals in metabolism.Water: Body composition – extra- and intra- cellular fluid; Physiological functions; water balance and its regulation; Requirement and the sources; Nutritional and health problems due to imbalance of water intake.

Reference and Textbooks:

HongKongNeiman,N. Catherine. (1990). *Nutrition*, Wm.C. Brown Publishers. USA.

Shubhangini,A. Joshi,(1992). *Nutrition and Dietetics*.Tata Mc Grow- Hill publishing Company Ltd, New Delhi.

Shakuntala Manay. (2008)*Foods Facts and Principles*. New Age International (P) Ltd, Publishers, Chennai.

Swaminathan, M. (2010). *Essentials of Food and Nutrition*. Vol. I and II, Ganesh and Company, Madras.

Srilakshmi,B.(2018).*Nutrition Science*(7thed.). New Age International (P) Ltd, Publishers, Chennai.

Outcomes	Enable the students to ➢ Know about major nutrients in food and the current trends in nutrition ➢ Understand the nutritional deficiencies and recommended dietary allowances to acquaint about nutritional requirement in special conditions

Course Teacher
R.Ramya
Prof.K. Manimekalai

Semester -III			
Coursecode:4533C2	**CORE VI - HUMAN PHYSIOLOGY**	**Credits: 5**	**Hours: 5**
Objectives	➢ To enable the students to understand the basic structure and functions of human body ➢ To inculcate knowledge about common diseases / disorders affecting each system		
Unit - I	**Digestive System** - Brief Description of Organs of the GI Tract. Accessory Organs of Digestion – Liver, Gall Bladder and Pancreas, Digestion and Absorption of Carbohydrates, Protein and Fats. NERVOUS SYSTEM- Elementary Anatomy of Nervous System, Functions of Different Parts of the Brain, Autonomic, Sympathetic and Parasympathetic Nervous System.		
Unit - II	**Respiratory System** - Respiratory Organs, Structure and their Functions, Mechanism of Respiration. Sense Organs – Eye: Structure and Functions. Ear: Structure and Functions. Nostrils: Structure and Functions		
Unit - III	**Blood**- Composition and Functions, Blood Clotting and its Significance, Blood Groups, Blood Transfusion and its Importance. Cardiovascular System-Structure of Human Heart and Functions – Cardiac cycle, Blood Pressure, Pulse Pressure. Lymphatic System - Lymph, Lymph Glands and its Functions.		
Unit – IV	**Skin** – Structure and Functions, Regulation of Body Temperature. Excretory System-Structure and Functions of Kidney, Mechanism of Urine Formation.		
Unit - V	**Reproductive System** - Reproductive system of Male and Female, Menstrual cycle, Menarche, Fertilization and Menopause. Endocrine System-Listing of endocrine glands and location, Functions of Thyroid, Parathyroid, Adrenal and Pituitary Glands.		

Reference and Textbooks:

Winwood (1988)*Sear's Anatomy and Physiology for nurses*.Edward Arnold, London

Chatterjee,C.C.(1988). *Text book of Medical physiology*.W B Saunder's Co, London.

Guyton,A.C, Hall, J.E.(1996). *Text book of Medical Physiology*.Prism Books (Pvt) Ltd, Bangalore.

Kumar, R and Kumar,M. (2004). *Guide to prevention of lifestyle diseases*, Deep and Deep publications, New Delhi.

Kelly A. Young,(2013). *Anatomy and Physiology,* California State University, Long Beach.

Outcomes	Enable the students to ➢ Understand the functions of the human organ system ➢ Gain knowledge on the common diseases and disorders

Course Teacher
P.Sindhuja
Prof.K.Manimekalai

Semester -III			
Coursecode:4533A3	**ALLIED: III FASHION DESIGNING**	**Credits:4**	**Hours:4**

Objectives	➢ To impart knowledge on Design, Fashion Production, elements and Principles of Design.
Unit - I	**Design:** Definition, Classification of Design: Structural and Decorative, Design Process: Concept, research, design exploration and conceptualization, design development.
Unit - II	**Fashion Production:** Sketching the fashion figure, Sketching Garments, Garments and Garment components, Construction design details, Trimmings and Treatments, Figure Types-Achieving Perfect Proportions
Unit - III	**Elements of Design:** Line: Classification, Types- Straight, Curved, Zigzag, Direction-Vertical, Horizontal, Diagonal, Application- Structural and Decorative, Psychological Association and visual effect of Lines. Shape: Define, Types of Shape, Visual effects of shape, Space and shape, Silhouette: Basic fashion Silhouettes, Texture: Define, Effect of Texture, Categories of Texture, Visual effects of texture.
Unit – IV	**Elements of Design** Colour: Understanding Colour, Colour dimensions: Hue, Value and Intensity, Colour Wheel: Munsell Colour system and Prang Colour System, Colour Warmth: Warm Colours, Cool Colours, Colour harmonies/Schemes, Colour forecasting, Expressive Qualities of Colour.
Unit - V	**Principles of Design:** Repetition, Gradation, Rhythm, Radiation, Harmony, Contrast, Dominanace, Proportion, Balance, Unity

Reference and Textbooks:

Pamela Stecker. (1996). *The Fashion Design Manual*, Macmillan Education AU.

Tim Jackson, David Shaw (2000). *Mastering Fashion Buying and Merchandising Management*.Macmillan International Higher Education.

Tate. (2004). *Inside Fashion Design*, Pearson Education India.

Richard Sorger, Jenny Udale (2006).*The Fundamentals of Fashion Design*. AVA Publishing.

Ann Marie Fiore (2010) *Understanding Aesthetics for the Merchandising and Design Professional.*

Outcomes	Enable the students to: ➢ Summarize the Design Process ➢ Describe the elements and Principles of design ➢ Discuss the fashion production process

Course Teacher
M.Priyanka
Prof.K. Manimekalai

Semester -III			
Coursecode:453S01	**SBS– I FABRIC ANALYSIS (P)**	**Credits: 2**	**Hours: 4**
Objectives	➢ To gain knowledge on fabric design and structure in woven and knitted fabrics		
	I. Woven Fabrics 1. Plain, Matt 2. Regular Twill 3. Honey comb 4. Extra Weft 5. Mock leno 6. Figured terry 7. Double Cloth 8. Sateen 9. Jacquard **II. Knitted Fabrics** 1. Single and Double jersey 2. Rib		
Reference and Textbooks: John T Andrews (1971). *Techniques of till fabric analysis*. Grorver E.B. (1980). *Hand Book of textile testing,* Wiley Easters University, Delhi. Booth JE (1996), *Principle of Textile Testing,* CBS Publication New Delhi. Ruth Clock & Grace Kunz (2000), *Apparel Manufacture- Sewn Product Analysis,* Upper Sadale River Publications, New York. Craig Huntington, (2013). *Tensile Fabric Structures*: Design Analysis and Construction.			
Outcomes	Enable the students to ➢ Aware about the fabric design and structure in woven and knitted fabrics ➢ Analyze the following fabric and draw design, draft and peg plan		

Course Teacher
Dr.A.K.P.Dhanakodi
Dr.P.Veeramani

Semester -III			
Coursecode:453S02	**SBS-I GARMENT DESIGNING AND CONSTRUCTION (P)**	**Credits: 2**	**Hours:4**
Objectives	➢ To design the garments for different occasions and seasons. ➢ To prepare patterns and construct garments		
	1. Designing garments for different occasions and seasons. 2. Preparing patterns and construction of garments (mens/womens/childrens)		
Reference and Textbooks: Gertrude Strict Land, (1956), *"A Tailoring Manual, Mac Million Company*, New York. Brockman, Magritha (1966), *The Theory of Fashion Design,* John Wiley Sons, New York. Mary Mathews (1986), *Advanced Clothing Construction, Part I & II,* Cosmic Press, Chennai. Peg Couch, *Garment Construction*: A Complete Course on Designing and Making Clothing. for Fit and Fashion (Illustrated Guide to Sewing) Sarah Veblen. *The Complete Photo Guide to Perfect Fitting*. Alison Smith. *The Sewing Book* (Dk)			
Outcomes	Enable the students to ➢ Design and construct the garments for different occasions and seasons ➢ Design and construct any type of garments including men's, women's and children's wear.		

Course Teacher

M.Priyanka

Prof.K. Manimekalai

Semester -IV			
Coursecode: 4534T4	**பொதுத்தமிழ் தாள் - 4 - பண்டைய இலக்கியமும் நாடகமும்**	**Credits:3**	**Hours:4**
நோக்கம்	கல்தோன்றி மண்தோன்றாக் காலத்து முன்தோன்றிய மூத்த மொழி தமிழ்மொழியாகும். ஆகையால் தமிழ் மொழி செம்மொழியாகக் கருதப்படுகிறது. இச்செம்மொழியில் தோன்றிய இலக்கியங்கள் சங்க இலக்கியங்கள், இடைக்கால இலக்கியங்கள், தற்கால இலக்கியங்கள் என வகைப்படுத்தப்படுகின்றன. சங்க இலக்கியங்களாக எட்டுத்தொகையும் பத்துப்பாட்டும் வகை செய்யப்படுகின்றன. இடைக்கால இலக்கியங்களாகக் காப்பியங்களும், பக்தி இலக்கியங்களும், சிற்றிலக்கியங்களும் வகைப்படுத்தப்படுகின்றன. இதைப்போல் தற்கால இலக்கியங்களாக உரைநடை, நாடகம், கவிதை என வகைப்படுத்தப்படுகின்றன. அவ்வகையில் சங்க இலக்கியங்களின் வழி மாணவர்களுக்கு அறங்களைப் போதிப்பதும், தற்கால இலக்கியங்களின் வழி நடிப்புடன் கூடிய அறக்கருத்துக்களை நாடகங்கள் வாயிலாக உணர்த்தப்படுவதும் இத்தாளின் நோக்கமாக அமைகின்றது.		
Unit - I	அ. பத்துப்பாட்டு - சிறுபாணாற்றுப்படை ஆ. நற்றிணை - வெள்ளிவீதியார் பாடல் எண்கள்: 70,335,348. இ. குறுந்தொகை பாடல் எண்.40 - யாயும் ஞாயும் எனத் தொடங்கும் பாடல் (குறிஞ்சி) செம்புலப்பெயல் நீரார் பாடல் எண்.43 - செல்வார் அல்லர் எனத் தொடங்கும் பாடல் (பாலை) ஔவையார் பாடல் எண்.49 - அணிற் பல்லன்ன எனத் தொடங்கும் பாடல் (நெய்தல்) அம்மூவனார் பாடல் எண்.61 - தச்சன் செய்த எனத் தொடங்கும் பாடல் (மருதம்) தும்பிசேர்கீரன் பாடல் எண்.110 - வாரார் ஆயினும் எனத் தொடங்கும் பாடல் (முல்லை) கிள்ளிமங்கலக்கிழார் ஈ. கலித்தொகை - பாடல் எண்.105. அரைசுபட எனத் தொடங்கும் பாடல் (முல்லை) சோழன் நல்லுருத்திரன். உ. அகநானூறு - திருமணச் சடங்குப் பாடல்கள் 2 (86,128) ஊ. புறநானூறு - பிசிராந்தையார் பாடல்கள் (பாடல் எண்கள். 67,184) எ. திருக்குறள் - பெரியாரைத் துணைக்கோடல், சிற்றினம் சேராமை ஆகிய இரு அதிகாரங்கள். ஏ. நாலடியார் - பாடல் எண்.135 - கல்வி கரையில எனத் தொடங்கும் பாடல். பாடல் எண்.215 - கோட்டுப் பூப்போல எனத் தொடங்கும் பாடல். பாடல் எண்.248 - நல் நிலைக்கண் தன்னை நிறுப்பானும் எனத் தொடங்கும் பாடல். ஐ. பழமொழி நானூறு பாடல் எண்.46 - நெடியாது எனத் தொடங்கும் பாடல். பாடல் எண்.47 - தோற்றத்தாலர் எனத் தொடங்கும் பாடல். பாடல் எண்.48 - மிக்குடையார் ஆகி எனத் தொடங்கும் பாடல்.		
Unit - II	**நாடகம்** - நீதிதேவன் மயக்கம் - அறிஞர் அண்ணா.		
Unit - III	**இலக்கணம்** அகப்பொருள், (7 திணைகள்), புறப்பொருள் (12 திணைகள்), களவும், கற்பும், உள்ளுறை, இறைச்சி (ஆ.சிவலிங்கனார், தமிழ் இலக்கண உணர்வுகள், கபிலன் பதிப்பகம், புதுச்சேரி.		
Unit – IV	**இலக்கிய வரலாறு** அலகு 1, அலகு 2ல் உள்ள பாடம் தொடர்பான இலக்கிய வகைகள் தொடர்பான இலக்கிய வரலாறு.		
Unit - V	**படைப்பாற்றல்** ஓரங்க நாடகம் படைத்தல்.		
பார்வை நூல்கள்			

நாற்கவிராச நம்பி (2004) -*நம்பியகப் பொருள்.*
பாக்கியமேரி (2013) - *வகைமை நோக்கில் தமிழிலக்கிய வரலாறு* (5வது பதிப்பு).
ஐயனாரிதனார் (2015) - *புறப்பொருள் வெண்பாமாலை.*
தமிழண்ணல் (2017) - *புதிய நோக்கில் தமிழிலக்கிய வரலாறு* (36வது பதிப்பு).
அறிஞர் அண்ணா (2019) - *நீதி தேவன் மயக்கம்* (7வது பதிப்பு).
(அலகு 1-ல் குறிப்பிடப்பட்டுள்ள பண்டைய இலக்கிய பகுதிகள்)

Course Teacher
Dept of Tamil

Semester -IV			
Coursecode:4534E4	**COURSE – IV- ENGLISH FOR ENRICHMENT – IV**	**Credits:3**	**Hours:4**
Objectives	➢ To make them familiar with the significant texts of great writers ➢ To give practice in group communication.		
Unit - I	**Drama** Pygmalion – G.B. Shaw		
Unit - II	**Fiction** Swami and Friends – R.K.Narayan		
Unit - III	**Tales from Shakespeare** 1.The Merchant of Venice 2. Romeo and Juliet 3. The Winter's Tale		
Unit – IV	**Grammar** 1. Concord 2. Question Tag 3. Kinds of Sentences 4. Direct and Indirect speeches		
Unit - V	**Composition** 1. Expansion of Proverbs 2. Group Discussion 3. Conversation (Apologizing, Requesting, Thanking)		
Reference and Textbooks: Narayan, R. K (2008*). Swami and Friends*.Chennai: Indian Thought Publications. Kriahnaswamy, N (2009). Modern English – *A Book of Grammar Usage and Composition*. Chennai: Lakshmi Publications. Nagarajan, V.et al (2013). *Tales from Shakespeare.* Chennai: Harrows Publications. Nagarajan,V. et al(2015*).One Act Plays*, Ed. By the Board of Editors, Harrows Publications, Chennai. Shaw, GB (2018).*Pygmalion. New Delhi:* Fingerprint! Publishing.			
Outcomes	➢ Can acquaintance with the literary works of great writers. ➢ Can understand formal and informal communication.		

Course Teacher
Dept of English

Semester -IV			
Coursecode:4534C1	**CORE – VII COMMUNITY NUTRITION**	**Credits: 5**	**Hours: 6**

Objectives	➢ To understand the Nutritional Problems of a Community ➢ To Learn the Methods used for assessing the Diet and Nutritional Status of a Community ➢ To know the On-going Intervention Programmes for Overcoming Malnutrition in the Community.
Unit - I	**Nutritional status in India**: Importance of Good Nutrition. Nexus between Poverty and Malnutrition, PEM, Micro Nutrient Deficiencies (Vitamin A, Nutritional Anaemia, and IDD), Prevalence of Diet related Non-Communicable Chronic Diseases.
Unit - II	**Assessment of Nutritional status of the Community**- Direct Methods – (a) Anthropometric Measurements (b) Clinical Examination (c) Bio-Physical Examination (d) Functional Assessment (e) Bio-Chemical Estimation. Indirect Methods - (a) Dietary Assessment (b) Vital Health Statistics.
Unit - III	**Nutrition Education**- Meaning, Need, Principles and Methods. Factors influencing Nutrition related Behaviour, Behavioural Changes related to Nutrition Education. Nutrition intervention for especially vulnerable groups, Awareness on Fortified Foods.
Unit – IV	**Intervention Programmes to Combat Malnutrition**- ICDS Programmes, Balwadi Nutrition Programmes, Noon Meal Programme. National Nutritional Anaemia Prophylaxis Programme (NNPP), Iodine Deficiency Disease Control Programme.
Unit - V	**Role of International Organizations** – UNICEF, FAO, WHO, CARE, World Bank.S

Reference and Textbooks:

National Nutrition Policy (1993). Govt. of India.

Swaminathan. M. (2010). *Essentials of Food and Nutrition*. Vol I and Vol. II Madras. Ganesh and company. Hyderabad.

SreeDevi.V. (2011). *Nutrition Education*. Discovery Publishing House. New Delhi.

Srilakshmi (2017). *Nutrition Science* (*6th ed*). New Age International Pvt.Ltd. New Delhi.

Srilakshmi (2018). *Food Science* (7th*ed*).New Age International Pvt.Ltd. New Delhi.

Outcomes	Enable the students to ➢ Aware about the nutritional problems and status in a community ➢ Gain knowledge on nutritional intervention programmes and acquire skill in Conducting Nutrition Education

Course Teacher

P.Venkateswari

Semester -IV			
Coursecode:4534C2	**CORE –VIIIPUBLIC HEALTH NUTRITION**	**Credits:5**	**Hours:6**
Objectives	➢ To understand the concept of public health and nutrition ➢ To be familiar with the current concerns in public health nutrition		
Unit - I	**Concept and scope of public nutrition -** Relation between health and nutrition - Definition and multidisciplinary nature of public nutrition - Concept and scope - Role of public nutritionist - Millennium Development goals - Current concerns in public health nutrition		
Unit - II	**Introduction to Nutritional deficiency diseases -** Causes, symptoms, treatment, prevention of the following: - Protein Energy Malnutrition (PEM), Vitamin A Deficiency (VAD), Iron Deficiency Anemia (IDA), Iodine Deficiency Disorders (IDD), Zinc Deficiency, Flurosis.		
Unit - III	**Social health problems -** Smoking, Alcoholism, Drug addiction, AIDS including AIDS Control Programme, community health care, national health care delivery system, population policy. Health economic and economics of malnutrition impact on productivity and national development.		
Unit – IV	**Nutrition for special conditions -** Introduction to Nutrition for physical fitness and sport, Feeding problems in children with special needs, Considerations during natural and man-made disasters e.g. floods, war- basic guidelines in disaster management.		
Unit - V	**Food Security** (a) Key terms, factors affecting food security, recent concern (b) Technologies for food and nutrition security (c) Health based interventions including immunization, provision of safe drinking water/sanitation, prevention and management of diarrhoeal diseases.		
Reference and Textbooks:			
Mudambi,S.R and Rajagopal,MV.(2007). *Fundamentals of Foods, Nutrition and Diet Therapy (5th ed).*New Age International Publishers. Wardlaw, GM. Hampl,J.S.(2007). *Perspectives in Nutrition (7th ed).* McGraw Hill.. Bamji MS, Rao NP, and Reddy V. (2009). Text Book of Human Nutrition. Oxford & IBH Publishing Co. Pvt Ltd. Khanna, K. et al. (2013). *Textbook of Nutrition and Dietetics*. Phoenix Publisher. Srilakshmi,B(2019*).Dietetics(8th ed).*New Age International (P) Ltd.			
Outcomes	Enable the students to ➢ Gain knowledge about public health and nutrition ➢ familiar with the current concerns in public health nutrition		

Course Teacher
R.Ramya
Prof.K. Manimekalai

Semester -IV			
Coursecode:4534A4	**ALLIED: IV INTERIOR DESIGN**	**Credits:4**	**Hours:5**
Objectives	➢ To understand the elements and principles of design. ➢ To learn about the elements of art in the creation of new design ➢ To learn the application of principles of design in creating beautiful interiors.		
Unit - I	**Development of Interior Design:** Concepts - A Historic Review. Place of Interior in the Modern Era- Changing trends and Salient Features, Objectives of Aesthetic Planning - Beauty, Expressiveness, Functionalism, Economy. Role of good Taste – Meaning and Importance. Need for developing skill in Aesthetics.		
Unit - II	**Design:** Definition, Meaning, Purpose. Types - Structural and Decorative Characteristics, Classification of Decorative Design - Naturalistic, Conventional, Geometric, Abstract, Historic, Biomorphic. Elements of Design - Line and Direction, Form and Shape, Size, Colour, Light, Pattern, Texture and Space - Application of elements to Form Designs.		
Unit - III	**Principles of Design:**Balance, Rhythm, Emphasis, Harmony, Proportion - Meaning and Application of Design Concepts in the Interior and Exterior Houses and other Commercial Buildings.Development of design from Motifs and Application.		
Unit – IV	**Flower arrangement**- Principles &Types of Flower Arrangement. Accessories in the Home – Classification and Selection.		
Unit - V	**Man as a Consumer of Design**:Qualities and Role of a Good Interior Designer, Interior Design - Career Options. Application of Design in Interior Decoration.		
Reference and Textbooks: Faulkner, S. and Faulkner, R. (1987) *Inside Today's Home*, Rine hart publishing company, Newyork. Pratap, R. M. (1988). *Interior Design principles and practice*, standard publisher's distribution, Delhi. Marcel van Dijk(2006).*Natural Interior Design* (Floral Interior Design). Goldstein, (2006). *Art in Everyday life*, Oxford and IBH publishing house. Seetharaman, P and Pannu, P (2013). *Interior Design and Decoration*, CBS publishers and Distributors, New Delhi			
Outcomes	Enable the students to ➢ Gain the basic knowledge on Interior design ➢ Aware about the New decorative design		

Course Teacher
Dr.A.K.P.Dhanakodi
Prof.K. Manimekalai

Semester -IV			
Coursecode:453S03	**SBS-IIFOODPROCESSING&PRESERVATION (P)**	**Credits:2**	**Hours:2**
Objectives	➢ To impact knowledge on basic and applied aspect of food processing and preservation. ➢ To provide the knowledge of basic principles and procedure in the production of important food product.		
	Practical 1. Preparation of jams and jellies 2. Preparation of syrups, sauce, squashes 3. Preparation of pickles, chutneys 4. Preparation of marmalades, candies 5. Preparation of crystallized fruits /vegetables 6. Visit to processing units 7. Visit to FCI and IIFPT (Tanjore)		
Reference and Textbooks: Olickle, J.K. (1990). *New Product Development and Value Added*. Food Development Division, Agriculture. Canada. Fuller, G.W. (1994). *New Food Product Development*: From Concept to Market place, CRC Press. New York. Man,C.M.D and James,A.A. (1994). *Shelf life Evaluation of Foods*. Blackie Academic and Professional. London. Sivasankar,B(2002).*Food Processing and Preservation*. Prentice Hall India Learning Private Limited International Journal of Food Science and Technology.			
Outcomes	Enable the students to ➢ Familiarize with the potential use of various by products of food industry. ➢ Gain knowledge on basic principles and procedure in the production of important food product.		

Course Teacher
P.Archana
Dr.P.Veeramani

Semester -IV			
Course code: 453S04	**SBS-II FUNDAMENTALS OF APPAREL DESIGNING (P)**	**Credits:2**	**Hours:2**
Objectives	➢ To study and prepare the basic components for apparel constructions. ➢ To enable the students to prepare the sample for fullness, necklines, pockets, yokes etc.,		
Practical	1. **Preparation of Samples for Temporary Hand Stitches** - (Even, Uneven, Diagonal, Slip Basting) 2. **Preparation of Samples for Permanent Hand Stitches** - (Running, Back, Run and Back, Hemming, Slip, Over – Casting, Over Handing, Whipping) 3. **Preparation of Samples for Seams** - (Plain, Single Top Stitch, Double Top Stitch, Welt, Lapped, Slot, French, Mantua Makers, Flat Fell-Machine or Hemmed, Piped and Corded Seam.) 4. **Preparation of samples for seam finishes** - (Pinked, Double Stitch, Edge Stitch, Herring Bone, Over Cast and Bound Seam Edge Finish.) 5. **Preparation of samples for Fullness** - (Darts, Tucks, Pleats, Godets, Gathers and Shirrs, Frills and Ruffles.) 6. **Preparation of samples for Faced Necklines** - (Bias, Shaped, and Decorative.), Binding Necklines - (Bias and French.) 7. **Preparation of samples for Pockets** - (Patch Pocket, Welt/Bound Pocket, Set-in Pocket/Side Seam Pocket.) 8. **Preparation of samples for yokes and its types**. 9. **Preparation of plain sleeves, Puff sleeve (anyone type**), raglan / kimono Sleeve. 10. **Preparation of Fasteners** - (Hook and Eye (Thread and Metal), Press Button, Velcro, Fancy Button, Shirt Button, Button Hole, Link/Kurta Buttons.)		
Reference and Textbooks: Anita A. Stamper, (1991).*Experimental Apparel Construction*: Laboratory Manual. Wendy Mullin, (2006).*Sew U*: The Built by Wendy Guide to Making Your Own Wardrobe. Wendy Mullin, (2008).*Sew U Home Stretch*: The Built by Wendy Guide to Sewing Knit Fabrics. Gokarneshan. N. (2009), *Fabric Structure and Design* - New Age International. Nora Abousteit and Alison Kelly, (2011).*The Burda Style Sewing Handbook.*			
Outcomes	Enable the students to ➢ Prepare the basic components for apparel constructions ➢ Prepare the sample for fullness, necklines, pockets ,yokes etc.,		

Course Teacher
M.Priyanka
Dr.P.Veeramani

Semester - V			
Course code:4535C1	**Clinical Nutrition**	**Credits: 5**	**Hours: 5**

Objectives	➢ To learn about the metabolism of carbohydrate, amino acid and lipids ➢ To understand the Biochemical changes due to metabolic disorders
Unit - I	**Metabolism -** Review of Digestion and Absorption. Carbohydrate – Glycogenesis, Glycogenolysis, Glycolysis, Gluconeogenesis, Kreb's Cycle and Pentose Phosphate Pathway. Amino Acid - Intermediary Metabolism and Urea Cycle.Lipid - Intestinal Resynthesis of TG, Transport, Oxidation of Fatty Acids, Biosynthesis of Cholesterol
Unit - II	**Genetic control of metabolism -** Nucleic Acids, DNA replication, RNA – Synthesis, Types and Functions, Genetic Code, Protein Biosynthesis, Recombinant DNA Technology
Unit - III	**Biochemical changes due to disorders of metabolism -** Diabetes Mellitus, Inborn errors of Metabolism with respect to Lactose, Galactose, Phenyl Alanine and Uric Acid (Gout)
Unit – IV	**Digestive System** - Diarrhoea, Constipation, Gastritis, Ulcers, Colitis, Malabsorption Syndrome -Nutritional Implications. Metabolic and Nutritional Implications of Hepatitis, Cirrhosis of Liver, Hepatic Coma. Pancreatitis, Cholecystitis and Cholelithiasis
Unit - V	**Renal System -** Metabolic and Nutritional implications of Nephritis, Nephritic Syndrome, Renal Failure. Renal Calculi, Dialysis. Water, Electrolyte Losses, and Replenishment. Effect of Dehydration and Water Intoxication

Reference and Textbooks

Shils. et al. (1994). *Modern Nutrition in Health and Disease*. Vol. I and II. New York, Lea and Febiger

Ramakrishnan, S. & Venkat Rao. (1995). *Nutritional Biochemistry*. Chennai, T.R.

Devlin, T.M... (1997). *Textbook of Clinical Biochemistry*. New York, John Wiley and Sons.

Harper, H.A. (1997). *Review of Physiological Chemistry(21st ed)*. Los Angeles, Lange Medical

Ramnik Sood(2010).*Textbook of Clinical Biochemistry*. CBS Publishers.

Outcomes	➢ Aware about the digestion and absorption of carbohydrate, amino acid and lipids. ➢ Gain knowledge about Biochemical changes due to metabolic disorders

Course Teacher
Dr.P.Veeramani
P.Archana

Semester - V			
Coursecode:4535C2	**Clinical Nutrition (P)**	**Credits: 4**	**Hours: 8**
Objectives	➢ To develop skills in Analysis of Urine and estimation in Serum		
Content	1. Analysis of Urine 2. Collection of Blood and separation of Plasma and Serum 3. Estimation of Blood Glucose 4. Estimation of Total Protein 5. Determination of A/G ratio 6. Estimation of Serum Urea 7. Estimation of Serum Creatinine 8. Estimation of Cholesterol 9. Estimation of Bilirubin		
Reference and Textbooks Oser, B.L. (1965). *Hawk's Physiological Chemistry*. New Delhi: Tata McGraw Hill ICMR. (1978). *Laboratory techniques in Nutrition*. Hyderabad: NIN. Chawla, R. (1995). *Practical Clinical Biochemistry*. Methods and Interpretations. New Delhi: Jaypee Brothers. Medical (P) Ltd. Pattabiraman, T.N. (1998). *Laboratory Manual in Biochemistry*.New Delhi: All India Publishers and distributors Ramnik Sood(2010).*Textbook of Clinical Biochemistry*. CBS Publishers.			
Outcomes	➢ Enable the students to estimate the Blood Glucose, Total Protein, Serum Urea, Serum Creatinine, Cholesterol and Bilirubin in the given sample		

Course Teacher
Dr.P.Veeramani
P.Archana

Semester – V			
Coursecode:4535C3	**Food Microbiology**	**Credits: 5**	**Hours: 5**

Objectives	➢ To study about the microorganisms in food ➢ To Study the Microbes in relation to food spoilage, food borne diseases and food preservation
Unit - I	**Microbiology of Foods:** Concept and its history. Role of microbiology in biotechnology, Role of microorganisms in fermented foods – soya bean products, vegetable and dairy foods, baked items, other preparations, uses of microbes in industry
Unit - II	**Incidence and growth of microorganisms in food** : Microbes in Air, water and soil, food contamination and its sources – fruits, vegetables, milk and meat Factors affecting the growth of microbes in food, control and its destruction – Physical and chemical methods
Unit - III	**Food spoilage** – its causes, spoilage and its chemical changes. Spoilage in cereal, milk and its products. Meat, poultry, fish and other sea foods, fruits and vegetables, fruit juices and preserves – by yeasts, moulds and bacteria. Prevention of spoilage
Unit – IV	**Hazards of food microbes**: Food borne diseases and its types - intoxications, infections, mycotoxins
Unit - V	**Examination of food borne diseases:** Receiving report and assessment, investigation, identifying causative factors, association between food items an illness, sample collection and reporting

Reference and Textbooks

Frazier, W.C & Westhoff, D.C. (1992). *Food Microbiology*. New Delhi: Tata McGraw Hill Hill Publishing Co., Ltd.

Annak. Joshua, (2001). *Microbiology.* Popular Book Depot. Chennai.

Ray, B. (2001). *Fundamental Food Microbiology, (2nd ed),* CRC press, Boca raton FL

Jay JM, Loessner DA, Martin J.(2005) *Modern Food Microbiology. (7th ed).* Springer

Prescott LM, Harley JP, Klein DA.(2008) *Microbiology. (6th ed*). WMC Brown Publishers.

Outcomes	➢ Gain knowledge on microorganisms and its identification in food ➢ Familiar with microbes in food, food borne diseases and food preservation

Course Teacher
Prof.K. Manimekalai
R.Ramya

Semester - V			
Course code: 453E1A	**Elective I (A) - Pattern Making (P)**	**Credits: 3**	**Hours: 5**

Objectives	➢ To enable the students to draft and grade the patterns for sleeves, collars and yoke ➢ To gain knowledge on designing, drafting and grading basic apparel for children , women and men
	I. Draft and grade the Patterns for the following a) Sleeve 1. Plain 2. Puff sleeve at top and bottom 3. Bell sleeve b) Collar 1. Shirt 2. Peter pan 3. Stand c) Yoke 1. Simple 2. Partial **II. Designing and Drafting and grading basic apparel for** a) Children Bib, Panty, Jabla Summer Frock Angel Top, A-line Frock, Sun Suit b) Women Saree Petticoat, Middi and Middi top, Blouse, Nighty, Salwar Kameez c) MenFull Sleeve Shirt, Pleated Trouser, S.B.Vest, Nehru Kurtha, Night Dress

Reference and Textbooks:

Mary Mathews (1986). *Practical Clothing Construction*. Part I and II. Chennai, Cosmic Press.

Pattern Grading for Women's Clothes. (1990). *The E Technology of Seizing* – Gerry Cooklin, Blackwell Science Ltd

Renbourn .E.T. (2004), *Physiology & Hygiene of materials& clothing,* Published by Wood Head Publishing Limited.

Zarapkar, K.R. (2005). *Zarapkar System of cutting*. Navneet India Ltd

Vedhamani and Gibson. G. (2007), *"Retail Management: Functional Principles and Practices"* Jaico Publishing House.

Outcomes	➢ Learn about draft and grade the patterns for sleeves, collars and yoke ➢ gain knowledge on designing, drafting and grading basic apparel for children , women and men

Course Teacher

Dr.A.K.P.Dhanakodi

Prof.K. Manimekalai

Semester - V			
Course code: 453E1B	**Elective: I (B) - Fashion CAD and Portfolio (P)**	**Credits: 3**	**Hours: 5**

Objectives	➢ To understand the various concepts, elements, principles and knowledge gained while studying fashion design course ➢ To gain knowledge on development of technical specification – measurements, grading, pattern making, cutting & sewing of the portfolio garment
	The students shall prepare: 1. Current trend for Mens/Women's/Kids wear in the market 2. Fashion Forecast for Spring/Summer 2020/Autumn/Winter 2020 3. Inspiration recording 4. Target Customer profiling 5. Forecast interpretation for target customer profile 6. Mind mapping / abstract development for design/ styles 7. Mood Board 8. Colour Board 9. Material Exploration 10. Theme /Story Board 11. Silhouette Development 12. Croqui model rendering with silhouette, colour & material textures 13. Develop measurement for the portfolio sketches 14. Present the Portfolio presentation – Collection submission

Reference and Textbooks:

Backman T.N. Maynard H.H and Davidson W.R (1970), *Principles of Marketing,* Ronald Press Company New York.

Pillai and Bagavathi R.S.N. (1987), *Marketings*. Chand & Co Ltd, Newdelhi.

Haroldcars / John Pomeroy, (1992), *Fashion Design & Product development,* Black Well Science, Inc., 238, Main Street, Cambridge.

Mike Easey (1995) – *Fashion Marketing*, Black well Science Ltd

Suresh, G.S. (2016). *Computer Aided Design Laboratory*. Lakshmi Publications

Outcomes	➢ Understand the trend analysis and fashion forecast ➢ Prepare the product based out of approved portfolio sketches using Fashion CAD ➢ Develop a technical specification – measurements, grading, pattern making, cutting & sewing of the portfolio garment

Course Teacher
M.Priyanka
Dr.P.Veeramani

Semester - V			
Coursecode:453E2A	**Elective: II (a)-Family Resource Management**	**Credits: 3**	**Hours: 3**

Objectives	➢ To understand family resource management ➢ To understand the basic principles of art in Interior decoration
Unit - I	**Home Management**: Meaning and definition of home management. Management process - planning, controlling and evaluation, factors influencing home management. Planning: Types and importance of controlling the plan - Adjusting, supervising checking and types of evaluation
Unit - II	**Values:** concept, characteristics, classification and factors influencing values. Goals - Concept, types, factors influencing goals. Standards – concepts, characteristics and classifications. Interrelatedness of values, goals and standards. Resources – Types, Characteristics of resources
Unit - III	**Money:**definition, types. Family budget - definition, importance, types, steps in making budget, factors affecting the budget, advantages of budgeting. Account keeping – importance, types of account systems, methods of handling money, family financial records. Time - definition, steps in making time plan, tools in time management.
Unit – IV	**Time management process:** planning, controlling and evaluation. Energy – definition, planning, controlling and evaluation. Fatigue-concept of fatigue, types of fatigue, avoidance of fatigue. Decision making: process of decision making, types of decisions, methods of resolving conflicts, Resources – Definition, meaning, classification of resources
Unit - V	**Work simplification**: Definition, importance, techniques – process chart, operation chart, Mundel's classes of changes. Concept of fatigue, types of fatigue, avoidance of fatigue

Reference and Textbooks:

Kaur, H., & Macneil, C. (1994). *Theory and Practice of Home Management*. Delhi: Surjeet.

Nickell,P., & Dorsey,J.M. (2002). *Management in Family Living*.CBS Publishers & Distributors

Nair, S.R. (2002). *Consumer Behaviour in Indian perspective*. Mumbai:,Himalaya

Varghesese, M.A.et al (2006) *Home Management*. New Age International

Seetharaman,P.et al(2019).*An Introduction to Family ResourceManagement*.CBSpublishers and distributors pvt ltd

Outcomes	➢ Gain knowledge on family resource management ➢ Will gain knowledge about the basic principles of art in Interior decoration

Course Teacher
P.Sindhuja
Dr.P.Veeramani

Semester - V			
Coursecode:453E2B	**Elective: II (B)-Genetically Modified and Organic Foods**	**Credits: 3**	**Hours: 3**
Objectives	➢ To aware about the concepts in genetically modified and organic foods. ➢ To learn the Genetic modification in Food industry and in Medicine		
Unit - I	**Genetic modification**: Concepts, methods of genetic modification, genetic modification of bacteria, plant & animal, controversies over genetic modification, policy around the world (USA, European Union, EU regulation, Japan, China & other developing countries)		
Unit - II	**Genetic modification in Agriculture:**Transgenic plants, genetically modified foods, application, future applications, ecological impact of transgenic plants		
Unit - III	**Genetically modified foods:** Organic foods, types of organic foods, identifying organic foods, organic food & preservatives		
Unit – IV	**Genetic modification in Food industry**: Background, history, controversies over risks, application, future applications		
Unit - V	**Genetic modification in Medicine:** Gene therapy, types of gene therapy, vectors in gene therapy, molecular engineering, human genetic engineering, problems & ethics		
Reference and Textbooks: Lillian E. Forman. (2009). *Genetically modified foods.*ABDO. Pamela C. Ronald. (2010). *Tomorrow's table: Organic Farming, Genetics and the Future of Food.* Oxford. Maria Rodale. (2010). *Organic Manifesto* –How organic farming can heal our planet, feed the world and keep us safe. Rodale. Natalie Regis. (2015). *Genetically modified crops and foods*. Britannica			
Outcomes	➢ Gain knowledge about the genetically modified and organic foods. ➢ Aware about genetic modification in Food industry and in Medicine		

Course Teacher
P.Venkateswari
Dr.P.Veeramani

Semester - V			
Coursecode: 453S05	**Skill Based Subject: III (A) – Bakery & Confectionary**	**Credits: 2**	**Hours: 2**
Objectives	➢ To understand the importance of baking and confectionery ➢ To develop skills and responsibility for setting up bakery and confectionery units		
Unit - I	**Introduction and Scope of Bakery and Confectionary:**Types of Flour and their Characteristics-Other Raw Materials Used in Bakery and their Role, Heat and Density Measurement, Conversion Tables, Dough and Cake Mixing Temperatures, Yeast Calculations		
Unit - II	**Types of Leavening Agents, Types of Yeast**: Nature & Characteristics of Dough-Types of Dough, Steps in Yeast Dough Production, Types of Dough Making Process, Controlling Fermentation, Bread Faults and their Causes		
Unit - III	**Biscuits:** Meaning and Preparation. Cookies-Production, Sandwich Cookies, Ice Box Cookies, bar Cookies, Macaroons, Wafers and Lace Cookies, Assorted Cookies, Factors affecting the texture, Colour and keeping quality of Biscuits and Cookies		
Unit – IV	**Different Types of Paste**: Short Crust, Sweet Crust, Rough Puff, Puff Paste, Choux Paste, Suet Paste Preparation, Faults and Products of the above pastes- Icings, Fillings and Glazes- Different Types of Icings		
Unit - V	**Sponges:** Basic Preparation Methods, Types of Sponges, Preparation of Syrups. Steps Involved in Preparing Large Cakes-Wedding Cakes, Birthday Cakes, Gateaux Basic Preparation		
Reference and Textbooks: Wilfred J. France. F. (1976). (ed.). *The New International Confectioner*. Virtue and Company. William J. Sultan. (1989)..*Practical Baking*.(5th ed)John Wiley & Sons Kinton & Ceserani. (1987*). Practical Cookery*. Holder and Stoughton Wayne Gisslen (2008). . *Professional baking*.(5th ed) John Wiley and Sons. Ashok Kumar.Y. (2009). *Text book of Bakery & Confectionary*. PHI learning.			
Outcomes	➢ Gain the basic knowledge on bakery and confectionary ➢ Develop the skills on setting up the bakery and confectionery unit		

Course Teacher
R.Ramya
Dr.P.Veeramani

Semester - V			
Coursecode: 453S06	**Skill Based Subject III (A) – Home Based Catering**	**Credits: 2**	**Hours: 2**
Objectives	➢ To understand the Home based catering ➢ To develop skills and responsibility for Food Service Unit		
Unit - I	**Introduction to Food Service -** Factors contributing to the growth of food service industry - Kinds of food service establishments		
Unit - II	**Food Production -** Menu planning: Importance of menu, Factors affecting menu planning, Menu planning for different kinds of food service units - Food Purchase and Storage - Quantity Food production: Standardization of recipes, quantity food preparation - techniques, recipe adjustments and portion control - Hygiene and Sanitation		
Unit - III	**Resources 3 -** Money – Manpower – Time - Facilities and equipment - Utilities		
Unit – IV	**Planning Of A Food Service Unit Preliminary Planning -** Survey of types of units, identifying clientele, menu, operations and delivery		
Unit - V	**Planning the set up -** Identifying resources - Developing Project plan - Determining investments- Project Proposal		
Reference and Textbooks Wood Levelle. (1988*). Food Service in Institutions* (6th ed.). New York: Macmillian. West B. Bessi Knight, J. B., & Kotschevar, L.H. (2000). *Quantity Food Production Planning & Management*. J (3rd ed.). ohn Wiley & Sons Taneja, S., & Gupta, S.L. (2001). *Entrepreneurship development*. Galgotia Mohini Sethi. (2005). *Institution Food Management*.New Age International Publishers Philip E. Thangam. (2008) *Modern Cookery for teaching and Trade Part I & II*. Orient Longmam			
Outcomes	Gain the basic knowledge on Menu planning Develop the skills on Food Service Establishments		

Course Teacher
Prof.K. Manimekalai
R.Ramya

Semester - V			
Coursecode: 453S07	**Skill Based Subject III (B)-Basics of Wet Processing**	**Credits: 2**	**Hours: 2**
Objectives	➢ To impart knowledge on Pretreatment, Dyeing and colouring, Machineries for dyeing, Printing and Finishing		
Unit - I	**Pretreatment:** Introduction, Fabric Preparation Processes: Desizing, Scouring, Mercerization, Carbonization, Heat Setting, Drying, Quality control in fabric preparation		
Unit - II	**Dyeing :** Dyeing-Definition, Classes of dyes for different fiber: Cellulosic fibers, Protein fibers, Polyamide fibers, Polyester fibers, Acrylic fibers, Fiber blends, Fluorescent brightening agents, Different Stages of Dyeing Process, Colour fastness, Assessment of Colour fastness		
Unit - III	**Machineries for Dyeing:** Machinery for Fiber Dyeing, Yarn Dyeing and Fabric Dyeing - Woven and Knitted – Hank, Beam, jigger, Winch, Jet, Soft Flow Dyeing Machines, Colour Matching Cabinet		
Unit – IV	**Printing**: Definition, Preparation of Printing Paste, Methods of Printing - Direct, Discharge, Resist, Screen Printing, Roller Printing, Rotary Printing, Transfer Printing, Stencil Printing, Block Printing, Flock Printing and Other Latest Printing Methods		
Unit - V	**Finishing**: Objective of Finishing, Classification of Finishes: Physical or Mechanical, Chemical Finishes, functional Finishes, Plasma finishes, Surface finishing, Repellent finishes, Flame- and Fire-retardant finishes, Soil release finishes		
Reference and Textbooks: Francis J. Kafka. (1973). Batik, Tie Dyeing, Stencilling, Silk Screen, and Block Printing: The Hand Decoration of Fabrics. Courier Corporation NallanKallai (2000), *Fabric Finishing,* New Age International. Clarke .W (2004), *An Introduction to Textile Printing*, Wood Head Publishing Limited, Cambridge, England Rose Sinclair. (2014). *Textiles and Fashion: Materials, Design and Technology.* Elsevier. Asim Kumar Roy Choudhury. (2017). *Principles of Textile Finishing* Woodhead Publishing.			
Outcomes	Summarize the Pretreatment process in wet processing Classify classes of dyes Describe the machineries used for fiber, yarn and fabric dyeing		

Course Teacher
Dr.A.K.P.Dhanakodi
Prof.K. Manimekalai

Semester - V			
Coursecode:453S08	**Skill Based Subject III (B)-Training for Community Development**	**Credits: 2**	**Hours: 2**
Objectives	➢ To conceptualize the training programmes for community development ➢ To evaluate sustainability of training programme for community development		
Content	1. Designing training programmes on specific goal that aims at the development of community. 2. Developing skills in selection and use of different training methods – case study, role play, psychodrama, buzz group, group discussion, transactional analysis, business games, etc. 3. Organizing and conducting training programmes for village people 4. Evaluating specific training programmes 5. Visit to training and development organizations that focus on development of community		

Reference and Textbooks

Prior, J. (1994). *Hand Book of Training and Development*. Bombay: Jaico.

Anne Hope & Sally Timmel. (2000). *Training for Transformation*: A Handbook for Community Workers. UK: Practical Action.

Jerry W. Robinson. (2010). *Introduction to Community Development*: Theory, Practice and Service Learning. SAGE Publications

Asha Ramagonda Pati,l(2013). Community Organization and Development: An Indian Perspective.PHI Publications.

Manohar Pawar,(2014).Social and Community Development Practice. SAGE India

Outcomes	➢ Become competent to frame and evaluate the sustainability of the training programme for community development

Course Teacher
Prof.K. Manimekalai
P.Sindhuja

Semester - VI			
Coursecode:4536C1	**Diet Therapy**	**Credits: 5**	**Hours: 6**
Objectives	➢ To understand the basic concepts in Dietetics and feeding methods in therapeutic diet ➢ To gain knowledge of Inborn errors of Metabolism		
Unit - I	**Basic concepts of Dietetics, Therapeutic adaptations of Normal Diet, Routine Hospital Diets -** Clear-Fluid, Full-Fluid, Soft, Regular Normal Diet. Special Feeding Methods -Parenteral and Enteral Feeding. Nutritional Care for Overweight, Underweight and Anemia		
Unit - II	**Febrile Conditions –** Typhoid, Tuberculosis and Malaria Infections.GI tract Diseases- Peptic Ulcer, Constipation, Diarrhoea, Steatorrhoea, Irritable Bowel Syndrome, Malabsorption Syndrome, Celiac Sprue, Tropical Sprue		
Unit - III	**Diseases of the Liver -** Hepatitis, Cirrhosis of Liver, Hepatic Coma. Diseases of Gall Bladder and Pancreas – Cholelithiasis, Cholecystitis, cholecystectomy, Pancreatitis, Diabetes mellitus-Prevalence, Types, Etiology, Symptoms, Diagnosis, Treatment and Complications		
Unit – IV	**Diseases of the Cardiovascular System-** Atherosclerosis and Hypertension, Hypercholesterolemia.Renal Diseases - Glomerulonephritis, Nephrotic Syndrome, Acute and Chronic Renal Failure, Dialysis and End Stage Renal Disease		
Unit - V	**Management of Cancer, Surgery, Trauma and Burns, Inborn errors of Metabolism –** Biochemical Basis and Nutritional Management of Phenyl Ketonuria and Maple Syrup Urine Disease, Allergies- Food Allergy, Reactions, Types of Allergens, Diagnosis and Treatment		

Reference and Textbooks

Mahaj, L. K., & Arlin, M. T. (1992).Kruse's Food, Nutrition and Diet Therapy, (8th ed.). London: W B Saunders

Joshi, S. A. (1992). *Nutrition and Dietetics.* New Delhi: Tata McGraw Hill

Williams, M.H. (2002). *Nutrition for health and fitness.* Boston: Mc-Graw Hill

Ruth. A. Roth. (2007). *Nutrition and Diet Therapy.*Australia: Thomson Delmar Learning

Gibney, M.J. et al. (2005). *Clinical Nutrition. (1st.ed.).* Blackwell Science

Townsend,C.E. (2000). *Nutrition and Diet therapy. (7th ed.).* Delmar Publisher Albany

Outcomes	➢ Aware about the routine hospital diets and special feeding methods in therapeutic diet ➢ Understand the inborn errors of metabolism, food allergy and its diagnosis and treatment

Course Teacher
Prof.K. Manimekalai
P.Archana

Semester - VI			
Course code: 4536C2	**Diet Therapy Internship (P)**	**Credits: 3**	**Hours: 6**

Objectives	➢ To familiarize the students with the principles of Planning Therapeutic Diets for various Disease Conditions ➢ To have practical experience in the Management of a Dietary Department and Patient Counselling for a Period of One Month
	Contents 1. Observation and study of organization and Management of the dietary department. 2. Understanding the medical history of the patients, study of case sheets and diagnostic tests used 3. Planning therapeutic diets and Computation of Nutritive value 4. Observation and Study of a. Purchase storage and issue b. Production c. Service d. Evaluation and Follow Up 5. Participation in Diet counselling units, experience in Imparting Diet counselling and understanding the Records Maintained in Diet Counselling Units

Reference and Textbooks

Joshi, S. A. (1992). *Nutrition and Dietetics.* New Delhi: Tata McGraw Hill

Mahaj, L. K., & Arlin, M. T. (1992). *Kruse's Food, Nutrition and Diet Therapy, (8th ed.).* London: W B Saunders

Joshi, S. A. (1992). *Nutrition and Dietetics.* New Delhi: Tata McGraw Hill.

Ownsend,C.E**.** (2000)**.** *Nutrition and Diet therapy.(7th ed.).* Delmar Publisher Albany

Ruth. A. Roth. (2007). *Nutrition and Diet Therapy.*Australia: Thomson Delmar Learning

Outcomes	➢ Gain knowledge about Appropriate Nutritional Care for Life Cycle ➢ Able to plan the therapeutic diets and computate the Nutritive value

Course Teacher
Prof.K. Manimekalai
P.Archana

Semester -VI			
Course code: 4536C3	**Extension Education in Home science**	**Credits: 5**	**Hours: 6**
Objectives	➢ To impart the areas of extension, development, management and communication ➢ To understand the importance and use of communication technology in extension work		
Unit - I	**Socio-Economic conditions in the Village, Analysis of the Rural Problems –** Illiteracy, Poverty, Poor Health and Malnutrition. Role of Women in Home and Farm. Role of Voluntary Agencies in Rural Upliftments – BGMS, KGNMI, AIWC, CSWB		
Unit - II	**Home Science Extension Education -** Meaning, Definition, Objectives and Principles of Extension Education. Need for Extension Education, Origin, Concept, Objectives and Characteristics of Home Science Extension, Steps in Home Science Extension Teaching, Role of Home Science Extension in Rural Development		
Unit - III	**Extension Administration and Functionaries:** Administrative set-up for Rural Development – Central, State, District, Block and Village Level, Extension personnel working at Block Level, Role and Functions of Women extension Workers, Qualities of an Extension Worker, Training Women Extension Workers		
Unit – IV	**Communication Methods and Teaching Aids used in Extension -** Meaning, Key Elements, Types, Problems, Importance of Communication in Extension. Audio Aids – Radio and Recorded Talks.Visual Aids – Flannel Graphs, Flash Cards, Posters and Charts.Audio Visual Aids – Films, Television and Video Cassettes. Recent Trends in Communication – Meaning and use of Website, e-Mail, Multimedia and e-Learning		
Unit - V	**Extension Programme Building-** Meaning and importance of Programme Planning, Principles of Programme Planning, Execution and Evaluation, Programme Development Process. Welfare Programmes for Women and Children - Programmes for Women and Children –Swadhar**,** Swa-ShaktiSABLA, Balika Samriddhi Yojana (BSY), Kishor Shakti Yojana (KSY), SGSY		

Reference and Textbooks

Dahama, O.P. & Bhat Nagar, O.P. (1985). *Extension and communication for development.* New Delhi: Oxford and IBH.

Reddy, A.A. (1987). *Extension Education.* Andhra Pradesh: Srilakshmi Press.

Kumar & Hansra, (1997) *Extension Education for Human Resource Development.* New Delhi: Concept Publishers.

Mikkelsen, Britha, (2002), *Methods for Development Work and Research.* New Delhi:Sage Publications 2.

Dale R, (2004) *Evaluating Development Programmes and Projects*. New Delhi: SagePublications. 3.

Outcomes	➢ Gain the basic knowledge in Extension Education, Management and communication ➢ Adopt the various communication technologies for the extension activity

Course Teacher
Prof.K. Manimekalai
P.Sindhuja

Semester - VI			
Coursecode:4536C4	**Food Service Management**	**Credits: 5**	**Hours: 5**
Objectives	➢ To inculcate the historical development and types of Food Services in India. ➢ To teach about the Resources required for Managing Food Outlets and Maximizing Resource Use		
Unit - I	**Organization of food service management -** Definition, Various types of Food Service institutions, their characteristics and functions		
Unit - II	**Planning a food service unit, layout design, planning of different work areas –** preparation, cleaning, storing, serving and dining areas.Lighting and ventilation, working heights in relation to equipment		
Unit - III	**Institutional Menu Planning -** Factors influencing menu planning, principles of menu planning, different kinds of menus		
Unit – IV	**Quality food Service –** types-Centralized, de-centralized objectives. Styles of service		
Unit - V	Importance of sanitation and hygiene in food, kitchen hygiene, Hygienic handling of Food, employee's health, hygiene of food service unit		

Reference and Textbooks

Drucker, P.F. (1975). *Management.* New Delhi: Allied Publishers

Hitchcock, M.J. (1980). *Food Service Systems Administration.* New York: Macmillan

Moore, C.L & Jaedicke, R.K. (1980). *Managerial Accounting, Cengage South Western.*

Boella, M.J. (1983). *Personnel Management in the Hotel and Catering Industry. (3rd ed.).* London: Hutchinson

Sethi, M., & Malhan, S. (1993). *Catering Management:* An integrated approach. New Delhi: Wiley Eastern

Outcomes	➢ Gain Knowledge about Food Service Management ➢ Learn about the importance of food quality, sanitation and hygiene in food service unit

Course Teacher
R.Ramya
Prof.K. Manimekalai

Semester - VI			
Coursecode:453E3A	**Food Sanitation & Hygiene**	**Credits: 3**	**Hours: 3**
Objectives	➢ To raise consciousness about Sanitation and Hygiene ➢ To impart skills in Handling of Food in the Food Service and Preparation Area		
Unit - I	**Introduction to Hygiene & Sanitation in Foods -** Relationship of Microbes to Sanitation, Public Health Hazards due to contaminated Water and Food; Role of Microorganisms - Water and Food Borne Illness; Physical, Chemical and Biological Contamination; Sources of Contamination, Vehicles and Routes		
Unit - II	**Kitchen and Food plant hygiene -** Safety in food Procurement, Storage Handling and Preparation; Safety of Leftover Food, HACCP- Need, Benefits and Principles		
Unit - III	**Personnel Hygiene -** Food Handlers, Habits, Clothes, Illness; Training & Education for safe Methods of Handling and Serving Food		
Unit – IV	**Cleaning -** Sterilization & Disinfection; Different Methods used – Detergents, Heat, Chemicals; Tests for Sanitizers Strength; Selecting and Installing Equipment; Cleaning of Equipment and Premises		
Unit - V	**Waste Product Handling -** Waste Disposal – Solid and Liquid Waste; Rodent and Insect Control; Uses of Pesticides		

Reference and Textbooks

Moore, C.L & Jaedicke, R.K. (1980). *Managerial Accounting, Cengage South Western.*

Schmidt, Ronald, H., & Rodrick, G.E. (2003). *Food Safety Handbook.* Wiley Interscience

Mehta, Rajesh & George, J. (2005). Food *Safety Regulations Concerns and Trade*: The Developing Country Perspective. Macmillan

Entis, Phyllis. (2007).*Food Safety*: Old Habits, New Perspectives. USA: ASM.

Paster, Tara. (2007). *The HACCP: Food Safety Training Manual*. Johnwiley

Outcomes	➢ Develop the skills in handling of Food in the Food Service and Preparation Area

Course Teacher
P.Venkateswari
Dr.P.Veeramani

Semester - VI			
Coursecode:453E3B	**Elective: III (B)-Regulations Of Food Product Development**	**Credits: 3**	**Hours: 3**

Objectives	➢ To understand the concept of new food product development ➢ To gain knowledge on regulatory aspects in product development and product commercialization
Unit - I	**Food Needs & Consumer Preference -** Market survey and its importance in designing a questionnaire to find consumer needs for a product. Developing a Product to Meet the Requirements of a needy people. Product life cycle.Creating brand value for the Product
Unit - II	**New Food Product -** Definition, Classification, New Food Product Development (NPD) process and activities, The Stage-Gate model NPD success factors, new product design, market-oriented NPD methodologies, organization for successful NPD; Reasons for new food product development (corporate, market place, technological and governmental influences) Assessing needs from various perspectives
Unit - III	**Generation of New Product Ideas Internal sources of ideas**; External sources of ideas; Market place analysis Screening; Objectives of screening; Criterion of screening Refining the Screening Procedure for the product Sensory Evaluation, Shelf life Testing, Product Integrity and conformance to standards
Unit – IV	**Quality, Safety & Regulatory Aspects** - Product Stability; evaluation of shelf life; changes in sensory attributes and effects of environmental conditions; accelerated shelf life determination; developing packaging systems for maximum stability and cost effectiveness; interaction of package with food; Regulatory Aspects; whether standard product and conformation to standards; Approval for Proprietary Product
Unit - V	**Product Commercialization, Launch, Evaluation & CaseStudies -** Outcomes and activities in product commercialization, Pre-launch trial, Steps in product launch, Evaluation of the Launch, product performance testing, developing test market strategies, Case Studies of some successes and failures, food choice models and new product trends

Reference and Textbooks

Drucker, P.F. (1975). *Management.* New Delhi: Allied Publishers

Moore, C.L & Jaedicke, R.K. (1980). *Managerial Accounting, Cengage South Western.*

Mary Earle & Richard Earle (2008). *Case studies in food product development*. USA :Wood head and CRC Press LLC

Howard R. Moskowitz, I., Sam Saguy & Tim Straus (2009*). An Integrated Approach to New Food Product Development.* USA: Taylor and Francis Group, LLC.

Jacqueline H. Beckley., & Michele Foley Elizabeth J. Topp & J. C. Huang Witoon Prinyawiwatkul. (2007). *Accelerating New Food Product Design and Development.* Blackwell Publishing Company. USA: IFT Press

Outcomes	➢ Become competent in new food product development. ➢ Aware about regulatory aspects in product development and product commercialization

Course Teacher
Prof.K. Manimekalai
P.Venkateswari

Semester - VI			
Course code:453S09	**Skill Based Subject: IV(A)-Quality Management in Food Processing and Preservation**	**Credits: 2**	**Hours: 2**
Objectives	➢ To teach different methods of food processing and preservation. ➢ To impart about the food quality assurance		
Unit - I	**Principles of fresh food storage: Nature** of harvested crop, plant, animal; product storage; effect of cold storage and quality – storage of grains		
Unit - II	**Processing and preservation by heat:** Blanching, pasteurization, sterilization and UHT processing, canning, extrusion cooking, dielectric heating, microwave heating, baking, roasting and frying		
Unit - III	**Processing and preservation by low Temperature** – refrigeration, freezing, CA, MA, and dehydro-freezing. Food irradiation, mechanism, the electro-magnetic spectrum, forms of radiant energy		
Unit – IV	**Processing and preservation by drying**- selection of methods based on characteristics of foods to be produced, advantages and disadvantages of different methods, sundrying, tray or tunnel drying, spray drying, drum drying, freeze drying, fluidized bed drying		
Unit - V	**Food Quality assurance** –Quality parameters- physical, chemical, functional, microbial; Rapid diagnostic methods of food quality – instruments and kits, Food standards – GMP, codex alimentations, ISO – 9000 serious, Food laws - PFA, FPO, AGMARK, MPO, BIS, Food safety and standards act, 2006		

Reference and Textbooks

World Health Organisation. (1998). *Guidelines for Drinking Water Quality, (2nd edition).* 1*(2), and (3)*, Geneva

Rick Parkar. (2002). *Introduction to Food Science,* Library of Congress Cataloging-in- Publication Data, First Edition,

Thoms Richardson & Johan, W. Finley. (2003). *Chemical Changes in Food during Processing*. New Delhi: CBS Publishers and Distributors.

Desrosier, N.W & James, N. (2007). *Technology of food preservation.* AVI.

Fellows, P.J. (2005). (2nd ed.). *Food processing technology: Principle and Practice.* CRC Publishers

Outcomes	➢ Gain knowledge on methods in food processing and preservation ➢ Understand the food quality assurance

Course Teacher
R.Ramya
Prof.K. Manimekalai

Semester - VI			
Coursecode:453S10	**Skill Based Subject: IV Fundamentals Of Tourism & Hospitality Management**	**Credits: 2**	**Hours: 2**
Objectives	➢ To gain knowledge on tourism and hospitality management		
Unit - I	**Tourism:**Meaning, Significance and History of Tourism in India Tourism industry,Systems, components, infrastructure Types of Tourism: Ecotourism, Heritage tourism, Medical tourism, Educational tourism etc		
Unit - II	**Travel Agent:** Types of travel agencies, Functions of travel agency, How to setup travel agency, sources of income for a Travel Agency. The tour operator: Types of tour operators, packages tour, guides and escorts. Formalities and regulations for tourism (in brief): Passport, Visa, Health Regulations for International Travel, Customs Regulations, Emigration and Immigration, Taxes Paid by Travellers & Travel Insurance. Itinerary Planning: Meaning, components, resources for planning, calculation of tour cost		
Unit - III	**Hospitality Management:** Meaning, Principles, Scope, Accountancy and Book Keeping		
Unit – IV	**Hospitality Institutions:** Meaning& types, understanding the basic culinary and catering terminologies, Classification of Hotels, Hotel divisions & departments, Functions & personnel in Front Office Management& House Keeping		
Unit - V	**Services:**Types of services offered in the hotel, Types of menu, Room setting, Table setting & different types of napkin folds		

Reference and Textbooks

Kamra, K.K., &.Chand, M. (2006). *Basics of Tourism-Theory (1st ed.).*Operation and Practice. New Delhi: Kanishka

Puri, M. & Chand, G. (2006). *Tourism Management (1st ed.).* New Delhi: Pragun

Metti, M.C. (2008). *Hotel Management and Catering*. New Delhi: Anmol

Roday,S., Biwal .A., & Joshi. V. (2009). *Tourism Operations and Management (1st ed.)*.. New Delhi: Oxford University press.

Mohini Sethi. (2016). *Institutional Food Management (2nd ed.).* New Age International

Outcomes	➢ Aware about the types of tourism, travel agent and tour operators ➢ Familiarize the students in hospitality management

Course Teacher
Prof.K. Manimekalai
Dr.P.Veeramani

Semester - VI			
Coursecode:453S11	**Skill Based Subject: IV(B) - Office Automation**	**Credits: 2**	**Hours: 2**

Objectives	➢ To Understand the basic concepts in Office Automation ➢ To enhance the knowledge on MS Office, MS Power Point and MS Access
Unit - I	**Computer Concepts –** Introduction to Computers – Definition - Advantage & Disadvantages – Classification of Computers (Desktop, Laptop, Notepad) Hardware features & Uses – Defining Hardware – Components of Computer –Diagram of Computer – Primary Storage Concept – Secondary Storage Devices – Input / Output, Devices – Software Concepts – Operating System (O/S)
Unit - II	**Management Information Systems** – An Overview – Introduction to MIS – Meaning and Role of MIS – Objectives of MIS – Elements of MIS – Characteristics of MIS – Applications of MIS (Briefly) in Accounting and Finance Management – Marketing Management – Materials Management - Production Management – Personnel Management – Role of Computers in MIS
Unit - III	**Introduction to MS Office 2000** – Characteristics & Its applications – Introduction to MS Word Creating, Editing, Formatting, Saving Documents – Types of Document Formats – Mail Merge Features – Spreadsheet Introduction to MS Excel – Definition, Characteristics, Mathematical functions in Excel, Creating and Working with Graph and Charts
Unit – IV	**Introduction to MS Power Point** – Creating, Formatting, Editing, Viewing Slideshow, Creating a Presentation using MS Power Point
Unit - V	**Introduction to MS Access** – Meaning, Creating of Database, Modification, Storing and retrieving of records from a database – Introduction to Business Process Outsourcing – Creation of email ID

Reference and Textbooks

Memoria, *C.B. (2006)MS Office* – Complete Reference. New Delhi: BPP

Microsoft Office (2007) Bible - John Walkenbach,Herb Tyson,Faithe Wempen,cary N.Prague,Michael R.groh,Peter G.Aitken, and Lisa a.Bucki -Wiley India pvt.ltd.

Malhotra, T.D. (2007)*Computer Awareness and Applications* New Delhi: Kalyani\

Gabriel Gurley R. (2008). *A Conceptual Guide to OpenOffice.org 3* - CreateSpace Independent Publishing Platform.

Mustafa, A. (2011) *Principles of Information Technology.* Madurai: Nagas

Outcomes	➢ Gain the basic knowledge on Computer concepts ➢ Familiarize with the MS Office, MS Power Point, MS Access

Course Teacher
Prof. K. Manimekalai
ALU Computer center

Semester - VI			
Course code: 453S12	**Skill Based Subject: IV (B) NGO and Corporate Social Responsibility**	**Credits: 2**	**Hours: 2**
Objectives	➢ To familiarize the students about the characteristics and functions of NGO ➢ To understand the role of CSR in improving the standard of living of the down trodden		
Unit - I	**Concept of NGO:** Meaning of NGO and GO, Difference between Government Organizations and NGO, Characteristics of good NGO, Structure of NGO, Functions of NGO, Historical Perspective of NGO, Advantages of NGO, Present status of NGO, Contribution of NGO in the Development. Role of International NGO in development		
Unit - II	**Establishing a NGO:** Steps for starting NGO, Registration of NGO, Selection of Personnel, Training of Personnel, Proposal writing under NGO, Identifying Funding agencies, Resource Mobilization, Planning, Implementation and Evaluation strategy under NGO. Accountability and impact assessment for NGOs		
Unit - III	**NGO Management**: Organizational types and structures, Managing people and teams in NGOs, NGO management competencies		
Unit – IV	**CSR:** Meaning, CSR Process, Steps in developing a CSR strategy and policy evolution**.** Tools, technical guidance and standards to be used for assessment		
Unit - V	**Indian Scenario:** Overview of CSR in India, CSR initiatives government and corporate establishments		
Reference and Textbooks: Lewis, D. (2001). (*2n ed.). Management of Non Governmental Development Organization* New York: Routledge, Chandra, S. (2003). *Guidelines for NGO Management in India.* New Delhi: Kanishka. Abraham, A. (2003). *Formation and Management of NGOs. (3rd ed.).* New Delhi: Universal Law Publishing Co. Pvt Ltd, Aggarwal, S. (2008). *Corporate Social Responsibility in India.* New Delhi: Sage Sundar, P. (2013). *Business and Community*: The Story of Corporate Social Responsibility in India. New Delhi: Sage.			
Outcomes	➢ Gain knowledge on characteristics, structure and functions of NGO ➢ Aware about the role of CSR in improving the standard of living of the down		

Course Teacher
Prof.K. Manimekalai
Dr.P.Veeramani

M.A INTEGRATED HOME SCIENCE (NUTRITION AND DIETETICS

Semester -I			
Coursecode: 453N01	**Advanced Food Science**	**Credits: 5**	**Hours: 5**
Objectives	➢ To impart Knowledge on functional Classification of Food ➢ To guide about the Composition and Nutritive value of Food		
Unit - I	Functional Classification of Foods, Basic Food Groups -4, 5, 7 & 9.Cooking Methods – Water, Oil and Air as Medium. Cereals – Structure, Composition, Nutritive value of Rice, Wheat, Ragi, Maize, Cereal Cookery - Cooking Methods, Effect of Moist Heat and Dry Heat, Factors Affecting Gelatinization. Fermentation – Definition, Advantages, Cereal Products.		
Unit - II	**Pulses** – Composition and Nutritive value of pulses and Legumes- Toxic constituents in Pulses, Pulse Cookery – Methods, Effect of Cooking, Factors affecting Cooking Quality, Germination – Definition and Advantages. Nuts and Oil Seeds - Classification, Composition and Nutritive value. Fruits and Vegetables – Classification, Composition, Selection and Nutritive value of some common Vegetables and Fruits, Pigments in Fruits and Vegetables, Fruit and Vegetable Cookery – Methods, Effect of Cooking on Nutrients and Pigments.		
Unit - III	**Fleshy Foods** – Structure, composition, nutritive value and selection of fleshy foods, post-mortem changes, tenderization of meat. Meat cookery –Cooking methods and changes during cooking. Egg – structure, composition, nutritive value, selection of egg, egg cookery, factors affecting foam formation. Fish – Classification, composition, nutritive value, selection, curing and smoking of fish.		
Unit – IV	**Milk** - Composition, Nutritive value, Effect of Heat, Acid and Enzymes on Milk Proteins. Milk Products - Pasteurized Milk, Whole Milk Powder and Cheese. Fats and Oils-Composition, Nutritive value, Types of Fatty Acids in Groundnut, Sunflower, Sesame, Coconut and Soybean. Hydrogenation Process- Types and Prevention of Rancidity, Changes in Fat on Cooking. Sugar - types of Sugar and Stages of Sugar Cookery, Artificial Sweeteners – Cyclamate and Saccharin.		
Unit - V	**Beverages** – Classification, Coffee and Tea – Processing and Preparation. Cocoa – Use of Cocoa in Chocolate Preparation. Spices – Uses of Spices in Cookery. Food adulteration – Definition, Common Adulterants and Methods to detect Adulterants. Food standards – BIS, AGMARK, FPO. Fortification and Enrichment - Definition, Objectives, Fortification of Cereal and Cereal Products.		

Reference and Textbooks:

Swaminathan M., (1979) *Food Science and Experimental Foods,* Ganesh and Co., Madras, Reprint.

Swaminathan M., (1996) *Essentials of food and nutrition,* Vol. I & II Bappco Publications,

Srilakshmi, B.(2002) *Nutrition Science,* New Age International [p] ltd, New Delhi,

Srilakshmi B. (2005) *Food Science*, New Age International (P) Ltd. Publishers, third edition,

Manay Shakunthala, N & Shadaksharaswamy M.(2008) *Food facts and principles*, New Age International (P) Ltd. Publishers, Reprint.

Outcomes	➢ Gain knowledge about the Functional Classification of Food ➢ Familiarize with the Nutritive value of Food

Course Teacher
Prof.K. Manimekalai

Semester -I			
Coursecode: 453N02	**Advanced Food Science (P)**	**Credits: 5**	**Hours: 10**
Objectives	➢ To guide students on Experimental Cookery on Healthy Foods ➢ To enlighten about the Formulation of Healthy Foods		
Unit - I	1. Experimental Cookery of Cereals a. Steaming, Boiling and pressure Cooking of Rice b. Preparation and Evaluation of Mixed Rice, Ragi Leaf Cake and Biscuit. 2. Experimental Cookery of Pulses c. Boiling, Steaming and Pressure Cooking of Pulses using Hard and Soft Water with and without Baking Soda. d. Preparation of Sprouted /Germinated Pulse Flour.		
Unit - II	3. Experimental Cookery of Fruits and Vegetables a. Effect of Acid and alkali on Color, texture and Flavor. b. Enzymatic and Non-enzymatic Browning and its Prevention. 4. Experimental Cookery on Milk and Milk Products. a. Effect of acid on Milk. b. Preparation and evaluation of Ice-Cream and Kheer.		
Unit - III	5. Experimental Cookery on Egg a. Effect of Sugar and Salt on Egg Foam Formation. b. Preparation and Evaluation of Poached, Scrambled Egg and Egg Custard.		
Unit – IV	6. Experimental Cookery on Sugar a. Stages of Sugar Cookery b. Preparation and Evaluation of Mysore Pak and Gulabjamun. 7. Experimental Cookery on Fats and Oils a. Smoking Point of Oils b. Preparation and Evaluation of Puri and Potato Chips.		
Unit - V	8. Formulation of Healthy foods a. Preparation and evaluation of (i) Ragi malt (ii) Kulandai Amudhu		

Reference and Textbooks:

Bahasahe & B. Dosa, (2002) *Hand book ofnutrition and diet.*

Wardlaw & Insel MG, Insel PM (2004*). Perspectives in Nutrition, (6rd ed)*. Mosby.

Srilakshmi (2007). *Food Science, (4th ed)* New Age International Ltd.

Bamji MS, Krishnaswamy K, Brahmam GNV (2009). *Textbook of Human Nutrition, (3rd ed).* Oxford and IBH Publishing Co. Pvt. Ltd.

Chadha R & Mathur P (eds). (2015) *Nutrition: A Lifecycle Approach.* Orient Black swan, Delhi.

Outcomes	➢ Understand knowledge about Experimental Cookery on Healthy Foods ➢ Gain knowledge about the Formulation of Healthy Foods

Course Teacher
R.Ramya
Prof.K. Manimekalai

Semester -I			
Coursecode: 453N03	**Community Nutrition and Education**	**Credits: 5**	**Hours: 5**
Objectives	➢ To develop competencies among students for reaching out to vulnerable group in community and facilitating them to secure rewarding careers and vocations in extension media and communication for Rural Development.		
Unit - I	**Community**- Definition,Characteristics and Types of Community.Assessment of Nutritional Status of the Community - Direct Methods – (a) Anthropometric Measurements (b) Clinical Examination (c) Bio-Physical Examination (d) Functional Assessment (e) Bio-Chemical Estimation. Indirect Methods - (a) Dietary Assessment (b) Vital Health Statistics		
Unit - II	Incidence, Prevalence, Epidemiology and Preventive Measures of Nutritional Problems in India – PEM, Vitamin A Deficiency, Anemia, Iodine Deficiency Disorder and Fluorosis. Food Security - Definition, Factors affecting Food Security System, National and International Approaches to Improve Food Security.Nutrition Education- Types and Methods of Education, Principles of Planning, Executing and Evaluating Nutrition Education Programmes, Problems of Nutrition Education.		
Unit - III	**Nutrition Intervention Policies and Programmes in India**- National Nutritional Policy, PDS, Objectives and Operation of Chief Minister Noon Meal Programme (CMNMP), Integrated Child Development Services (ICDS), Primary Health Center (PHC). National organizations – ICMR, NIN, NNMB, CFTRI, DFRL, ICAR, NIPCCD, NSI, NFI and IDA.International Organizations– FAO, WHO, UNICEF, UNESCO, UNDP and World Bank. International Voluntary Services – CARE, CRS, IDRC, Micronutrient Initiative (MI), IFPRI, WFS, WFP, AUSAID, CIDA, SIDA, DANIDA, USAID.		
Unit – IV	**Extension** - Meaning, Philosophy, Objectives, Principles, Functions, Components and Dimension of Education. Extension Education – Meaning, Process and Principles of Learning in Extension Education. Extension Models- Technology, Innovation Transfer Model, Social Education Model, Indigenization Model, Social Action / Conscientization Models, Empowerment Participation Model, Combination Models. Extension Approaches – Agricultural Extension, Commodity Specialized, Training and Visit, Participatory, Project, Farming Systems Development, Cost Sharing, Educational Institution, Integrated Approach, Cluster and Target Approach.		
Unit - V	**National Extension Systems** – Genesis, Growth, Objectives, Principles, Critical Appraisal of the Community Development Programme. ICAR Extension System- Agricultural Universities, KVK, TTCs. Extension Systems of Ministry of Rural Development, Department of Science and Technology, Department of Industries and Department of Women and Child Development. Development work by NGOs, Government-NGO Collaboration.Support Structures and their Functions – Panchayat, Panchayat Union and DRDA, Central Social Welfare Board, State Social Welfare Board, National Level Voluntary agencies like CAPART, KVIC, Local Level Voluntary Agencies. People's Organization at Gross Roots – SHGs, Elected Panchayats.		

Reference and Textbooks:

Park & Park, (2000), Park's *Textbook of Preventive and Social Medicine, (18th ed)* M/S BanarasidsBhanot, Jabalpur.

Srilakshmi, B. (2002), *Nutrition Science,* New Age International [p] ltd, New Delhi.

Swaminathan, M. (2003) *Handbook of Food and Nutrition,* the Bangalore Printing and Publishing Co. Ltd,

Reddy, R.S. (2004*), Nutrition Education*, (*1st ed),* Common Wealth Publisher.

Padmini Gupta, RuchiThakkar, (2015) *Nutritional Disorders and Community Health*, Pointer Ltd Publishers, Jaipur.

Outcomes	➢ Develop Competencies in Community Nutrition and Nutrition Education ➢ Familiar with Extension Media and communication for Rural Development

Course Teacher
P.Arhcana
Prof.K. Manimekalai

Semester -I			
Coursecode: 453N04	**Food Product Development and Marketing**	**Credits: 5**	**Hours: 5**

Objectives	➢ To understand the concept of development of a New Product and Prepare New Products Based on Special Dietary Requirements, Functionality, Convenience and Improvisation of Existing Traditional Indian Foods.
Unit - I	New Food Products – Definition, Classification, Factors Shaping New Product Development – Social Concerns, Health Concerns, Impact of Market Place Influence and Technology, Reason for New Food Product Development- Corporate, Market Place, Technological and Governmental Influences.
Unit - II	Steps in Food Product Development- Market and Literature Survey to Identify the Concepts of New Products Based on Special Dietary Requirements, Functionality, Convenience and Improvisation of Existing Traditional Indian Foods, Screening of Product Concept on the Basis of Techno-Economic Feasibility, Development of Prototype Product and Standardization of Formulation Process, Proximate Analysis of New Product, Packaging, Labelling and shelf-Life Studies, Cost Analysis and Final Project Report.
Unit - III	Case Studies of Food Processing, Business and its Aspects, Business Opportunity, Identification and Assessment Techniques, Business Idea Generation and Evaluation Exercise, Market Assessment Study, Analysis of Competitive Situation, SWOT Analysis for Business and for Competitors, Preparation of Business Plan, Preparation of Project Report, Methods of Arrangement of Inputs – Finance and Material.
Unit – IV	Criteria for Formulation of New Food Products for Infants, Pre-School Children, Adolescents, Pregnant and Nursing Mothers, Old Age, Sports Persons, Armed Sources Personnel, Emergencies and Therapeutic Uses. Standard and Safety of Food and Food Products; Criteria for Ingredients and Finished Products Laid down in the FSSAI Regulations, 2011.
Unit - V	Concept of Market and Marketing –Market Promotion and Positioning of Food Products, Role of Advertisement and Technologies in Promotion of New Products. Export Potential for selected Indian Food Products, Role of Export Promoting Agencies and Export Quality Control and Inspection Act, 1963.

Reference and Textbooks:

Sivarama Prasad. A, (1985*) Agricultural Marketing in India* – Mittal Publications, New Delhi,

Acharya. S.S. & N.L. Agarwal, (1992) *Agricultural Marketing In India* – Oxford and IBH Publishing Co. Pvt. Ltd., New Delhi,

Fuller, Gordon W. (2001) *New Product Development*- FromConcept to Marketplace*,* CRC Press.

Gordon. W. Fuller., (2004) *New Food Product Development,* From concept of market place CRS Press.

Aaron, L. Brody, John B. Lord,(2005) *Developing New Food Products for A ChangingMarket Place, (2nd ed*), CRS Publication, New Delhi,

Outcomes	➢ Gain knowledge about Food Product Development and Marketing ➢ Understand the new Products Based on Special Dietary Requirements, Functionality, Convenience and Improvisation of Existing Traditional Indian Foods.

Course Teacher
Prof.K. Manimekalai
Dr.P.Veeramani

Semester -I			
Coursecode:453EC1	**(A) Home Textiles**	**Credits: 4**	**Hours: 4**

Objectives	➢To teach Introduction to Home Textiles ➢To impart knowledge in Finishes and quality parameters used in Home Textiles.
Unit - I	**Introduction to Home Textiles** – Fabrics used for Living area, Kitchen, Dining, Bed room, Bath room accessories and general household textiles. Textile for Furnishing and Upholstery applications – scope, fixed upholstery, non-stretch loose covers & stretch covers.
Unit - II	**Window/ Curtain Fabrics** - Sun filters (Sheers and nets), Semi-sheers, Reflective textiles, drapes and Blinds. Bed Room Textile - Sheets & Pillow Cases, Quilted Textile, and Blankets & Rugs - Jacquard blankets, Printed blankets, Fire proof blankets, Baby blankets. Bed Spreads, Mattress covers.
Unit - III	**Dining Room Textiles Tablecloths** – Colour – Woven & Printed type, jacquard types, embroidered types, non-woven types. Table mats – Colour -woven, Printed jacquard, embroidered. Kitchen Textiles - Aprons, Dish cloth, Tea cosy, Bread bag, Mittens, Pot Holders, Table Mats – Construction & manufacturing.
Unit – IV	**Bath Room Textiles** – bath towels, bathrobes, floor mats, handtowels, beach towels, towels for swimming pools. Terry Towels – types and applications. Floor Coverings – Introduction, Pile fabrics, Backing fibres & fabrics – Tufted carpets, woven carpet.
Unit - V	**Finishes and quality parameters used in Home Textiles** - Protection against unpleasant odour-Temperature regulated beddings - Antimicrobial finish - Moisture management - Test Method for towels, Rug and Home textiles and - Performance specifications of different home textiles.

Reference and Textbooks:

Talukdar, Ajagaonkar, Sriramulu (2004) *Weaving Machine, Mechanisms, Management.*

Anbumani, (2006) *KnittingManufacture Technology,* New Age International, Chennai.

Gokarneshan N, (2009) *Fabric Structure and Design* - New Age International.

Subrata Das, (2010) *Performance of Home Textiles,* Wood head Publications India Pvt Ltd.

Global Organic *Textile Standard, version 3* (2011), www.globalstandard.org.

Outcomes	➢ Gain the knowledge describes various types of home textiles and their manufacturing methods. ➢ Understand requirements of different home textiles ➢ Learn to understand various finishes and evaluation methods of home textile.

Course Teacher
Prof.K. Manimekalai
Dr.P.Veeramani
M.Priyanka

Semester -I			
Coursecode:453EC2	**(B) Women and Society**	**Credits: 4**	**Hours: 4**
Objectives	➢ To impart about the Different Sectors of Development, with Special Reference to Women. ➢ To introduce the Issues and Concerns of Women.		
Unit - I	**Women in Society** – Ancient India and Modern India - Concept of Women's Movements – Basic Elements – Classifications Social Reform Movement – Status of Women through Ages.		
Unit - II	**Women's Role: Post Independence Period** - Indian Independence Movement and Women's Movements - Narmada Valley Project Environmental Movement - Chipko Movement- Peasant Movement.		
Unit - III	**Position of Women in: Hinduism, Islam, Christianity** - Social Reformers worked for the Emancipation of Women – Rajaram Mohan Roy, DayanadaSaraswathi, Mahatma Phule, PanditRamabai, Annie Besant, Sarojini Naidu, Muthulakshmi Reddy, DurgabaiDeshmukh, ArunaAsaf Ali.		
Unit – IV	**Women and Political Participation** - Women and Panchayat Raj Institutions – International, National and Local Women in Parliament through Ages - Women's Bill.		
Unit - V	**Rise of NGOs in Women Organization for Women's Development** – All India Women's Conference (AIWC) – Women's India Association (WIA) - National Council of Women in India (NCWIE) - Indian Association of Women's Studies.		

Reference and Textbooks:

Geraldine Forbes (1998) *Women in Modern India,* New York: Cambridge University.

Leslie L. Heywood (2005) *The Women's Movement* Today, Jaipur: Rawat Publications Vol. I & II.

Goel, A, Kaur, A & Sultana, A (2006). V*iolence against women: Issues and Perspectives.* New Delhi, Deep & Deep Publishers.

SumitSarkar & TanikaSarkar (2007) *Women and Social Reform in Modern India*, New Delhi: Permanent Black Vol. I & II.

Suguna B (2009) *Women's Movements,* New Delhi: Discovery Publishing House Pvt. Ltd.

Outcomes	➢ Gain knowledge about the status of women in the society. ➢ Can set up their own NGOs for the development of women.

Course Teacher
Prof.K. Manimekalai
Dr.P.Veeramani

Semester –II			
Coursecode:453N05	**Advanced Nutrition And Itermediary Metabolism**	**Credits: 5**	**Hours: 5**
Objectives	➢ To highlight metabolic role of nutrients and their relationship to human health and wellbeing. ➢ To gain knowledge on advanced nutrition.		
Unit - I	**Energy metabolism Basal and resting metabolism (BMR and RMR) –** influencing factors.Methods to determine energy requirements and expenditure. Thermogenesis, adaptation to altered energy intake, latest concepts in energy requirements and RDA - ICMR and WHO		
Unit - II	**Carbohydrates:** Classification and functions, Digestion and absorption process, Metabolism and regulation, dietary fibre meaning and types, Physiological role and health benefits of dietary fibre. Resistant starch -meaning and its physiological benefits-Requirements and food sources. Glycemic index of foods. **Proteins –** Classification and Functions, Review of metabolism of Protein, Concepts of essential and non-essential amino acids – their role in growth and development. Physiological functions of proteins. Requirements, nitrogen balance concept. Methods of evaluating protein quality. Protein malnutrition. Muscle wasting – clinical features and biochemical changes. RDA-ICMR and WHO.		
Unit - III	**Lipids –** Classification and Functions, Review of metabolism of Lipid, Concepts of visible and invisible fats, EFA, SFA, MUFA, PUFA – sources and physiological functions. Role of lipoproteins (Chylomicrons, VLDL, IDL, LDL and HDL), cholesterol, triglycerides in health and disease.**Vitamins:** Classification – fat and water soluble; Fat soluble vitamins (A,D,E and K): Functions, Requirements and food sources; Physiological, pharmacological and therapeutic effects, toxicity and deficiency of fat soluble vitamins; Water soluble vitamins: Thiamine, riboflavin, niacin, biotin, pyridoxine, folic acid, pantothenic acid, ascorbic acid, cyano-cobalamin, choline, inositol functions, requirements, food sources; Deficiency and toxicity of water soluble vitamins.		
Unit – IV	**Minerals: Macro minerals**: Calcium, phosphorus, Magnesium, sodium, potassium and chloride functions, requirements, food sources, deficiency and toxicity; Micro minerals: Iron, copper, zinc, manganese, iodine, fluoride. Trace Minerals: Selenium, cobalt, chromium, vanadium, silicon, boron, nickel functions, requirements, food sources, deficiency and toxicity. Interrelationship between vitamins and minerals in metabolism.		
Unit - V	**Water:** Body composition – extra- and intra- cellular fluid; Physiological functions; water balance and its regulation; Requirement and the sources; Nutritional and health problems due to deficiency or excess of water intake. Phytochemicals:Non nutritive food components and their potential health benefits: polyphenols, tannins, phytate, phytoestrogens, cyanogenic compounds, lectins and saponins.		
Reference and Textbooks: Charley, H. (1982). *Food Science, (2nd ed)* John Wiley & Sons, New York. Julians, B.O. (1985). *Rice Chemistry and Technology, (2nd ed),* American Association Chemists, St. Paul Mimesota, USA. Potter, N. & Hotch Kiss, J.H. (1996): *Food Science, (5th ed),* CBS Publishers and Distributors, New Delhi Srilakshmi, B. (2005) *Food Science,* New Age International (P) Ltd., Publishers, New Delhi Srilakshmi (2008). *Nutrition Science.* New Age International Pvt. Ltd, New Delhi.			
Outcomes	➢ Know about the metabolic role of nutrients and their relationship to human health and wellbeing. ➢ Understand advanced nutrition.		

Course Teacher
Prof.K. Manimekalai

Semester -II			
Coursecode: 453N06	**Therapeutic Nutrition**	**Credits: 5**	**Hours: 5**
Objectives	➢ To educate the students on Causes, Symptoms, Dietary Management and Drug Treatment Interaction with Nutrients in Various Disorders.		
Unit - I	**Therapeutic Diets –** Principles & Objectives of Diet Therapy.Review of Hospital Diets - Regular Diet, Liquid Diet, Light Diet, Soft Diet, Pre and Postoperative Diet.Diet Planning and Use of Exchange List in Nutrient Calculation.Gastrointestinal Disorder – Etiology, Symptoms and Modifications of Diet in Diarrhea, Constipation, Peptic Ulcer, Malabsorption Syndrome and Ulcerative Colitis.		
Unit - II	**Disorders of Liver and Gall Bladder -** Etiology, Classification and Dietary Regimen in Jaundice, Hepatitis, Cirrhosis, Hepatic Coma, Cholecystitis and Cholelithiasis.Nutrition and Alcoholism.Diseases of Kidney – Etiology and Dietary Management in Glomerulonephritis, Nephrosis, Renal Failure, Nephroscelerosis, Nephrolithiasis, Dialysis and Transplantation.		
Unit - III	Diseases of the Heart & Circulatory System- Risk Factors of Cardiac Diseases, Causes, Prevention and Dietary Management of Hypertension, Atherosclerosis, Congestive Heart Failure, Hyperlipoproteinemia, Hypercholesterolemia, Role of Antioxidants in the Prevention and Treatment.Obesity and Underweight – Etiology,Types,Dietary Modifications in the Management of Obesity and Underweight. Anemia – types, Signs & Symptoms, Clinical Manifestation, Dietary Interventions.		
Unit – IV	**Disorders of Pancreas -** Pancreatitis.Diabetes mellitus – Classification, Etiology Symptoms, Metabolic Changes, Long Term & Short Term Complications, Types of Insulin, Dietary Modifications for Diabetes Mellitus, Glycemic Index of Foods, Nutritive and Non- Nutritive Sweeteners.Nutrition in Cancer- Causes of Cancer Cell Development, Metabolic and Nutritional alterations in Malignancy, Bodies' Defense System, Cancer Therapy & nutrition, eating problems in cancer, feeding and blend preparation for cancer.		
Unit - V	**Etiological Factors & Dietary Modifications in Fever, Injury & Burns, Allergy, Dental Diseases –** Dental Caries & Peritonitis.Management of Nervous Disorders- Alzheimer Disease, Parkinson's disease, Autism.Management of Musculoskeletal System Disorders - Osteoporosis, Osteomalasia, Osteoarthritis and Rheumatoid Arthritis.HIV Infection and AIDS- Clinical Manifestations, HIV infection and other disease, Immunity and AIDS Virus, Dietary Management.		

Reference and Textbooks:

Passmore, D.P & Break, J.P, (1986*) Human Nutrition& Dietetics,* English language Book Society, Livingeston,

Gopal, C. Kamalakrishnaswamy, (2000) *Nutrition in Major Metabolic Disease*, Oxford India Paper Backs Publisher,

LoryA. Smolin & Mary B.Grosvenor, (2000) *Nutrition Science and Application,* (3rd ed), Saunders College Publishing New York,

MahtabS.Bamji, Prasad Rao, N.Vinodini Reddy. (2003) *Textbook of Human Nutrition*, (2nd ed), Oxford and IBH Publishing Co. Pvt .Ltd,

Mahan, L.K., Stump, S.E & Krause, S, (2004)*Food Nutrition & Diet therapy, (11th* ed), W.B. Saunders Co,

Outcomes	➢ Gain Knowledge about Dietary Management and Therapeutic Nutrition

Course Teacher
P.Archana
Prof.K. Manimekalai

Semester -II			
Coursecode: 453N07	**Therapeutic Nutrition Internship**	**Credits: 8**	**Hours: 10**

Objectives	➢ To familiarize the students in planning the Diet for an Infant, Preschool Children, School Children, Adolescent Boys and Girls, Adult Men and Women, Old Aged Person and macro and micro deficiency diseases. ➢ To enable the students to interpret the Patient Data and Diagnostic Tests
Unit - I	Market survey of Commercial Nutritional Supplements and Nutritional support substrates. Commonly used test for Diagnosis of Various diseases- system wise.
Unit - II	Interpretation of Patient Data and Diagnostic Tests. Drawing up of Diet Prescription using a Case Study Approach.
Unit - III	Follow up – Acceptability of Diet Prescription, Compliance and Discharge Diet Plan. Preparation of Diet Counseling Aids for common Physiological conditions in Life Cycle and Disorders.
Unit – IV	Planning balanced Diet for an Infant, Preschool Children, School Children, Adolescent Boys and Girls, Adult Men and Women, Old Aged Person.
Unit - V	Planning Diet for Deficiency conditions of Macro and Micronutrients. Planning Diet for Patients with common Multiple Disorders, Complications and Discharge Diet Plans.

Reference and Textbooks:

Passmore, D.P & Break, J.P,(1986*) Human Nutrition& Dietetics,* English language Book Society, Livingeston,

Shills, E.M & Olson, S.J & SMC, (1994) *Modern nutrition in Health and Diseases*, *Volume II, (8th)ed)*, Lea & Febringes, philadelphia.

Lory. A & Mary.B Grosvenor (2002*) The Management of Nutrition in Major Emergencies,* (1st *ed).* A.I.T.D. SPublishers and Distributors Delhi,

MahtabS, Bamji, Prasad Rao, N.Vinodini Reddy. (2003) *Textbook of Human Nutrition*, .(2st*ed)* Oxford & IBH Publishing Co. Pvt .Ltd,

Mahan, L.K., Stump, S.E & Krause, S, (2004)*Food Nutrition & Diet therapy, (11thed)*, W.B. Saunders Co,

Outcomes	➢ Plan the Diet for an Infant, Preschool Children, School Children, Adolescent Boys and Girls, Adult Men and Women, Old Aged Person and macro and micro deficiency diseases. ➢ Interpret the Patient Data and Diagnostic Tests

Course Teacher
P.Archana
Prof.K. Manimekalai

Semester -II			
Coursecode:453EC3	**A) Women and Health**	**Credits: 4**	**Hours: 4**

Objectives	➢ To introduce the concept of health and the issues related to Gender and health ➢ To study about the national and international initiatives in the promotion of Women Health.
Unit - I	**Health: Concept, Definition and types of Health** – Parameters of Community Health - Gender and Health - Health Care Needs of Women, Men - Health Status of Women - Factors influencing women's health - Health Indicators of Women - Mortality - Morbidity - Sex ratio, Child sex ratio, Life expectancy – MMR – IMR - Fertility
Unit - II	**Women's Reproductive Health –** Prenatal and Postnatal Health Care- Menstrual Hygiene Management – Menopause – Hormonal Replacement Therapy – Sex Selective Abortion – MTP – NRT – Amniocentesis Test – Clinical Trials and Gender Bias – Infertility: IVF & Surrogacy
Unit - III	**Gender and Nutrition–** Nutrition for pregnant and lactating mothers - Nutrition education for girls – Anemia - Gender bias in nutrition intake -Protein Malnutrition – Gender Discrimination - Breast Feeding – Health Seeking Behaviour – Initiatives on Nutrition Program – National –NRC – RishoriSakthi - International - WHO – ICDS – UNICEF - RTI and STI.
Unit – IV	**Gender and Health issues -** Feminization of HIV/AIDS – Private vs Public Spheres – Balancing Drudgery and Health Needs – Pollution and Health –Health issues of LBGTQ– Environmental & Occupational Health hazards- Health issues relating to: sex workers, women with disabilities, women elderlies.
Unit - V	**Programme and Policies for Health for women -** International Initiatives – ICPD – Cairo- National Health Policies, NRHM, ASHA, ICDS - Population Policy 2000 – MCH to RCH Approaches-Impact of Population Policies – Tamil Nadu State

Reference and Textbooks:

Sarma, suranjan, (2004) *Public Investment Primary Health Care*, New Delhi: Mittal publications,

Srivastava Sanjay, Brasted Howard (ed) (2004) *Sexual sites, Seminal Attitudes Sexualities*, Masculinities and Culture in South Asia, New Delhi Sage: publications,

Rao, Mohan, (2004)*From Population Control to reproductive health*, New Delhi Sage Publication,

Singh, Lakshmi Abha,(2005*) Rural Women: Work and Health*, New Delhi: The women press,

Census of India Report-(2011) *Ministry of Home Affairs, Government of India*, New Delhi.

Outcomes	➢ Gain knowledge on the concept of health and the issues related to Gender and health ➢ Able to acquire information on National and International Initiatives in the promotion of Women Health.

Course Teacher
Prof.K. Manimekalai
Dr.P.Veeramani

Semester -II			
Coursecode:453EC4	**(B) Food Service Management**	**Credits: 4**	**Hours: 4**

Objectives	➢ To inculcate the Knowledge on Institutional Food Administration Strategies and Management.
Unit - I	Introduction to Food Service Systems.Evaluation of the Food Service Industry.Characteristics of the various types of Food Service Units - **Commercial** - Hotel, Motel, Restaurant, Cafeteria and Chain Hotels; **Welfare** - Hospital, School Lunch, Residential Establishment and Industrial Catering; **Transport** - Air, Rail, Sea and Space; **Miscellaneous** - Contract and Outdoor.
Unit - II	Management - Approaches to Catering Management, Principles and theories of Management, Functions of Management, and Styles of Management, Management Tools- Organisation Chart and its Types.Strategies in Planning- Conceptual Strategy, Marketing Strategy, Financial Strategy, types of Plans.
Unit - III	**Management of Resources** – Finance- Determining the Finance needed to Establish or Run an Unit, Budgets, Sources of Finance, Planning Adequate Cash Flow.Space & Equipment Management- Steps in Planning Layouts, Determining Equipment, Selection and Placement, Layout Analysis.
Unit – IV	**Material Management-** Menu Planning, Planning the Material Needed, Methods of Selection, Purchasing, Receiving and Storage, Quantity Food Production, Service and Modes of Delivery, Planning Menus for Banquet, Outdoor Catering, Packed Meals and Restaurant.Personnel Management- Manpower Planning, Placement, Recruitment, Induction, Training, Motivation and Performance Appraisal.Time and Energy Management- Measures for Utilisation and Conservation.
Unit - V	**Financial Management -** Definition, Cost Concept, Component of Cost, Food Cost Control, Pricing, Book Keeping and Accounting, Techno-Economic Feasibility of Food Production/Service Enterprise.Marketing and Sales Management-Market Survey, Sales Analysis, Market Promotion.Quality Assurance- Food Quality, Total Quality Management, Analysis of Processed and Finished Products.

Reference and Textbooks:

Coltman, M.M. (1977*): Food and Beverage Cost Control*. Printice Hall Series.

Avery, A.C. (1980): *Modern Guide to Food Service Equipment*. Boston CBI Publishing Company.

Brodner, J., Maschal, H.T., Carlon, H.M. (1982): *ProfitableFood and Beverage Operation (4th*ed), Hayden Book Company, New Jersey.

Kumar, H.L. (1986) *Personnel Management in Hotel and Catering Industries,* Metropolitan Book Company N. Delhi.

Brichfield, J. (1988*): Design and Layout of Food Service Facilities,* New York. Van Nortrand. Reinhold.

Outcomes	➢ Gain the Knowledge about Food Service Management ➢ Learn about Management of human and non human resources in food service management.

Course Teacher
R.Ramya
Dr.P.Veeramani

Semester -III			
Coursecode: 453N08	**Advanced Biochemistry**	**Credits: 5**	**Hours: 6**

Objectives	➢ To understand the composition and function of blood, water and electrolyte ➢ To study about the classification, Structure and properties of macro nutrients
Unit - I	**Components of blood** - Composition and function of blood, Plasma and blood corpuscles, Structure and function of haemoglobin, abnormal haemoglobins. Blood coagulation – mechanism and regulation, Blood groups. Water and electrolyte - Regulation of water and electrolyte balance, Role of hormones and kidneys in their maintenance, Hydrogen ion homeostasis and acid-base balance.
Unit - II	**Carbohydrates** - Occurrence, Classification and Structure, Physic-chemical properties andbiological importance of carbohydrates. Monosaccharide and related compounds, disaccharides and Polysaccharides.Metabolism of carbohydrates - Aerobic and anaerobic degradation, Glycogenesis and Glycogenolysis, Glycolysis and Gluconeogenesis, Cori's cycle, Pyruvate Dehydrogenase complex, Krebs-cycle and Pentosephosphate pathway, Regulation of carbohydrate metabolism, Sugar derivatives of biomedical importance and Inter conversion of Hexoses.
Unit - III	**Proteins** - Classification, structure and properties of amino acids,Classification, properties and structure of proteins - Primary, secondary, tertiary and quaternary structure.Metabolism of Proteins - General reactions of protein metabolismEssential amino acids.Inborn errors of protein metabolism.Metabolism of amino acids- Decarboxylation, Transamination, Deamination, Glycine, Tyrosine,Tryptophan, Methionine and urea cycle.Nucleic acids-Biosynthesis and degradation of purines and pyrimidines andtheir regulation.
Unit – IV	**Lipids** - Structure and Biological importance and distribution of fats and fatty acids, Chemical properties and characterization of fats.Metabolism of Lipids - Biosynthesis of saturated and unsaturated fatty acids, β-Oxidation of fatty acid. Biosynthesis of glycerides, phospholipids and cholesterol, Regulation of lipid metabolism and ketone bodies, Disorders of lipid metabolism, lipoproteins and their significance.Role of free radicals and antioxidants in health and disease.
Unit - V	**Vitamins -** Structure and Biochemical properties of water soluble and fat solublevitamins and their coenzyme activity.**Hormones -** Mechanism of hormone action and its regulation - Hormones of Pancreas, Pituitary, Adrenal, Thyroid and Sexhormones.**Bioenergetics -** Electron transport chain, Oxidative Phosphorylation and synthesis of ATP Enzymes, Enzymes in differential diagnosis of diseases and their clinical significance

Reference and Textbooks:

Devlin, T.M. (1997*), Text book of Biochemistry with Clinical (4th ed).* Correlations,Wiley Liss Inc.

Stryer, L. (1998), *Biochemistry*, *4th Ed.* W.H. Freeman and Co.

Murray, R.K., Granner, D, K., Mayes, P.A. and Rodwell, VW. (2000): *Harpers* B*iochemistry*, (*25th ed).* Macmillan worth Publishers.

Nelson D.L. & Cox, M.M. (2000): *Lehninger's Principles of Biochemistry,Macmillan (3rd ed).* Worth Publishers.

Conn, E.E., Stumpt. P.K. Bruening, G. & Doi, R.H. (2001*): Biochemistry*, (*5th ed)*, Elsevier Publishers.

Outcomes	➢ Gain knowledge about composition and function of blood, water and electrolyte. ➢ Know about the classification, Structure and properties of macro nutrients

Course Teacher
P.Venkateswari
Prof.K. Manimekalai

Semester -III			
Coursecode: 453N09	**Food Microbiology**	**Credits: 5**	**Hours: 6**

Objectives	➢ To study about Microorganisms and its Identification in Food. ➢ To gain knowledge on prevention methods in contamination and spoilage of food
Unit - I	**Introduction** – Types of Microorganisms, Classification and Nomenclature, Morphology, Structure, Reproduction and their Importance in Food (Bacteria, Fungi, Viruses and Prions, Protozoans and Others).
Unit - II	**Microbial growth-** Bacterial Growth Curve, Intrinsic and Extrinsic Parameters of Food affecting the Microbial Growth. Food Spoilage - Types of Food Spoilage, Factors responsible for Microbial Spoilage, Acidity, Temperature, Time, Oxygen, Moisture, Water Activity, Changes caused by Micro-Organisms during Spoilage (Breakdown of Proteins, Carbohydrates, Fats and Other Constituents).
Unit - III	**Water –** Sources, Bacteriology of Water Supplies, Bacteriological Examination, Purification of Water, Water Borne Diseases and Prevention.Microbiology of Food and its Products.Food Borne Diseases-Food Infection and Intoxication.
Unit – IV	**Isolation and Identification of Microorganisms,** - Methods for the Microbiological Examination of Foods – Indicator Organisms, Direct Examination, Cultural Techniques, Enumeration Methods – Plate Counts, Most Probable Number Counts, Alternative Methods – Dye Reduction Tests, Electrical Methods, ATP Determination.Rapid Methods for the Detection of Specific Organisms and Toxins.
Unit - V	**Prevention of Contamination and Spoilage** – Principles of Preservation, Classification of Food Preservation.Physical Methods - High Temperature, Low Temperature, Irradiation and Microwave Heating, Concentration, Drying and Dehydration.Chemical Methods- Preservatives.Biological - Fermentation, Hurdle Technology.Non Destructive Method of Preservation – High Pressure Processing, Pulse Electric Field, Pulse Light Field, Ultrasound, MAP, CAP and Vacuum Packaging.

Reference and Textbooks:

Frazier, W.C, (1988) *Food Microbiology,* (*4th ed)*, MC Graw Hill Inc.

Banwart, (1989) *Basic Food Microbiology, (2nd ed)* CBS Publisher.

Bensaon, H.J, (1990) *Microbiological applications,* C. Brown publishers, U.S.A,

Pelczar, M.I & Reid, R.D, (1993) *Microbiology,* (5th ed), MC Graw Hill Book Company, New York.

Atlas M.Ronals (1995) *Priniciples of microbiology, (1st ed),* Mosby – year book Inc,Missouri, U.S.A,

Outcomes	➢ Gain knowledge on Microorganisms and its Identification in Food ➢ Aware about the prevention methods in contamination and spoilage of food

Course Teacher
P.Archana
Dr.P.Veeramani

Semester -III			
Coursecode: 453N10	**Research Methodology And Statistical Analyses**	**Credits: 5**	**Hours: 6**
Objectives	➢ To teach Research Methodology ➢ To inculcate different Statistical Methods and its Applications		
Unit - I	**Meaning of Research-** Process of Research, Objectives of Research. Developing a Research Proposal, Presentation of a Problem, Identifying the Problem - Defining and delimiting the Problem. Types of Research - Fundamentals, Applied, Action, Animal Experimentation, Descriptive, Tools for Research, Different types of Experimental Designs.		
Unit - II	**Measure of Central Tendency-** Mean, Median, Mode and their uses with examples and their Advantages and Disadvantages. Measure of Dispersion: Significance and Methods used in studying Dispersion (Range, Quartile Deviation, Mean Deviation and Standard Deviations) with their Uses, Advantages and Disadvantages.		
Unit - III	**Test of Relationship** - Meaning, Types and Methods used to Study Correlation (Simple Co-efficient of Correlation, Rank Correlation, Co-efficient of Concurrent Deviation and other Methods), Regression along with their uses, Advantages and Disadvantages. Testing of Hypothesis, Meaning, and Basic Concept Concerning Testing of Hypothesis.		
Unit – IV	a) **Sampling-** Type Probability, Non-Probability Sampling, Probability Sampling- Simple Random Sampling, Systematic Random Sampling, Two Stage and Multi Stage Sampling, Cluster Sampling. Non- Probability Sampling, Quota Sampling, Volunteer Sampling.b) Data Analysis- Quantitative, Qualitative, Presentation of Data - Tables, Graphs, Illustrations, Foot Notes and Bibliographical Presentations.		
Unit - V	**Research Reports** - Basic concepts of Research Report - Preliminaries - Title Page, acknowledgement, List of Tables, List of Figures, Index - Main text. - Bibliography, Glossary, Appendices. Role of computers and its applications in Textile Research. Student- "t" Test, Analysis of Variance- One way and Two way Classification- Characteristics of ANOVA, Chi-Square: General Features of Chi-square- 2*2 type tables, Duncan's Table.		

Reference and Textbooks:

Kothari.C.R. (1991) *Research Methodology, Methods and Techniques,* Wiley Eastern Ltd, New Delhi.

Gupta S.P. (1993), *Statistical Methods, Sultan Chand and sons*, Daryagang, New Delhi.

Bandarkar, P.L. & Wilkinson T.S (2002) *Methodology And Techniques Of Social Research,* Himalaya Publishing House, Mumbai.

Harman E & Montages, I (Eds) (2007) *The Thesis And The Book*, New Delhi: Vistar.

Wilkinson & Bhandarkar.P.L.C. (2010) *Methodology and Techniques of Social Research,* Himalaya Publishing House, Bombay.

Outcomes	➢ Gain knowledge on types of Research and Able to develop a Research Proposal. ➢ Get familiar with various Statistical Tools used in the Research Methodology.

Course Teacher
R.Ramya
Dr.P.Veeramani

Semester -III			
Course code:453EC5	**(A) Fabric Sourcing**	**Credits: 4**	**Hours: 6**

Objectives	➢To enable the students to Knowledge of different fabric clusters in India. ➢ To implant knowledge about Lead Time for Fabric Sourcing
Unit - I	**Fundamentals of Fabric Sourcing** - How to buy Fabrics - Basic knowledge requisite for fabric sourcing - Vendor Analysis & Development Selection of Vendor.
Unit - II	**Knowledge of different fabric clusters in India** - Types of fabrics produced, and end use of different fabrics produced in India - Factors to be considered while sourcing from domestic suppliers - Overseas Fabrics sourcing locations - Knowledge of different overseas fabric locations - Sourcing from nominated vendors - Factors to be considered while sourcing from overseas.
Unit - III	**Lead Time for Fabric Sourcing** - Time and Action Calendar for Sourcing - Lead time calculation for Fabric Sourcing - Cost Benefit Analysis while fabric sourcing - Just in Time Sourcing - Import verses Domestic Sourcing - advantages and disadvantages.
Unit – IV	**Import of Fabrics** - Policy Procedure and Documentation for Importing Fabrics from global sources - Licenses required for importing fabrics from overseas markets - Clearing and Forwarding, port handling procedure while importing fabrics - Transportation for import of fabrics.
Unit - V	**Supply Chain Management & Inventory** - In-bound logistics & Out-bound logistics - Transportation - Internal and external - Warehouse management - storing & inventory systems - Stock Plans and issuing materials to operations - Procedure to issue materials to internal customers - Supermarket or Store operations in garment industry.

Reference and Textbooks:

Booth, JE (1996*) Principle of Textile Testing by CBS* Publication New Delhi.

Susheela Dantyagi, (1998*) Fundamentals of Textiles and their Care, Orient Longman (5thed),* Ltd New Delhi.

Ruth Clock & Grace Kunz, Upper by Sadale (2000). *Apparel Manufacture- SewnProduct Analysis River* Publications, New York.

Anbumani, (2006) *Knitting Manufacture Technology*, New Age International, Chennai.

Gokarneshan N. (2009*) Fabric Structure and Design* - New Age International.

Outcomes	➢ Gain knowledge the fundamentals of Fabric sourcing methods and lead time ➢ Become familiar with different fabric clusters in India

Course Teacher
Prof.K. Manimekalai
Dr.P.Veeramani
Dr.A.K.P. Dhanakodi

Semester -III			
Coursecode 453EC6	**(B) Functional Foods And Nutraceuticals**	**Credits: 4**	**Hours: 6**

Objectives	➢ To learn the Concepts on Functional Foods and Nutraceuticals ➢ To instruct the Isolation of photochemical from plant materials.
Unit - I	**Functional Foods andNutraceuticals**–Definition, concept, history and market; Evolution ofnutraceuticals and functional foods market.Classification of nutraceuticals and functional foods.Significance and relevance of nutraceuticals and functional foods in the management of diseases and disorders.
Unit - II	**Natural occurrence of certain phytochemicals-** Antioxidants and flavonoids: omega – 3 fatty acids, carotenoids, dietary fiber, phyto estrogens; glucosinates; organo sulphur compounds. Dosage for effective control of disease or health benefit with adequate safety; studies with animals and humans; acute and chronic studies. Regulatory issues.
Unit - III	**Isolation of phytochemicals from plant materials:** Care in handling and storage of raw materials with minimal damage to sensitive bioactive compounds; Extractive methods for maximum recovery and minimal recovery and minimal destruction of active material; stability studies. Recent developments in the isolation, purification and delivery of phytochemicals.
Unit – IV	**Prebiotics, probiotics and symbiotics: Probiotics-**Definition, types and relevance; Usefulness in gastro intestinal health and other health benefits; development of probiotic products; recent advances in probiotics; Challenges and regulatory issues related to probiotic products. Prebiotics: Prebiotic ingredients in foods, types of prebiotics and their effects on gut microbes; health benefits of prebiotics; recent development in prebiotics. Symbiotic.
Unit - V	**Functional foods** - Definition, development of functional foods, use of bioactive compounds in appropriate form with protective substances and activators; Effect of environmental condition and food matrix; Effects of processing conditions and storage; Development of biomarkers to indicate efficacy of functional ingredients; Research frontiers in functional foods; delivery of immunomodulators /vaccines through functional foods. Nutrigenomics-concept of personalized medicine.Use of nanotechnology in functional food industry.

Reference and Textbooks:

Cupp J & Tracy TS. (2003)*Dietary Supplements: Toxicology and Clinical Pharmacology*. Humana Press.

Brigelius-Flohé, J & Joost HG. (2006) *Nutritional Genomics: Impact on Health and Disease.* Wiley VCH. Goldberg I.

Wildman, R.E.C. (2007) *Handbook of Nutraceuticals and Functional Foods, (2nd ed).* CRC Press.

Yashwant Vishnupant Pathak ,(2011) *Hand Book of Nutraceutical, Volume* CRC Press.

Dilip Ghosh et.al (2012) *Innovation in health and functional food*, CRC Press.

Outcomes	➢ Gain knowledge on Natural occurrence of certain photochemical ➢ Understand about the Prebiotics, probiotics and symbiotics.

Course Teacher
P.Venkateswari
Prof.K. Manimekalai

Semester -IV			
Coursecode: 453N11	**Macro And Micro Nutrients**	**Credits:5**	**Hours: 5**
Objectives	➢ To study about the micronutrients& macronutrients and its utilization		
Unit - I	**Energy** – Energy value of Foods and its Determination, Energy Expenditure, Components, Basal Metabolism, Factors affecting BMR, Role of Carbohydrate, Protein and Fat in Energy Metabolism. Dietary Fibre - Definition, types of Fibre in Plant Foods, Sources, Role of Dietary Fibre and Resistant Starch, Effect of over Consumption of Fibre		
Unit - II	**Carbohydrates** – Chemistry, Classification, Function, Digestion, Absorption, Utilization - Glycolysis, TCA Cycle, HMP Shunt and Energy Production, Gluconeogenesis. Pentose Phosphate Pathway, Glycogenolysis, Glycogenesis. Proteins - Chemistry, Structure, Classification, Function and Utilization.		
Unit - III	**Amino Acids** – Classification, Function, General Pathways of Protein Metabolism, Denaturation, Transamination, Deamination, Decarboxylation and Urea Formation, Evaluation of Protein Quality. Fats – Chemistry, Classification, Function, Utilization, Transport, Deposit, Biosynthesis and Oxidation of Saturated and Unsaturated Fatty Acids, Cholesterol, Phospholipids and Bile Pigments.		
Unit – IV	**Fat Soluble and Water solubleVitamins** - Chemistry, Digestion, Absorption, Metabolism, Physiological Functions, RDA, Sources, Deficiency Disorders, Toxicity, Method of Assessment and Interrelationship.		
Unit - V	**Macro minerals** (calcium, phosphorous, potassium, sodium and chloride) and Micro minerals (iodine, iron, copper, fluorine, zinc, magnesium, manganese, chromium and selenium) - Chemistry, digestion, absorption, metabolism, physiological functions, RDA, sources, deficiency disorders, toxicity, method of assessment and interrelationship between minerals. Water and Acid base balance. Interrelationship between vitamins and minerals.		
Reference and Textbooks:			
Gardon, M. Wardlaw., Paul, M. Iunset &Marcia, F.Seyler, (1994) *Contemporary Nutrition,* Mosby Publications, Geissler, C. & Powers, H., (2007)*Human Nutrition, (11thed)*, Elsevier Publication, Lehninger, A.L, (2000)*Biochemistry,* Worth Publishers Inc, New York, Powar & Chatwal, (2000) *Biochemistry*, Himalaya publishing house, Srilakshmi, B., (2002) *Nutrition Science,* New Age International,			
Outcomes	➢ Attain knowledge about macronutrients and its utilization ➢ Understand knowledge about Micronutrients and its Utilization		

Course Teacher
R.Ramya
Dr.P.Veeramani

Semester -IV			
Coursecode:453N12	**Nutrition Through Life Cycle**	**Credits: 5**	**Hours: 5**
Objectives	➢ To understand the growth, development and nutritional requirements in different stages of life cycle.		
Unit - I	**Different food Groups -** Recommended Dietary Allowances for Indians, Basis for requirement, computation of Allowances. Nutrition in Pregnancy-Stages of Gestation, Physiological Adjustments, and Weight gain during Pregnancy and Nature of Weight Gain, Nutritional Requirements, Storage of Nutrients, Physiological cost of Pregnancy and Complications of Pregnancy.		
Unit - II	**Nutrition in lactation**- physiological adjustments during lactation, hormonal controls & reflex action, physiology of milk production, nutritional components of colostrum and mature milk, special foods during lactation, nutritional requirements during lactation. Nutrition in infants – rate of growth, weight as the indicator, breast Vs bottle feeding, nutritional allowances, feeding premature and low birth weight infants, supplementary feeding and weaning foods.		
Unit - III	**Nutrition in preschool Children –** Growth and development, Food Habits, Nutritional Requirements, Supplementary Foods.Nutrition in School Age - Early and Middle Childhood, Growth and Development, Food Habits, Nutritional Needs and Packed Lunch. Nutrition during adolescence - physical growth, pubertal changes, nutritional needs and eating disorders- anorexia nervosa and bulimia, adolescent pregnancy and its complications.		
Unit – IV	**Nutrition during adulthood –** Type of work, adult consumption unit, meal planning and nutritional needs.Nutrition during old age - physiological and psychological changes during old age, nutritional requirements, factors affecting food intake, common nutritional problems in old age.		
Unit - V	**Nutrition in yoga and exercise-** Role of carbohydrate, protein, fat, vitamins, minerals, water and electrolyte in exercise and its changes before, during and after exercise. Nutrition in Sports, Nutrition in space, defence, high altitudes, extreme hot and cold climates and submarines.		

Reference and Textbooks:

Gopal,C. Kamalakrishnaswamy, (2000) *Nutrition in Major Metabolic Disease,(1st ed)* Oxford India Paper backs Publisher

Srilakshmi, B. (2002*) Nutrition Science,* New Age International [p] ltd, New Delhi,

MahtabS.Bamji, Prasad Rao, N.Vinodini Reddy. (2003) *Textbook of HumanNutrition*,(2nd ed) Oxford and IBH Publishing Co. Pvt .Ltd, ,

Swaminathan,M. (2004) *Advanced text book on Food and Nutrition,*(2nd ed), Anmol Publication Pvt,Ltd,

VenkataiahS.D., (2004) *Nutrition Education*, Anmol Publication Pvt. Ltd, Revised

Outcomes	➢ Aware about the growth, development and nutritional requirements in different stages of life cycle

Course Teacher
P.Archana
Prof.K. Manimekalai

Semester -IV			
Coursecode: 453N13	**Computer Application in Nutrition and Dietetics (P)**	**Credits: 5**	**Hours: 10**
Objectives	➢ To teach the basics of Computer and its Applications. ➢ To educate skills to apply Computer Based Technology in Food Science and Nutrition.		
Unit - I	**Introduction to Computers -** History of Development of Computers, Main Frame, Minis, Micros and Super Computer Systems, Binary Numbers, Bits, Bytes, CPU, Input and Output Devices, Main and Auxiliary Stage Device, Software and Hardware		
Unit - II	**Operating Systems and MS Office -** Introduction to Operating Systems, Windows Applications MS Word, MS Excel, MS Access and MS Power Point **Practical:**MS Word, Excel, Access and Power Point		
Unit - III	**Computer Networks -** LAN, WAN, Intranet, Extranet, Service Providers, Modem, Fibre Optics Basic of HTML, WWW, URL, TCP/IP **Practical:** Introduction to Computer Networks		
Unit – IV	**Multimedia -** Basic Elements, Hardware, Application of Multimedia, Introduction Multimedia, Authorizing Tools. **Practical:** Introduction to Macro Flash Player, Adobe Photoshop, Corel Draw.		
Unit - V	**Application of Computers in Food Science and Nutrition -** Power Point Presentation, Nutrient and Diet Calculation, Nutrition Education and Counseling, Nutrition Software and Websites, E-Journals in Food Science and Nutrition, Use of SPSS. **Practical -** Developing Mini Projects in Food Science and Nutrition using MS Word, MS Excel and MS Power Point.		

Reference and Textbooks:

Bansal.S.K (2004*) Text Book of Information Technology,* APH, Publishing Corporation.

Balagurusamy. E (2008) *Computing Fundamentals and C programming,* TataMcGraw Hill Education Private Limited, New Delhi.

James F. Kurose & Keith W Ross (2008) *Computer Networking, A Top-Down Approach Featuring the Internet, Pearson Education and Dorling Kindersley(3rd ed)*Publishers, Delhi

Andrew S. Tanenbaum (2009) *Computer Networks, Pearson Education and Dorling Kindersley (4th ed),* Publishers, Delhi.

Ralf Steinmetz & Klara Nahrstedt (2011) *Multimedia – Computing, Communications and Applications,* Person Education and Dorling Kindersley Publishers, Delhi

Outcomes	➢ Understand the Basics Knowledge in Computer Applications ➢ Gain knowledge about the Application of ICT in Food Science

Course Teacher
P.Venkateswari
Prof.K. Manimekalai

Semester - IV			
Course code: 453N14	**Project/Dissertation**	**Credits: 6**	**Hours: 10**
Objectives	Help the students to lean the knowledge about the projects		

Preliminary

1. Title Page- title, Authors Name
2. Certificate of Originality by the Guide
3. Declaration by the Author
4. Table of Contents
5. List of Tables
6. List of Figures
7. Acknowledgement
8. Abstract

I. Introduction: Statement of the Problem, Significance, Need for the Study, Objectives, and Definitions.
II. Review of Literature
III. Methodology: Tools used, Procedures, Hypothesis.
IV. Results and Discussion: Tables and Figures, Statistical Presentations, Hypothesis Testing.
V. Summary and Conclusion
VI. Suggestion for the Future Study
VII. References

Outcomes	They will gain in-depth knowledge about the research ethics.

Course Teacher
Prof.K. Manimekalai
R.Ramya
P.Venkateswari

SPECIALISATON II

M.A. INTEGRATED HOME SCIENCE
TEXTILES & CLOTHING

SEMESTER – I			
Coursecode: 453T01	**CORE I –FASHION AND CLOTHING PSYCHOLOGY**	**Credits: 5**	**Hours: 5**
Objectives	➢ To teach about Fashion Psychology ➢ To illuminate experience in different types of Display Techniques		
Unit - I	**Factors influencing Fashion Changes** – Psychological needs of Fashion, Social Psychology of Fashion, Technological, Economical, Political, Legal and Seasonal influence. Role of costume as a status Symbol, Personality and Dress, Clothes as Sex Appeal, Cultural value Fashion Cycles, Repetition of Fashion		
Unit - II	**Fashion Evolution –** Fashion Cycles, Length of Cycles, Consumer groups in Fashion Cycles – Fashion Leaders, Fashion Innovators, Fashion Motivation, Fashion Victim, Fashion Followers Adoption of Fashion – Trickle Down, Trickle up and Trickle Across Theory Fashion Forecasting – Market Research, Evaluating the Collection, Fashion services and Resources (Fashion services, Colour Services, Video Services, Newsletter Services, Websites, Directories and References), Design - Historic and Ethnic Costume.		
Unit - III	**Visual Merchandising of Fashion, Types of Displays -** Window Displays, Interior Displays, Elements of Display – The Merchandise, Mannequins and Forms, Props, Signage, Lighting. Merchandising Presentation – Tools and Techniques – Back Drop, Forms, Fixtures. Fashion Show – Definition, Planning, Budgeting, Location, Timings, Selection of Models, Collection, Set Design, Music, Preparing the Commentary, Rehearsal.		
Unit – IV	**Understanding fashion designer** – types – classicist, idealist, influenced, realist, thinking poet. Indian Fashion Designers – Haute couture – ohithosla, GitanjalKashyap, Hemant Trivedi, J.J.Valaya, James ferrerira, Ritu Kumar, RohitBal, Tarun Tahiliani Minimalists – Himanshu and Sonali Sattar, Sangeethe Chopra, Wendell Rodricks Village India – BhaminiSubramaniam, Anjumodi, Indiar, Broker, MadhuJain.Studio line – BhairaviJaikishan, Kishan Mehta, Ravi Bajaj, RituBeri, Rockys.		
Unit - V	World fashion Centres - France, Italy, America, far east contributions of well known designers from France, America, Italy, Britain and Far east countries.		
Reference and Textbooks: Harriet T, McJimsey (1973) – *Art and Fashion in Clothing Selection*, The Iowa state University Press, Ames, Iowa. Drake et-al (1991) – *Retail Fashion,* Promotion and Advertising Macmillan publishing Company, New York. Bennett (1998) – *Inside the Fashion Business,* Coleman & Co, Mumbai. Gini Stephens Frings *(1999) – Fashion, From Concept to Consumer, 6th Edition* – Prentice Hall. Renbourn E.T. (2004), Physiology & Hygiene of materials& clothing, Published by Wood Head Publishing Limited.			
Outcomes	Enable the students to ➢Get basic knowledge on Fashion Psychology ➢Through industry visit students get hands on experience in types of display Techniques.		

Course Teacher
Prof.K. Manimekalai
Dr.A.K.P.Dhanakodi

SEMESTER – I			
Coursecode: 453T02	**CORE II- CARE FOR TEXTILES AND CLOTHING**	**Credits: 5**	**Hours: 5**
Objectives	➢To teach about care and Maintenance for Textile and Clothing ➢To instruct about various techniques in Textile Care and Clothing		
Unit - I	**Water-** Types- Soft Water-Hard Water- Purification of Water.Soaps, Detergents – Types– Manufacturing – Properties. Reasons for non use of Banned Chemicals and solvents like CFC, CTC & white petrol		
Unit - II	**Laundering –** Objects – Methods - Laundering of White - Coloured, Cotton, Silk, Wool, Synthetic Fabrics, Lace Materials, Carpets. Clothing Care - Mending, Darning and Seasonal Care, Wardrobe Planning and Storage.		
Unit - III	**Dry Cleaning -** Chemicals Used - Sequence - Method - Commercial Dry Cleaning. Washing Machines – Types - Working Principles.		
Unit - IV	**Ironing and Pressing -** Hot Air, Steam, Permanent Pressing, Tumble Drier –Hydro Extractor - Working Principle.		
Unit - V	**Stain Removal –** Methods of Washing- Application of Friction- Hard Friction, Rubbing and Scrubbing - Application of Light Pressure, Squeezing, Suction-Washing-Use of Washing Machine, Methods of Finishing-Draping- Ironing - Dry Cleaning and Storage, Care Labeling. Details of care instructions		

Reference and Textbooks:

Tate of. Glesson (1961), *Family Clothing,* Latest Edition, John Wiley &Sons Inc., Illinois.

Dantyagi. S. (1980), *Fundamentals of Textiles & their Care.* Orient Langnass Ltd. New Delhi.

Wingate I.B. (1986), *Textile Fabrics & Their Selection,* Allied Publishers Pvt. Ltd. Madras

Neomi D'Souza (1998), *Fabric Care,* New Age International Publisher.

Susheela Dantyagi (1998), *Fundamentals of Textiles and their Care 5th edition,* Orient Longman Ltd, New Delhi.

Renbourn E.T. (2004), *Physiology & Hygiene of materials& clothing,* Published by Wood Head Publishing Limited.

Durgadeulkar (2011), *Household Textiles and Laundry Work,* Alma Ram & Sons. Delhi.

Outcomes	Enable the students to ➢Develop their knowledge on various types in Maintenance and Care of Textiles and Clothing. ➢Expand their knowledge on various techniques adopted for Textile Care.

Course Teacher
Dr.P.Veeramani
M.Priyanka

SEMESTER – I			
Coursecode: 453T03	**CORE-III ADVANCED WET PROCESSING (T)**	**Credits: 5**	**Hours: 5**

Objectives	➢To moralize knowledge in the field of Preparation Methods ➢To impart knowledge about Dying Process ➢To teach about Finishing Process
Unit - I	**Pre-treatment Process -** Singeing, Desizing, Scouring, Bleaching, Mercisation, Wet Processing Equipments - Kier, J-Box, Stenter, Wool Carbonizing, Weighing of Silk, Heat Seating.
Unit - II	**Dyeing-**Definition, Classification of Colorants, Different Stages of Dyeing Process, Dyes suitable for different Fibres - Cotton, Viscose Rayon, Silk, Wool, Nylon and Polyester.
Unit - III	**Machinery for Fibre Dyeing, Yarn Dyeing and Fabric Dyeing –** Woven and Knitted – Hunk, Beam, Tigger, Winch, Jet, Soft Flow Dyeing Machines, Colour Matching Cabinet.
Unit - IV	**Printing -** Definition, Preparation of Printing Paste, Methods of Printing, Styles of Printing Methods of Printing - Direct, Discharge, Resist, Screen Printing, Roller Printing, Rotary Printing, Transfer Printing, Stencil Printing, Block Printing, Flock Printing and Other Latest Printing Methods.
Unit - V	**Finishing –** Classification of Finishing- Mechanical and Chemical Finishing- Enzymes, Wrinkle Free, Water Repellent, Proof Finishing, Bio polishing, Bio washing, Stone wash, Acid wash, Flame Retardant Finishes, Mildew Finish. Finishing Defeats and Remedies.

Reference and Textbooks:

Shenai V. A, *Vol 3, 4, 6, and 10* (1996), *Technology of Textile Processing*, Sevak Publications, Mumbai.

Marsh. J. T. (1999*), An Introduction to Textile Finishing*.

NallanKallai (2000), *Fabric Finishing,* New Age International.

SusheelaDantyagi*, 5th Edition,* (2000), *Fundamentals of Textiles and their care*, Abhishek Publication.

Clarke .W. (2004)*, An Introduction to Textile Printing*, Wood Head Publishing Limited, Cambridge, England.

Outcomes	Enable the students to ➢Gain knowledge on Pre-treatment process and equipments used in Wet Processing. ➢Aware about the Dying Process Suitable for Different Fibres. ➢Get familiar on types and Methods in Finishing Processes.

Course Teacher
Prof.K. Manimekalai
Dr.A.K.P.Dhanakodi

SEMESTER – I			
Coursecode: 453T04	**CORE-IV ADVANCED WET PROCESSING (P)**	**Credits: 5**	**Hours: 5**
Objectives	➢ To impart knowledge in advanced technology of wet processing		
	PREPARATION OF SAMPLES FOR FIBRE, YARN AND FABRIC **Pre-treatment Process** 1. Desizing 2. Scouring 3. Bleaching 4. Mercerization **Dyeing** 1. Dyeing of cotton fabric with direct dye; Reactive dye, Vat,. 2. Dyeing of silk, wool with acid and basic dyes 3. Dyeing of polyester fabric with disperse dyes **Printing** 1. Hand screen printing 2. Block printing using different techniques 3. Stencil printing using different techniques 4. Tie and Dye 5. Batik printing		
Reference and Textbooks: Shenai V. A, (1996) ,*Technology of Textile Processing Vol 3, 4, 6, and 10*, Sevak Publications, Mumbai. Marsh. J. T. (1999*), An Introduction to Textile Finishing*. SusheelaDantyagi*, 5th Edition,* (2000), *Fundamentals of Textiles and their care*, Abhishek Publication. NallanKallai (2000), *Fabric Finishing,* New Age International. Clarke .W (2004)*, An Introduction to Textile Printing*, Wood Head Publishing Limited, Cambridge, England.			
Outcomes	➢ Enable the students to ➢ Gain in depth knowledge on advanced technology of wet processing.		

Course Teacher
Prof.K. Manimekalai
M.Priyanka

SEMESTER – I			
Coursecode:453EC1	**ELECTIVE –I (A) HOME TEXTILES (T)**	**Credits: 4**	**Hours: 4**
Objectives	➢ To teach Introduction to Home Textiles ➢ To impart knowledge in Finishes and quality parameters used in Home Textiles		
Unit - I	**Introduction to Home Textiles –** Fabrics used for Living area, Kitchen, Dining, Bed room, Bath room accessories and general household textiles. Textile for Furnishing and Upholstery applications – scope, fixed upholstery, non-stretch loose covers & stretch covers		
Unit - II	**Window/ Curtain Fabrics** - Sun filters (Sheers and nets), Semi-sheers, Reflective textiles, drapes and Blinds. Bed Room Textile - Sheets & Pillow Cases, Quilted Textile, Blankets& Rugs - Jacquard blankets, Printed blankets, Fire proof blankets, Baby blankets. Bed Spreads, Mattress covers		
Unit - III	**Dining Room Textiles Tablecloths –** Colour – Woven & Printed type, jacquard types, embroidered types, non-woven types. Table mats – Colour -woven, Printed jacquard, embroidered. Kitchen Textiles Aprons, Dish cloth, Tea cosy, Bread bag, Mittens, Pot Holders, Table Mats – Construction & manufacturing		
Unit – IV	**Bath Room Textiles –** bath towels, bathrobes, floor mats, handtowels, beach towels, towels for swimming pools.- Terry Towels – types and applications. Floor Coverings – Introduction, Pile fabrics, Backing fibres& fabrics – Tufted carpets, woven carpet.		
Unit - V	**Finishes and quality parameters used in Home Textiles -** Protection against unpleasant odor, - Temperature regulated beddings - Antimicrobial finish, - Moisture management - Test Method for towels, Rug and Home textiles and - Performance specifications of different home textiles		
Reference and Textbooks: Tate of. Glesson (1961), *Family Clothing, Latest Edition,* John Wiley &Sons Inc., Illinois. Dantyagi. S. (1980), *Fundamentals of Textiles & their Care.* Orient Langnass Ltd. New Delhi Talukdar, Ajagaonkar, Sriramulu (1998), *Weaving Machine, Mechanisms, Management*. Neomi D'Souza (1998), *Fabric Care,* New Age International Publisher. Subrata Das (2010), Performance of Home Textiles, Wood head Publications India Pvt Ltd.			
Outcomes	Enable the students to ➢Gain the knowledge describes various types of home textiles and their manufacturing methods. ➢Understand requirements of different home textiles ➢Learn to understand various finishes and evaluation methods of home textile.		

Course Teacher
Dr.P.Veeramani
M.Priyanka

SEMESTER – I			
Coursecode:453EC1	**LECTIVE –I (B) WOMEN AND SOCIETY**	**Credits: 4**	**Hours: 4**
Objectives	➢To impart about the Different Sectors of Development, with Special Reference to Women. ➢To introduce the Issues and Concerns of Women. ➢To Teach the Strategies for Empowerment of Women.		
Unit - I	**Women in Society –** Ancient India and Modern India - Concept of Women's Movements – Basic Elements – Classifications Social Reform Movement – Status of Women through Ages.		
Unit - II	**Women's Role:** Post Independence Period - Indian Independence Movement and Women's Movements - Narmada Valley Project Environmental Movement - Chipko Movement- Peasant Movement		
Unit - III	**Position of Women in:** Hinduism, Islam, Christianity - Social Reformers worked for the Emancipation of Women – Rajaram Mohan Roy, DayanadaSaraswathi, Mahatma Phule, PanditRamabai, Annie Besant, Sarojini Naidu, Muthulakshmi Reddy, DurgabaiDeshmukh, ArunaAsaf Ali.		
Unit – IV	Women and Political Participation - Women and Panchayat Raj Institutions – International, National and Local Women in Parliament through Ages - Women's Bill		
Unit - V	**Rise of NGOs in Women Organization for Women's Development** – all India Women's Conference (AIWC) – Women's India Association (WIA) - National Council of Women in India (NCWIE) - Indian Association of Women's Studies		
Reference and Textbooks: Geraldine Forbes (1998) *Women in Modern India,* New York: Cambridge University Sharma. A. (2002). *Women in Indian religions.*New Delhi:Oxford University press. Leslie L. Heywood (2005) *The Women's Movement Today*, Jaipur: Rawat Publications Vol.I & II SumitSarkar & TanikaSarkar (2007), *Women and Social Reform in Modern India*, New Delhi: Permanent Black Vol I &II. Suguna. B. (2009) *Women's Movements*, New Delhi: Discovery Publishing House Pvt. Ltd 4.			
Outcomes	Enable the students to ➢Gain knowledge about the status of women in the society. ➢Can set up their own NGOs for the development of women		

Course Teacher
Prof.K. Manimekalai

SEMESTER – II			
Coursecode:453T05	**CORE V– APPAREL MERCHANDISING**	**Credits: 5**	**Hours: 5**
Objectives	➢To enlighten the students about the Field of Apparel Merchandising ➢ To impart the Process to flow in Apparel Industry ➢ To know about Merchandiser Role in Apparel Industry		
Unit - I	**Merchandising: Introduction, Meaning -** Apparel Merchandising – Concepts of Six Rights – Organization Structure of an Apparel Industry – Classification of Exporters – Rating or Grading of Export Houses – Classification of Buyers – Export merchandising and Retail Merchandising.		
Unit - II	**Merchandising Process flow in Apparel Industry** – Buyer sourcing and Communication – Enquiry – Order Confirmation – Order Review and its Importance – Planning and Programming: Master Planning, Scheduling or Route Card – Factors for Route Card – Programming for Yarn, Knitting, Dyeing, Stitching, Sampling, Accessories – Samples: Meaning and Importance – Types of Samples – Expedition of Samples.		
Unit - III	**Quality Standards and Specifications –** Quality Assurance – Quality control – Quality Inspection and its Types – Testing – Check Points before Cutting – Pilot Run or Trial Run and its Importance – Approvals – Types of Approvals – Shipping Marks – Final Inspection Procedures – Self, Second and Third Party Inspection – Effective Expedition Procedures – Order Sheet and its Contents – Packing List and its Contents – Document Formats: Order Sheet, Packing List, Invoice, Inspection and Testing Reports - Assortment and its Types.		
Unit – IV	**Types of Merchandiser –** Functions of a Merchandiser – Essential Requisites of a Good Merchandiser – Vendor Sourcing, Evaluation and Development – Global Sourcing – Vendor Nomination by Buyers – Reasons for Vendor Nomination – Documents Recording and Maintenance – Claims and Reasons for Claims – Factory Audits – Buyers Code of Conducts		
Unit - V	**Export Associations –** Apparel Export Promotion Council – Journals and Magazines related to Apparel and Textiles –Trade Shows and Fairs – Participation in Trade Shows – Advantages of Trade shows and Fairs – Apparel & Textile Trade shows and Fairs in India.		

Reference and Textbooks:

Chas A, (1961), *Historic Costume*, Bernard and Co, Illinois.

Jeannette Jarnow, Kitty G. Dickerson (1996), *Inside the Fashion Business*. McMillan Publishing Co.

Renbourn .E.T. (2004), *Physiology & Hygiene of materials& clothing,* Published by Wood Head Publishing Limited.

Daragho' Reilly, Jullian J. Gibbs (2004), *Building Buyer Relationships*.

Vedhamani &Gibson. G.(2007), *"Retail Management:Functional Principles and Practices"* Jaico Publishing House.

Outcomes	Enable the students to ➢Gain basic knowledge on Apparel Merchandising in General. ➢Get aware about the various processes in Apparel Industry and able to run own Industry. ➢Get conscious about the Rolesof Merchandiser in Apparel Industry.

Course Teacher
Dr.P.Veeramani
M.Priyanka

SEMESTER – II			
Coursecode: 453T06	**CORE-VI ADVANCED GARMENT CONSTRUCTION (T)**	**Credits: 5**	**Hours: 5**
Objectives	➢To inculcate knowledge on Standardized Body Measurements ➢To guide about the Various Patterns in Garment Construction ➢To get familiar with Drafting and its Types.		
Unit - I	**Standardized body Measurements -** Observation of head Theory		
Unit - II	**Preparation of Fabric for Cutting -** Developing blocks for basic Front, Basic Back, Basic sleeve and Basic Collar. - Interfacing – Purpose, Types, Steps involved in applying interfacing on Garments- NeckLine.		
Unit - III	**Designing, Drafting and Construction details of the following** - School Uniforms – Maternitywear - Industrial Uniforms - Women Formal wear		
Unit – IV	**Designing, Drafting and Construction details of the following** - T-Shirt - Men's Shirt- Half/Full Sleeve - Trousers - Narrow bottom or Bell bottom		
Unit - V	Selection and use of Home Furnishings, Curtains, Draperies, Pillow Covers & Cushion Covers.		
Reference and Textbooks: Mabel, D. Erwin (1945), *Practical Dress Design*, The Macmillan Company, New York. Gertrude Strict Land, (1956), *"A Tailoring Manual, Mac Millon Company*, New York. Brockman, Magritha (1966), *The Theory of Fashion Design,* John Wiley Sons, New York. Mary Mathews (1986),*Advanced Clothing Construction, Part I & II,* Cosmic Press, Chennai. Marsh. J. T. (1999), An *Introduction to Textile Finishing*.			
Outcomes	Enable the students to ➢Gain knowledge on standardized body measurements used in Garment Construction ➢Elevate knowledge on various pattern used in advanced Garment Construction Unit. ➢Able to Design Draft and Sew Dresses for Themselves.		

Course Teacher
Dr.P.Veeramani
Dr.A.K.P.Dhanakodi

SEMESTER – II			
Coursecode:453EC3	**ELECTIVE – II (A) WOMEN AND HEALTH**	**Credits:4**	**Hours: 4**
Objectives	➢To introduce the concept of health and the issues related to Gender and health ➢To study about the national and international initiatives in the promotion of Women's Health.		
Unit - I	**Health:** Concept, Definition and types of Health – Parameters of Community Health - Gender and Health - Health Care Needs of Women, Men - Health Status of Women - Factors influencing women's health - Health Indicators of Women - Mortality - Morbidity - Sex ratio, Child sex ratio, Life expectancy – MMR – IMR - Fertility		
Unit - II	**Women's Reproductive Health –** Prenatal and Postnatal Health Care- Menstrual Hygiene Management – Menopause – Hormonal Replacement Therapy – Sex Selective Abortion – MTP – NRT – Amniocentesis Test – Clinical Trials and Gender Bias – Infertility: IVF & Surrogacy		
Unit - III	**Gender and Nutrition–** Nutrition for pregnant and lactating mothers - Nutrition education for girls – Anemia - Gender bias in nutrition intake -Protein Malnutrition – Gender Discrimination - Breast Feeding – Health Seeking Behaviour – Initiatives on Nutrition Program – National –NRC – RishoriSakthi - International - WHO – ICDS – UNICEF - RTI and STI.		
Unit – IV	**Gender and Health issues:** Feminization of HIV/AIDS – Private vs Public Spheres – Balancing Drudgery and Health Needs – Pollution and Health –Health issues of LBGTQ– Environmental & Occupational Health hazards- Health issues relating to: sex workers, women with disabilities, women elderlies.		
Unit - V	**Programme and Policies for Health for women :** International Initiatives – ICPD – Cairo- National Health Policies, NRHM, ASHA, ICDS - Population Policy 2000 – MCH to RCH Approaches-Impact of Population Policies – Tamil Nadu State.		

Reference and Textbooks:

Human Development Report (1995), Delhi: UNDP.

Reducing Malnutrition (1995)-*A Call for Urgent Action,* India: UNICEF.

United Nations Economic & Social Commission for Asia and the Pacific(1997). *Women In India:* A Country Profile, *Statistical Profile no.12.* United Nations,New York.

World Health Organization. (2000) *"Women of South East Asia-A health profile"* WHO.

Rao, Mohan (2004), *From Population Control to reproductive health,* Sage Publication, New Delhi.

Singh, Lakshmi Abha, Rural Women (2005): *Work and Health,* The women press, New Delhi.

Neera Desai & UshaThakkar (2006). *Women in Indian Society*, National Book Trust, New Delhi.

Census of India Report-(2011), *Ministry of Home Affairs*, Government of India, New Delhi.

Outcomes	Enable the students to ➢Gain knowledge on the concept of health and the issues related to Gender and health ➢Able toacquire information on National and International Initiatives in the promotion of Women Health.

Course Teacher
Prof.K. Manimekalai

SEMESTER – II			
Coursecode:453EC4	**ELECTIVE II (B) FOOD SERVICE MANAGEMENT**	**Credits: 4**	**Hours:4**
Objectives	➢To inculcate the Knowledge on Institutional Food Administration Strategies and Management.		
Unit - I	**Introduction to Food Service Systems**. Evaluation of the Food Service Industry. Characteristics of the various types of Food Service Units - Commercial - Hotel, Motel, Restaurant, Cafeteria and Chain Hotels; Welfare - Hospital, School Lunch, Residential Establishment and Industrial Catering; Transport - Air, Rail, Sea and Space; Miscellaneous - Contract and Outdoor.		
Unit - II	**Management of Resources** – Finance- Determining the Finance needed to Establish or Run an Unit, Budgets, Sources of Finance, Planning Adequate Cash Flow.Space & Equipment Management- Steps in Planning Layouts, Determining Equipment, Selection and Placement, Layout Analysis.		
Unit - III	**Material Management-** Menu Planning, Planning the Material Needed, Methods of Selection, Purchasing, Receiving and Storage, Quantity Food Production, Service and Modes of Delivery, Planning Menus for Banquet, Outdoor Catering, Packed Meals and Restaurant.Personnel Management- Manpower Planning, Placement, Recruitment, Induction, Training, Motivation and Performance Appraisal.Time and Energy Management- Measures for Utilisation and Conservation.		
Unit – IV	**Management -** Approaches to Catering Management, Principles and theories of Management, Functions of Management, and Styles of Management, Management Tools- Organisation Chart and its Types.Strategies in Planning- Conceptual Strategy, Marketing Strategy, Financial Strategy, types of Plans.		
Unit - V	**Financial Management -** Definition, Cost Concept, Component of Cost, Food Cost Control, Pricing, Book Keeping and Accounting, Techno-Economic Feasibility of Food Production/Service Enterprise.Marketing and Sales Management-Market Survey, Sales Analysis, Market Promotion.Quality Assurance- Food Quality, Total Quality Management, Analysis of Processed and Finished Products.		

Reference and Textbooks:

Coltman, M.M. (1977): *Food and Beverage Cost Control*. Printice Hall Series.

Green, E.F., Drake, G.G. Sweeny, J.F. (1978) *Profitable Food and Beverage Management;* Planning, Operations, Hayden Book Company, New Jersey.

Avery, A.C. (1980): *Modern Guide to Food Service Equipment.* Boston CBI Publishing Company.

Brodner, J., Maschal, H.T., Carlon, H.M. (1982): *Profitable Food and Beverage Operation 4th Edition,* Hayden Book Company, New Jersey.

Desseler, Garry (1987) *Personnel Management Modern Concepts and Techniques,* Prentice Hall, New Jersey.

Brichfield, J. (1988): *Design and Layout of Food Service Facilities*, VanNortrand Reinhold.New York.

Outcomes	Enable the students to ➢Gain the Knowledge about Food Service Management ➢Learn about Management of human and non human resources in food service management

Course Teacher
Dr.P.Veeramani

SEMESTER – III			
Coursecode: 453T08	**CORE VIII - TEXTILE TESTING AND QUALITY CONTROL**	**Credits: 5**	**Hours: 5**

Objectives	➢To study the Textile Testing Methods involved in Textile & Apparel Industry. ➢To study the Quality system used in Apparel Industry.
Unit - I	**INTRODUCTION TO TEXTILE TESTING** – FIBER - General aspects in Textile Testing. Routine tests performed in the Textile Industries. Textile testing equipments used in Spinning, Weaving, Knitting and Garment Units. Fibre testing: Fibre length, Strength Fineness, Maturity and Trash Analysis Testing of Cotton Manmade Fibre& Blends.
Unit - II	**YARN** - Yarn Testing; Yarn Numbering, Single Yarn Strength, Twist, Uster Class Mat Evenness, Crimp, Yarn Appearance for FIB. Standards and Specifications used in Textile Industries.
Unit - III	**FABRIC** - Fabric Testing; Length, GSM, Seam Strength - Tensile, Tearing and Elemendrofom and Bursting. Abrasion Resistance- Yarn and Fabric, Pilling Resistance, Crease Recovery, Bending Length, Drape Co- efficient, Air Permeability, Water Permeability.
Unit – IV	**FABRIC QUALITY** - Quality control: definition and importance. Textile quality standards and specifications.KES & FAST.Moisture and Thermal Relationship to the Fabric Properties. Colour Fastness Tests- Shrinkage and Dimentional Stability, Washing, Crocking, Pressing, Perspirations and Sunlight.
Unit - V	**GARMENT -** Garment Quality Tests; Cutting Sewing, Pressing, Finishing and Packaging Defects. Spec Sheet Analysis- Pattern Making and Marking. Customer Returns and Inspection Procedures. Care Labelling System. (Self Study).

Reference and Textbooks:

Grorver Wiley. E.B. (1980). *Hand Book of textile testing* by Easters University, Delhi.

Booth JE (1996), *Principle of Textile Testing*, CBS Publication New Delhi.

Sara J. Kadolph.(1998), *"Quality Assurance for Textiles and Apparels"*, Fair child publications, New York,

Ruth Clock & Grace Kunz (2000), *Apparel Manufacture*- Sewn Product Analysis, Upper Sadale River Publications, New York.

Slater.K. (1993), *Physical Testing& Quality,Vol.23, No1/2/3,* Textile Institute.

Outcomes	Enable the students to ➢Gain knowledge on general aspects of Textiles Testing and Quality Control. ➢Able to find the tests used to Identify various Textile Fiber ➢Get aware about the Quality Control and color Fastness Tests in Textiles.

Course Teacher
Prof.K. Manimekalai
Dr.A.K.P.Dhanakodi

SEMESTER – III			
Coursecode: 453T09	**CORE-IX RESEARCH METHODOLOGY AND STATISTICAL ANALYSES (T)**	**Credits: 5**	**Hours: 5**
Objectives	➢To teach Research Methodology ➢To inculcate different Statistical Methods and its Applications.		
Unit - I	**Meaning of Research-** Process of Research, Objectives of Research. Developing a Research Proposal, Presentation of a Problem, Identifying the Problem - Defining and delimiting the Problem. Types of Research - Fundamentals, Applied, Action, Experimental and Descriptive, Tools for Research, Different types of Experimental Designs.		
Unit - II	**Measure of Central Tendency**: Mean, Median, Mode and their uses with examples and their Advantages and Disadvantages. Measure of Dispersion: Significance and Methods used in studying Dispersion (Range, Quartile Deviation, Mean Deviation and Standard Deviations) with their Uses, Advantages and Disadvantages.		
Unit - III	**Test of Relationship**; Meaning, Types and Methods used to Study Correlation (Simple Co-efficient of Correlation, Rank Correlation, Co-efficient of Concurrent Deviation and other Methods), Regression along with their uses, Advantages and Disadvantages. Testing of Hypothesis, Meaning, Basic Concept Concerning Testing of Hypothesis.		
Unit – IV	**a) Sampling**- Type Probability, Non-Probability Sampling, Probability Sampling-Simple Random Sampling, Systematic Random Sampling, Two Stage and Multi Stage Sampling, Cluster Sampling. Non- Probability Sampling, Quota Sampling, Volunteer Sampling. **b) Data Analysis-** Quantitative, Qualitative, Presentation of Data - Tables, Graphs, Illustrations, Foot Notes and Bibliographical Presentations.		
Unit - V	**Research Reports - Basic concepts of Research Report** a. Preliminaries - Title Page, acknowledgement, List of Tables, List of Figures, Index. b. Main text. c. Bibliography, Glossary, Appendices. Role of computers and its applications in Textile Research. Student- "t" Test, Analysis of Variance- One way and Two way Classification-Characteristics of ANOVA, Chi-Square: General Features of Chi-square- 2*2 type tables, Duncan's Table. Survey on Textile related topics (Self study), Standardization of body measurements for a particular age group (Self study).		

Reference and Textbooks:

Kothari.C.R. (1991) *Research Methodology, Methods and Techniques,* Wiley Eastern Ltd, New Delhi.

Gupta S.P. (1993), *Statistical Methods,* Sultan Chand and sons, Daryagang, New Delhi.

Fink, A., (2009). *Conducting Research Literature Reviews*: From the Internet to Paper. Sage Publications.

Anthony, M., Graziano, A.M. and Raulin, M.L., (2009). *Research Methods*: A Process of Inquiry, Allyn and Bacon.

Wilkinson & Bhandarkar.P.L.C (2010). *Methodology and Techniques of Social Research*, Himalaya Publishing House, Bombay.

Outcomes	Enable the students to ➢Gain knowledge on types of Research and Able to develop a Research Proposal. ➢Get familiar with various Statistical Tools used in the Research Methodology.

Course Teacher
Dr.P.Veeramani

SEMESTER – III			
Coursecode: 453T10	**CORE-X COMPUTER APPLICATION IN TEXTILES AND CLOTHING (P)**	**Credits: 5**	**Hours: 10**
Objectives	➢To impart in computer application skill in textile and fashion designing through adobe photoshop and adobe illustrator.		
	DESIGNING WITH ADOBE PHOTOSHOP Exercise – 1 Creating Stripe, Plaid and Print Repeats with different textured looks Exercise -2 Generating Color Combinations, Multi Color Prints & Tie Dyes, Rendering a Black and White Sketch with patterns clean sketch. Exercise-3 Creating New Prints: Any Type of Florals, Nautical, Ethnics or Any Type depending on the Current Trend of the Year. Exercise-4 Create Men's wear, Women's wear and Kids wear items with all types of Prints, Plaid patterns, Stripes and different textured looks Exercise-5 Create Presentation Board using Adobe Photoshop **DESIGNING WITH ADOBE ILLUSTRATOR** Exercise-6Sketch industry standard fashion flats Exercise-7Add bitmap graphics to vector files **DESIGN YOUR OWN VECTOR GRAPHICS** Exercise-8 Quickly render sketches with color and fabric patterns Trace, modify and recolor vector graphics Exercise-9 Create Presentation boards with text, background and shadow effects using illustrator.		
Reference and Textbooks: Kathleen Colussy .M. (2006), *Rendering Fashion, Fabric and Prints with Adobe Illustrator, Steve Greenberg,* Pearson Prentice Hall. Kothari, V. K, (2010). *Progress in Textile Science, Vol I, II and III, IAFL Publications*, New Delhi. John Wiley & Sons, (02-Nov-2011), *Fashion Designer's Handbook for Adobe Illustrator*, Marianne Centner, Frances Vereker. Frederick L Chipkin, Origin inc, (20-Apr-2014), *Adobe Photoshop for Textile Design*: for Adobe Photoshop CC (creative cloud). Robert Hume (30-Apr-2019), *Fashion and Textile Design with Photoshop and Illustrator*: Professional Creative PracticeBloomsbury Publishing USA.			
Outcomes	Enable the students to ➢Able to create new fabric prints and new colour combinations ➢Illustrate Apparel design with Photoshop and illustrator ➢Create presentation board with Photoshop and illustrator		

Course Teacher
Prof.K. Manimekalai
M.Priyanka

SEMESTER – III			
Course code: 453EC5	**ELECTIVE –III (A) FABRIC SOURCING**	**Credits: 4**	**Hours: 4**
Objectives	➢To teach the students on different fabric clusters in India. ➢To implant knowledge about Lead Time for Fabric Sourcing ➢To teach about the Import procedures and documentation used while importing fabrics and other inputs ➢To educate the importance of Supply Chain Management		
Unit - I	**Fundamentals of Fabric Sourcing** - How to buy Fabrics - Basic knowledge requisite for fabric sourcing - Vendor Analysis & Development - Selection of Vendor.		
Unit - II	**Knowledge of different fabric clusters in India -** Types of fabrics produced, and end use of different fabrics produced in India - Factors to be considered while sourcing from domestic suppliers - Overseas Fabrics sourcing locations - Knowledge of different overseas fabric locations - Sourcing from nominated vendors - Factors to be considered while sourcing from overseas		
Unit - III	**Lead Time for Fabric Sourcing** - Time and Action Calendar for Sourcing - Lead time calculation for Fabric Sourcing - Cost Benefit Analysis while fabric sourcing - Just in Time Sourcing - Import verses Domestic Sourcing - advantages and disadvantages		
Unit – IV	**Import of Fabrics** - Policy Procedure and Documentation for Importing Fabrics from global sources - Licenses required for importing fabrics from overseas markets - Clearing and Forwarding, port handling procedure while importing fabrics - Transportation for import of fabrics		
Unit - V	**Supply Chain Management &Inventory** - In-bound logistics & Out-bound logistics - Transportation - Internal and external - Warehouse management - storing & inventory systems - Stock Plans and issuing materials to operations - Procedure to issue materials to internal customers - Supermarket or Store operations in garment industry.		

Reference and Textbooks:

Grorver E.B. (1980). *Hand Book of textile testing,* Wiley Easters University, Delhi.

Booth JE (1996), *Principle of Textile Testing,* CBS Publication New Delhi.

SusheelaDantyagi, (1998), *Fundamentals of Textiles and their Care,5th edition*, Orient Longman Ltd New Delhi.

Ruth Clock & Grace Kunz (2000), *Apparel Manufacture- Sewn Product Analysis,* Upper Sadale River Publications, New York.

Anbumani (2006), *KnittingManufacture Technology,* New Age International, Chennai.

Gokarneshan. N. (2009), *Fabric Structure and Design* - New Age International.

Outcomes	Enable the students to ➢Gain knowledge on the fundamentals of Fabric sourcing methods and lead time ➢Become familiar with different fabric clusters in India ➢Learn the procedure of importing textiles ➢Learn the supply Chain Management system

Course Teacher
Dr.P.Veeramani
Dr.A.K.P.Dhanakodi

SEMESTER – III			
Coursecode:453EC6	**ELECTIVE –III (B) FUNCTIONAL FOODS AND NUTRACEUTICALS**	**Credits: 4**	**Hours: 4**

Objectives	➢To learn the Concepts on Functional Foods andNutraceuticals ➢To instruct the Isolation of photochemical from plant materials.
Unit - I	**Functional Foods andNutraceuticals–** Definition, concept, history and market; Evolution of nutraceuticals and functional foods market.Classification of nutraceuticals and functional foods.Significance and relevance of nutraceuticals and functional foods in the management of diseases and disorders.
Unit - II	**Natural occurrence of certain phytochemicals-** Antioxidants and flavonoids: omega – 3 fatty acids, carotenoids, dietary fiber, phytoestrogens; glucosinates; organosulphur compounds. Dosage for effective control of disease or health benefit with adequate safety; studies with animals and humans; acute and chronic studies.Regulatory issues.
Unit - III	**Isolation of phytochemicals from plant materials:** Care in handling and storage of raw materials with minimal damage to sensitive bioactive compounds; Extractive methods for maximum recovery and minimal recovery and minimal destruction of active material; stability studies.Recent developments in the isolation, purification and delivery of phytochemicals.
Unit – IV	**Prebiotics, probiotics and symbiotics- Probiotics:** Definition, types and relevance; Usefulness in gastro intestinal health and other health benefits; development of a probiotic products; recent advances in probiotics; Challenges and regulatory issues related to probiotic products. Prebiotics: Prebiotic ingredients in foods; types of prebiotics and their effects on gut microbes; health benefits of prebiotics; recent development in prebiotics. Symbiotic.
Unit - V	**Functional foods** - Definition, development of functional foods, use of bioactive compounds in appropriate form with protective substances and activators; Effect of environmental condition and food matrix; Effects of processing conditions and storage; Development of biomarkers to indicate efficacy of functional ingredients; Research frontiers in functional foods; delivery of immunomodulators /vaccines through functional foods. Nutrigenomics- concept of personalized medicine.Use of nanotechnology in functional food industry.

Reference and Textbooks:

Gibson GR & William CM (2000). *Functional Foods* - Concept to Product.

Goldberg I (2004). *Functional Foods: Designer Foods, Pharma Foods.*

Paresh, C. Dutta (2004), *Phytosterols as Functional Food Components and Nutraceticals,* Marcel Dekker Inc, New York.

Brigelius-Flohé, J & JoostHG (2006). *Nutritional Genomics: Impact on Health and Disease.* Wiley VCH.

Wildman, R.E.C. (2007) *Handbook of Nutraceuticals and Functional Foods*, second edition. CRC Press.

Outcomes	Enable the students to ➢Gain knowledge on Natural occurrence of certain photochemical ➢Understand about the Prebiotics, probiotics and symbiotics. ➢Use the nanotechnology in functional food industry.

Course Teacher
Prof.K. Manimekalai

SEMESTER – IV			
Coursecode: 453T11	**CORE - XI TECHINICAL TEXTILES (T)**	**Credits: 5**	**Hours: 5**

Objectives	➢To inculcate the knowledge of application of textile materials in various technical areas. ➢To make the students to understand the requirements of fibre and fabrics, specific mechanical and chemical processing requirements, ➢To understand the quality evaluation for the particular applications of textiles in industrial areas.
Unit - I	**Introduction to technical textiles:** Definition - Classification of technical textiles - Market growth and potential - application of technical textiles in various fields. Technical Fibres: High – Strength and high-modulus organic fibres, High chemical and combustion resistant organic fibres, High performance inorganic fibres, Ultra-fine and novelty fibres.
Unit - II	**Medical Textiles:** Classification of medical textiles - Non-implantable materials, extracorporeal devices, implantable materials, healthcare and hygiene products. Design and characteristics required in textiles for medical and hygiene applications.
Unit - III	**Protective Textiles**: Definition – Basic requirements – Garment design - choice of materials in protection from hazards due to thermal, flame, chemical, mechanical, ballistic, nuclear and extreme climate - Footwear textiles.
Unit – IV	**Filtration Textiles:** Introduction, selection considerations - Fiber, fabric, design. Principle mechanism and types of filtration - Textiles in dry filtration, wet filtration, and filtration equipment - Design consideration - Filter media for paper industry – Design and manufacturing techniques, cigarette filters.
Unit - V	**Automobile Textiles** – Introduction - fibres used – types – functions – properties – characteristics and applications. **Sports Textiles** – Introduction - fibres used – types – functions – properties – characteristics and applications.

Reference and Textbooks:

Peters R.H. (1967), *Textile Chemistry, Vol. II by*.

Trotman E.R. (1968), *Textile Scouring and Bleaching*.

Jarmila S Vedova (1990), *"Industrial Textiles",* Elsevier Science Publishing Inc, New York.

Harrison P W (1998), *"The Design of Textiles for Industrial Applications",* The Textile Institute, Manchester.

David B Wootton (2001), *"The Application of Textiles in Rubber",* Rapra Technology Ltd., UK.

Outcomes	Enable the students to ➢Gain knowledge ontextile materials in various technical areas. ➢Able to find the quality evaluationin industrial areas.

Course Teacher
Dr.P.Veeramani
Dr.A.K.P.Dhanakodi

SEMESTER – IV			
Coursecode: 453T12	**CORE-XII FASHION MARKETING**	**Credits: 5**	**Hours: 5**
Objectives	➢To teach about the Fashion Marketing ➢To impart different types of Marketing ➢To guide the promotional activities of Government Organizations.		
Unit - I	**Meaning and classification of Marketing**, Fashion Marketing, Fashion Market Size and Structure, Marketing Environment - Micro Marketing Environment, Macro Marketing Environment. Trends in Marketing Management.		
Unit - II	**Marketing Function** – Definition and Classification.Product Planning and Development.The Fashion Industry and New Product Development.Product Line Policies-Product Mix and Range Planning. Fashion Product Cycles.		
Unit - III	**Fashion Advertising** - Preparation, Medias, Various Departments & Agencies. Fashion Sales Promotional Programme for Apparel Marketing, Communication and Promotion, Personal Selling, Point of Purchase, Sales Promotion Objectives & Its Types.		
Unit – IV	**Merchandising** – Definition, Role of Merchandiser, Types. Domestic & International Export Market Environmental.Channels of Distribution.		
Unit - V	Promotional Activities of Government Organization Buyers & Sellers Meet. Costing – Basics & its Importance of Costing.		
Reference and Textbooks: Philip C.F & Duneon D.T Irwin; *3rd edition* (1956), *Marketing:* Principle and methods Irwin publications. Backman T.N. Maynard H.H & Davidson W.R (1970), *Principles of Marketing*, Ronald Press Company New York. Pillai & Bagavathi. R.S.N. (1987), *MarketingS. Chand & Co Ltd,* Newdelhi. Haroldcars / John Pomeroy, (1992), *Fashion Design & Product development,* Black Well Science, Inc., 238, Main Street, Cambridge. Mike Easey (1995) – *Fashion Marketing*, Black well Science Ltd.			
Outcomes	➢Students get familiar with Fashion Merchandising in General ➢Students gain knowledge on different types in Fashion Merchandising ➢Students become familiar with promotional activities of Government Organizations		

Course Teacher
Prof.K. Manimekalai
M.Priyanka

SEMESTER – IV			
Coursecode: 453T13	**CORE – XIII PORTFOLIO PRESENTATION (P)**	**Credits: 5**	**Hours: 10**
Objectives	➢To teach students in Portfolio Presentation		
	Prepare the theme based designer garments. 1. Trend Analysis and Fashion Forecasts 2. Ideation and concept formation 3. Inspiration Board 4. Mood Board- With Research Work. 5. Colour Palette 6. Customer Profile. 7. Story Board 8. Material Exploration 9. Fabric Sourcing – Presentation. 10. Design Development 11. Illustration with Back Drops. 12. Technical Specification Sheet. 13. Pattern making and garment construction for children, men and women – three woven and three knitted garments. 14. Final presentation 15. Designer Show. (1) 16. Exhibition. (5)		
Reference and Textbooks: Philip C.F & Duneon D.T Irwin; (1956), *Marketing: Principle and methods 3rd edition*, Irwin publications. Backman T.N. Maynard H.H & Davidson W.R (1970), *Principles of Marketing,* Ronald Press Company New York. Pillai & Bagavathi R.S.N. (1987), *MarketingS.* Chand & Co Ltd, Newdelhi. Haroldcars / John Pomeroy, (1992), *Fashion Design & Product development,* Black Well Science, Inc., 238, Main Street, Cambridge. Mike Easey (1995) – *Fashion Marketing*, Black well Science Ltd.			
Outcomes	Enable the students to ➢ Get aware about the importance of Portfolio Presentation and able to present their own Portfolio.		

Course Teacher
Prof.K. Manimekalai
M.Priyanka

SEMESTER – IV			
Coursecode: 453T14	**DISSERTATION**	**Credits: 6**	**Hours: 10**
Objectives	➢ Help the students to learn the knowledge about the project		

Preliminary

1. Title Page- title, Authors Name
2. Certificate of Originality by the Guide
3. Declaration by the Author
4. Table of Contents
5. List of Tables
6. List of Figures
7. Acknowledgement
8. Abstract

I. Introduction: Statement of the Problem, Significance, Need for the Study, Objectives, and Definitions.
II. Review of Literature
III. Methodology: Tools used, Procedures, Hypothesis.
IV. Results and Discussion: Tables and Figures, Statistical Presentations, Hypothesis Testing.
V. Summary and Conclusion
VI. Suggestion for the Future Study
VII. References

Outcomes	➢ They will gain in depth knowledge about the research ethics.

Course Teacher
Prof.K. Manimekalai
Dr.A.K.P.Dhanakodi
M.Priyanka

NON- MAJOR ELECTIVE FOR OTHER DEPARTMENTS			
Coursecode: 453NME15	**PUBLIC HEALTH NUTRITION**	**Credits: 2**	**Hours: 3**
Objectives	➢ To understand the concept of public health and nutrition ➢ To be familiar with the current concerns in public health nutrition		
Unit - I	**Concept and scope of public nutrition -** Relation between health and nutrition - Definition and multidisciplinary nature of public nutrition. -Concept and scope - Role of public nutritionist -Millennium Development goals - Current concerns in public health nutrition		
Unit - II	**Introduction to Nutritional deficiency diseases -** Causes, symptoms, treatment, prevention of the following:Protein Energy Malnutrition (PEM), Vitamin A Deficiency (VAD), Iron Deficiency Anemia (IDA), Iodine Deficiency Disorders (IDD), Zinc Deficiency, Flurosis,		
Unit - III	**Social health problems -** Smoking, Alcoholism, Drug addiction, AIDS including AIDS Control Programme, community health care, national health care delivery system, population policy. Health economic and economics of malnutrition impact on productivity and national development.		
Unit – IV	**Nutrition for special conditions -** Introduction to Nutrition for physical fitness and sport, Feeding problems in children with special needs, Considerations during natural and man-made disasters e.g. floods, war.- basic guidelines in disaster management.		
Unit - V	**Food Security -** (a) Key terms, factors affecting food security, recent concern- (b) Technologies for food and nutrition security - (c) Health based interventions including immunization, provision of safe drinking water/sanitation, prevention and management of diarrhoeal diseases.		

Reference and Textbooks:

Srilakshmi B (2002); *Dietetics;* Fourth Ed; New Age International (P) Ltd.

Sharma S, Wadhwa A (2003); *Nutrition in the Community- A textbook;* Elite Publishing House Pvt. Ltd.

Wardlaw GM, Hampl; (2007), JS. *Perspectives in Nutrition; Seventh Ed;* McGraw Hill.

Mudambi, SR &Rajagopal, MV. *Fundamentals of Foods, Nutrition and Diet Therapy; Fifth Ed;* (2007); New Age International Publishers.

Sethi P &lakra P Aahaar vigyaan ,Poshan evam Suruksha, Elite Publishing house, (2015) Khanna K et al. *Textbook of Nutrition and Dietetics;* (2013); Phoenix Publisher.

Outcomes	Enable the students to ➢ Gain knowledge about public health and nutrition ➢ familiar with the current concerns in public health nutrition

Course Teacher
Dr.P.Veeramani

NON- MAJOR ELECTIVE FOR OTHER DEPARTMENTS			
Coursecode: 453NME16	**HOME TEXTILES**	**Credits: 2**	**Hours: 3**
Objectives	➢ To teach Introduction to Home Textiles ➢ To impart knowledge in Finishes and quality parameters used in Home Textiles		
Unit - I	**Introduction to Home Textiles** – Fabrics used for Living area, Kitchen, Dining, Bed room, Bath room accessories and general household textiles. Textile for Furnishing and Upholstery applications – scope, fixed upholstery, non-stretch loose covers & stretch covers		
Unit - II	**Window/ Curtain Fabrics** - Sun filters (Sheers and nets), Semi-sheers, Reflective textiles, drapes and Blinds. Bed Room Textile - Sheets & Pillow Cases, Quilted Textile, and Blankets& Rugs - Jacquard blankets, Printed blankets, Fire proof blankets, Baby blankets. Bed Spreads, Mattress covers		
Unit - III	**Dining Room Textiles Tablecloths** – Colour – Woven & Printed type, jacquard types, embroidered types, non-woven types. Table mats – Colour -woven, Printed jacquard, embroidered. Kitchen Textiles Aprons, Dish cloth, Tea cosy, Bread bag, Mittens, Pot Holders, Table Mats – Construction & manufacturing		
Unit – IV	**Bath Room Textiles** – bath towels, bathrobes, floor mats, handtowels, beach towels, towels for swimming pools.- Terry Towels – types and applications. Floor Coverings – Introduction, Pile fabrics, Backing fibres & fabrics – Tufted carpets, woven carpet.		
Unit - V	**Finishes and quality parameters used in Home Textiles** A. Protection against unpleasant odour, B. Temperature regulated beddings C. Antimicrobial finish, D. Moisture management E. Test Method for towels, Rug and Home textiles and F. Performance specifications of different home textiles		

Reference and Textbooks:

Tate of. Glesson (1961), *Family Clothing, Latest Edition,* John Wiley &Sons Inc., Illinois.

Dantyagi. S. (1980), *Fundamentals of Textiles & their Care.* Orient Langnass Ltd. New Delhi.

Talukdar, Ajagaonkar, Sriramulu (1998), *Weaving Machine, Mechanisms, Management.*

Neomi D'Souza (1998*), Fabric Care,* New Age International Publisher.

Subrata Das (2010), *Performance of Home Textiles,* Wood head Publications India Pvt Ltd.

Outcomes	➢ Students grain the knowledge describes various types of home textiles and their manufacturing methods. ➢ Students able to understand requirements of different home textiles ➢ Students learn to understand various finishes and evaluation methods of home textile.

Course Teacher
Prof.K. Manimekalai
M.Priyanka

BROAD BASED BOARD OF STUDIES MEMBE

CURRICULAM VITAE

Dr.K.MANIMEKALAI
Professor & Director
Department of Women's Studies
Alagappa University, Karaikudi
70100 83223
gomathimanimekalai@gmail.com

ACADEMIC QUALIFICATION

- **Ph.D.** 2002, Highly Commended, Women's Studies, Alagappa University, Karaikudi
- **M.Phil**. 1987, Commerce, Alagappa University, Karaikudi
- **M.Com**. 1986, Commerce, Sri Parasakthi College For Women (An Autonomous College) Courtallam, Madurai Kamaraj University, Madurai

ACADEMIC EXPERIENCE / SERVICE DETAILS

- **Vice-Chancellor,** 20th July 2012 to 19th July 2015, Mother Teresa Women's University, Kodaikanal
- **Registrar i/c**. 7th October 2010 to 19th July 2012, Alagappa University, Karaikudi
- **Director**, from5th November 2007 Centre for Women's Studies, Alagappa University, Karaikudi

PRIZE, MEDALS AND OTHER HONOURS

- Life Time Achievement Award, Academy of Universal Global Peace – USA, 2017.
- Appreciation Award – 2017, Alagappa University, Karaikudi
- Innovative B –School Award for Outstanding Contribution to Education, – 2016, 8th DNA and Stars of the Industry Group, Mumbai
- Innovative B –School Award for Outstanding Contribution to Education, – 2015, 7th DNA and Stars of the Industry Group, Mumbai

MEMBERSHIP IN ACADEMIC BODIES

- Governor's Nominee – Selection Committee, Bharathiyar University
- Governor's Nominee – Selection Committee, University of Madras
- Vice Chancellor's Nominee – Selection Committee, Bharathiyar University

CURRICULUM VITAE

Dr.P.VEERAMANI

Assistant Professor
Centre for Women's Studies
Alagappa University
Karaikudi-630 003
Sivagangai District
84897 94777
veeramws@gmail.com

ACADEMIC QUALIFICATION

- Ph.D. 2016, Highly Commended, Women's Studies, Alagappa University, Karaikudi
- M.Phil. 2005, Women's Studies Mother Teresa Women's University, Kodaikanal
- M.W.S & C.A. 2003, Master of Women's Studies and Computer Application, Alagappa University, Karaikudi

ACADEMIC EXPERIENCE / SERVICE DETAILS

- Assistant Professor, 27th June 2013 to till date, Centre for Women's Studies, Alagappa University, Karaikudi

MEMBERSHIP IN PROFESSIONAL

- Life Member - Indian Association for Women's Studies
- Life Member –Indian Academic Researchers Association.
- Member – Centre on Swasch Bharat Swasch Abiyan, Alagappa University, Karaikudi
- Hub Co-ordinator of Alagappa University in Entrepreneurship Development and Innovation Institute, Government of Tamil Nadu, Chennai
- SWAYAM Course Co-ordinator, Department of Women's Studies, Alagappa University, Karaikudi.

SEMINARS/WORKSHOPS/CONFERENCES

- Articles Published : 30
- International Seminar (Paper Presented) : 7
- National Seminar (Paper Presented) : 30
- Participated : 56

CURRICULUM VITAE

Dr.Meera Kaur, M.Sc., Ph.D., R.D., C.D.E.
Professor, Family Medicine
Max Rady College of Medicine, University of Manitoba
Meera.kaur@umanitoba.ca
CANADA

Dr. Meera Kaur is Associate Professor of Family Medicine, Max RadyCollege of medicine, Rady Faculty of Health Sciences at the University ofManitoba Canada, Faculty Research Supervisor at the Regent UniversitySchool of Medicine and Allied Sciences, Accra, Ghana and Conference Co-Chair of International Conference on Food Nutrition and Health (ICFNH).

Previously, she served as Adjunct Professor of Biochemistry, Faculty ofScience, University of Winnipeg and Research Consultant at the Instituteof Cardiovascular Sciences, St. Boniface General Hospital Research Centre,Winnipeg, Canada. Dr. Kaur holds a Ph.D. in Food Science specializing inMedical Nutrition (Enteral Food) from the leading Central FoodTechnological Research Institute (CFTRI), Mysore, India. Before immigrating to Canada, sheserved as Associate Professor in the department of Food and Nutrition at the Assam AgriculturalUniversity, Jorhat, India for over 15 years, where she has introduced several courses atundergraduate and graduate levels in the areas of Food Science and Clinical Nutrition.

Dr. Kaur's research interests lie in areas of Normal and Therapeutic Nutrition, Food Science andFamily Medicine. She has published research papers in prestigious journals like ClinicalLipidology, Journal of Functional Food, and Fundamental and Clinical Pharmacology, and haspublished several encyclopedia articles for SAGE and Springer publication, USA. She is also arecipient of three patents for her seminal research in the area of Enteral Foods, one of which islicensed to a renowned multinational pharmaceutical company. Based on her research, she haswritten a book entitled *Medical Foods from Natural Sources* published by Springer, USA(http://www.springer.com/food+science/book/978-0-387-79377-1). Springer recognizes thisbook as an extremely valuable asset in their program. Dr. Kaur is also a frequently soughtspeaker and has been presenting several research and invited papers at national andinternational conferences.

Dr. Kaur has received several awards in recognition of her contribution towards teaching,research and outstanding service to patients' care. A few of her awards worth mentioning arethe VAR's Nominee Award from the American Society of Parenteral and enteral Nutrition(ASPEN), USA; Merit Award from the Department of Immigration and Multiculturism,Government of Manitoba, Canada; Health Promotion Award from Seven Oaks General Hospital,Canada; Jawaharlal Nehru Award and Woman Scientist Award from the Government of India; Dr.A.K. Bose Memorial Award from Indian Dietetic association and Nutrition Society of India; and,Excellent Performance Award from Claris Life Sciences Ltd., India. Dr. Kaur can be contacted atmeera.kaur@umanitoba.ca or visit http://home.cc.umanitoba.ca/~kaur for further information.

CURRICULUM VITAE

S. LAKSHMI MANOKARI M.Sc., M.Phil., PhD
Professor and Head
Department of Textiles and Apparel Design
Periyar University, Salem-636 011
lakshmimanokari@yahoo.co.in
9943997453

EXPERIENCE
Teaching -21years
Research -17 Years
Administrative Experience (Head of the Department) -17 Years
Bachelor of Vocation Coordinator -3 Years

ON GOING FUNDED PROJECTS
Major Research Project Funded by UGC (Rs.9,91,000) for the Title "Synthesis of phytonano particles using Wrightia tinctoria and developing wound dressing with viscose non-woven for diabetic wound"

ON GOING SCHEMES
Received Grant from UGC to establish Bachelor of Vocation in Textiles and Apparel Design for 85 lakhs

TOTAL PUBLICATIONS
Journals -14, Conference Proceedings-03, Book Chapters-06 and Book-01

AWARDS AND ACKNOWLEDGEMENTS
2007- Certificate of Excellence for Contribution in Education by Lions Club of Erode Metro
2014- Best Teacher Award by Kongu Arts and Science college
2014- III Prize and Cash Award for Best Research Paper Presentation by Meditex International Conference, South India Textile Research Association, Coimbatore

COMPLETED PROJECTS

Coordinator 2006	To develop Virtual training module For Fashion designing, Beauty care and embroidery for Kwicksoft solutions pvt Ltd, Chennai **(Rs.94, 390)**
2006-2007	To Train Backward class Beneficiaries- in Fashion designing, Embroidery and Beauty care funded by National Backward Classes Finance Development Corporation **(Rs.2,67,000)**
2011	For Skill development training in Beauty care under SJSRY (Step up) funded by Erode City Municipal Corporation **(Rs. 1,58,650)**

CURRICULUM VITAE

Name : Dr.K.SANGEETHA
Qualification : M.Sc., M.Phil., Ph.D.,
Designation : Professor and Head
Department : Textiles and Apparel Design
Institution : Bharathiar University, Coimbatore – 641046
E-Mail : sangashreeram@gmail.com
Mobile : 98940-13466
Date of Birth : 20-07-1978
Service : 18 Years
Field ofSpecialization : Sustainable Textiles:Natural Fibres, Natural Dyes, Medical Textiles
TeachingExperience : 18Years
ResearchExperience : 15 Years
No. of Patents Applied : 2
Supervised : M.Sc. – 36;
M. Phil – 10;
Ph. D –2 (Completed); 7 (on going)

Books/Chapters Published:

Books - 6
Text Books for Diploma courses (Tamil Nadu Open University, Chennai) - 3
Course Materials for E-PG Pathsala - 2
Chapter in Books - 1
Edited Books - 3

Research Articles Published :

International Journals : 26

Conference Proceedings with ISBN:

International : 16
National : 20

Presentation in Conferences / Seminars:

International : 23
National : 50

Seminar / Conference Organized: 7

WorkshopsOrganized : 10

CURRICULUM VITAE

G. Somasundaram
Communication Address:
7/1, 2ndFloor, 6th Cross,
Venkateswara Layout,
S. G. Palaya,Dharmaram College Post
Bangalore - 560029
Mobile: +91 9343756560
E-mail: soms28@gmail.com

Current position:
South Zone Head of IL & FS Clusters, Bangalore

Key Skills

✓Build, Manage and Operate Learning & Development for Fashion/Apparel industry
✓Talent Acquisition & Talent Engagement
✓Curriculum & Content Development
✓Campus to Corporate Talent Development
✓MS Office & Internet communication
✓HR Compliance Audit & implementation

BBA *VHNSN College, Virudhunagar,*
Madurai Kamaraj University,
Regular – I Class

MBA *Department Of Management Studies,*
Madurai Kamaraj University,
Regular –I Class

PG Dip inInternationalBusiness
Pondicherry University,
DEP - I Class

Courses Attended & Professional Training:

a. Administrative Staff College of India
b. Retail Summit 2000 by Retailers Association of India, Bangalore
C. Indian Institute of Management (Lucknow): TrainingMethods & Skill Development June 2007
d. Productivity Improvement Techniques – JukiResearch Institute, Japan & Juki India
e. Retail Leadership Summit 2011, Mumbai – RetailersAssociation of India

Countries visited:

Japan: in October, 1999 5th Asian Textile Conferenceheld in Kyoto thro' NIFTHong Kong: 1999 research survey work for Dr. SriramKhanna on "Competitiveness of Indian Textile and Clothing Industry in post- MFA era"- for Min ofTextiles, Government of India.

Professional Experience:
Bangalore (since April 2016 till date)

✓IL & FS Cluster Development Initiative ltd,
✓ **Senior Manager – HR (Learning and Development)**in Page Industries (Jockey) (April 2014 till Dec, 2015)

✓*Adjunct Professor (on Contract) National Institute of Fashion Technology, Bangalore(June 2013 to Jan 2014)*

✓***Senior Manager HR*** **,Gokaldas Exports Ltd. (April**2012 to May 2013)

✓*Fashion & Retail Marketing Project Consultant*
PEARL ACADEMY OF FASHION, Chennai (Feb 2011 to April 2012)

✓*Regional Head- Skill Project & Head – Industry Linkages* ESDL, Chennai (Jun 2010 to Feb 2011)

✓ **Regional Head – Skill Development & IndustryLinkages, IL & FS Clusters** New Delhi (Ministry ofRural Development – Project SEAM for GarmentIndustry Pan India) (Apr 2007 to April 2010)

✓General Manager (Business Development – ERP) Reach Sewn Technologies and Consulting Pvt. Limited (Oct 2006 to March 2007)

✓ **Officiating Principal & Associate Professor, ArmyInstitute of Fashion & Design,** Bangalore (UnderArmy Welfare Education Society) (June 2005 toSept 2006)

✓**Assistant Professor & Centre Head for Fashion Management Department NationalInstitute ofFashion Technology, (NIFT)** (Min of Textiles, Govt. of India) Bangalore (July 1997 to Apr 2005)

✓**Head of the Department, Business Management,**

Kongu Arts & Science College, Erode (Aug 1995 to June 1997)

✓Export Merchandising Service for major garment manufactures (Jan 1994 to July 1995)

✓Merchandiser (Fabric) A& E Trading Pvt Ltd, Kumbakonam, (Apr 1993 to Jan 1994)
✓Export Executive, Meera Enterprises Pvt Ltd., Chennai (Sept 1989 to Mar 1993)
***Member of National HRD network (Hosur-Bangalore South Chapter)**

Signed by: G. Somasundar